ChronoMorph: Dual-View Generative Classification for Few-Shot Time Series

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Abstract

Few-shot time series classification (TSC) is challenging because each class provides only a few labeled sequences, yet reliable prediction often requires capturing both temporal dynamics and waveform morphology, as human analysts do when combining numerical evolution with shape-based evidence. Recent pretrained causal language models offer a promising interface for data-efficient transfer, but directly applying them to TSC exposes two mismatches: continuous time-series signals are not naturally aligned with the language-token space, and unconstrained autoregressive decoding is poorly suited to closed-set classification. We propose **ChronoMorph**, a dual-view generative transfer framework for few-shot TSC. ChronoMorph encodes each sequence through a temporal branch that captures ordered dependencies and a morphology branch that converts each waveform into a morphology map. The resulting continuous tokens are fused and projected into a pretrained causal language model via stage-wise transfer pretraining. Instead of unconstrained decoding, ChronoMorph performs supervised one-pass scoring over dataset-specific class tokens, which can be initialized and regularized by label-verbalizer prototypes when class names are semantically meaningful. Extensive few-shot experiments on the UCR benchmark and additional external datasets show that ChronoMorph consistently outperforms strong task-specific, Transformer-style, and LLM/foundation baselines. Further analyses validate the contributions of dual-view modeling, transfer pretraining, and the class-token interface.

1 Introduction

Time series classification (TSC) is a fundamental problem in healthcare monitoring, industrial diagnosis, sensor analytics, and scientific discovery [Ismail Fawaz et al., 2019]. In many practical scenarios, however, labeled time series are difficult to obtain: annotation often requires domain expertise, new categories may appear with only a few examples, and class distributions can be highly imbalanced. Few-shot TSC therefore requires more than fitting a classifier on a small support set. A successful model must retain transferable temporal evidence while inducing a stable decision rule from very limited supervision [Tang et al., 2020].

A key challenge is that discriminative evidence in time series is inherently heterogeneous. Some classes are distinguished by chronological dynamics, such as local trends, amplitudes, phase shifts, and long-range evolution. Others depend more strongly on waveform morphology, such as recurring peaks, valleys, cycles, motifs, or discriminative subsequences. This dual nature is reflected in the history of TSC methods: shapelet-based approaches search for local waveform patterns predictive of class membership, convolutional and patch-based models capture ordered temporal context through receptive fields, and time-series imaging methods transform sequences into visual structures to expose global morphology and cross-local relations [Ye and Keogh, 2009b, Dempster et al., 2020, Ismail Fawaz et al., 2020, Wang and Oates, 2015a]. Under few-shot supervision, this heterogeneity

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becomes especially problematic. With only a few labeled examples, a model that emphasizes a single form of evidence may latch onto unstable cues and miss the complementary temporal or morphological structure needed to separate fine-grained classes.

Recent studies have explored adapting pretrained language models to time-series analysis through numerical serialization, reprogramming, embedding alignment, semantic anchors, and parameter-efficient fine-tuning [Brown et al., 2020, Gruver et al., 2023, Zhou et al., 2023, Jin et al., 2024b, Sun et al., 2023, Jin et al., 2024a, Chang et al., 2025, Zhang et al., 2024, Jiang et al., 2024]. These models are attractive for few-shot TSC because they provide reusable pretrained computation, conditional-generation objectives, and token-level scoring interfaces. However, many language-model-based time-series methods are designed around forecasting, reconstruction, or unconstrained text generation, while few-shot TSC requires closed-set discrimination over dataset-specific labels. Moreover, time-series evidence is continuous, domain-specific, and structurally different from the discrete token sequences on which language models are pretrained. As a result, the central challenge is not merely to adapt a language model to time series, but to design an interface that preserves task-relevant temporal evidence, aligns it with the language-model input space, and supports reliable closed-set prediction.

This perspective suggests three requirements for LLM-based few-shot TSC: (i) the representation should preserve both chronological dynamics and morphology-oriented evidence, since either form may carry the decisive class signal; (ii) continuous time-series features should be aligned with the embedding space expected by the pretrained language model before downstream adaptation; and (iii) prediction should compare valid class candidates directly, rather than relying on unconstrained text generation [Raffel et al., 2020, Hokamp and Liu, 2017, Koo et al., 2024]. Failing to satisfy any of these requirements can weaken the role of the language model as a reliable few-shot classifier and reduce it to a loosely connected generative component.

We propose **ChronoMorph**, a dual-view generative transfer framework for few-shot TSC. For each sequence, ChronoMorph constructs two complementary views. A temporal branch models ordered local relations and long-range dynamics, while a morphology branch transforms the waveform into a two-dimensional morphology map that makes shape-level structure accessible to a frozen vision encoder. The two views are fused into continuous tokens and projected into a pretrained causal language model through stage-wise transfer pretraining, which reduces the mismatch between time-series representations and the language-model input space. For downstream classification, ChronoMorph replaces unconstrained decoding with supervised one-pass scoring over dataset-specific class tokens, turning the generative interface into a discriminative closed-set classifier. When class names carry meaningful semantics, label verbalizers initialize and regularize the corresponding class-token embeddings, allowing semantic information to help the same single-token decision rule without requiring multi-token phrase decoding.

Our main contributions are: (i) we formulate LLM-based few-shot TSC as an interface-design problem that jointly considers complementary time-series evidence, continuous-to-language alignment, and closed-set prediction; (ii) we introduce a dual-view time-series-to-language encoder that preserves chronological and morphology-oriented evidence before transferring continuous tokens into a pretrained causal language model through stage-wise pretraining; and (iii) we propose a supervised class-token decision interface for one-pass scoring over valid labels, and extend it with label-verbalizer initialization and regularization when class names are semantically meaningful. Across 128 UCR datasets, ChronoMorph achieves the best average accuracy and average rank under 1-, 2-, 5-, and 10-shot settings; additional semantic-label experiments further show when label semantics can improve the same class-token interface.

2 Related Work

LLMs and foundation models for time series. Recent surveys show that foundation-model research for time series has expanded from adapting pretrained language models to building native temporal foundation models Zhang et al. [2024], Jiang et al. [2024], Ye et al. [2024], Shi et al. [2025]. Representative adaptation-based methods include TEST Sun et al. [2023], which aligns time-series embeddings with the LLM embedding space through instance-, feature-, and text-prototype-aware contrastive learning; Time-LLM Jin et al. [2024a], which reprograms time-series patches into a frozen LLM via text prototypes; and LLM4TS Chang et al. [2025], which uses a two-stage alignment-and-task fine-tuning strategy. OpenTSLM Langer et al. [2025] moves toward time-series language models that treat time series as a native modality through soft-prompt or cross-attention interfaces. At the same time, recent ablation studies caution that LLM components do not automatically improve time-series performance when the modality interface is poorly designed Tan et al. [2024], Zheng et al. [2024]. The closest few-shot classification line is LLMFew Chen et al. [2025], which introduces a patch-wise temporal convolution encoder and LoRA-tuned decoder for few-shot multivariate TSC. In contrast, our work focuses on dual-view temporal evidence, explicit time-series-to-language transfer, and generative label-token scoring rather than an external classifier head.

Temporal and morphological evidence in TSC. Classical and deep TSC methods support the view that discriminative evidence can appear at multiple structural levels. Shapelet methods identify local subsequences with high class-discriminative power Ye and Keogh [2009a], Tang et al. [2020], convolutional methods such as InceptionTime and ROCKET exploit local receptive fields and kernel responses at multiple scales Ismail Fawaz et al. [2020], Dempster et al. [2020], and patch-based Transformers preserve local subseries information while enabling longer-context attention Nie et al. [2022]. These methods primarily operate in the chronological-dependency space. In parallel, time-series imaging methods transform one-dimensional signals into two-dimensional structures such as Gramian angular fields and Markov transition fields, enabling vision models to capture global shape and transition patterns Wang and Oates [2015b]. Recent work further shows that visualization can improve LLM reasoning over time series and that frozen vision transformers applied to imaged time series can produce strong classification representations Liu et al. [2025], Roschmann et al. [2025]. These two lines motivate our dual-space hypothesis: few-shot TSC labels may depend on both chronological dependencies and morphological structures.

Generative classification and semantic-prior label spaces. Casting classification as generation is widely used in text-to-text learning, where labels are generated as target tokens under the same sequence objective as other tasks Raffel et al. [2020]. However, unconstrained decoding is mismatched with closed-set classification because many generated strings are irrelevant to the valid label set. Prior work on constrained decoding motivates restricting generation to valid lexical or structural outputs Hokamp and Liu [2017], Koo et al. [2024]. ChronoMorph adopts the same closed-set principle but implements the final decision as supervised one-pass class-token scoring: the model learns dataset-specific class-token rows and compares their next-token logits rather than performing unconstrained generation. For datasets with meaningful label names, we further use label verbalizers as semantic priors over the same class-token rows. This keeps the decision aligned with the autoregressive interface while avoiding the length bias, shared-prefix artifacts, and language-prior sensitivity of direct phrase-likelihood classification.

3 Method

3.1 Unified Time-Series-to-Language Formulation

Given a time series $x \in \mathbb{R}^T$, ChronoMorph casts time-series prediction as conditional language modeling with continuous time-series tokens inserted into a textual prompt. Let $\tau(\cdot)$ be the tokenizer, E_{lm} the LLM input embedding matrix, P_ψ the projector into the LLM hidden space, and $Z_x \in \mathbb{R}^{N_x \times d_e}$ the token sequence produced by the time-series encoder. For a prefix p^{pre}, an optional time-series description p^{ts}, and a suffix p^{post}, the mixed input embeddings are

$$H_x = \left[E_{\mathrm{lm}}(\tau(p^{\mathrm{pre}})); E_{\mathrm{lm}}(\tau(p^{\mathrm{ts}})); P_\psi(Z_x); E_{\mathrm{lm}}(\tau(p^{\mathrm{post}}))\right]. \tag{1}$$

For an answer sequence $a = (a_1, \ldots, a_M)$, ChronoMorph optimizes the standard causal objective but masks the loss outside the answer span:

$$\mathcal{L}_{\mathrm{LM}}(a \mid x, p) = -\sum_{m=1}^{M} \log p_\Theta(a_m \mid H_x, E_{\mathrm{lm}}(a_{<m})). \tag{2}$$

Thus question answering, captioning, and classification share the same conditional-generation interface; classification is implemented as label-token prediction rather than as a separate discriminative head. Figure 1 summarizes the pipeline.

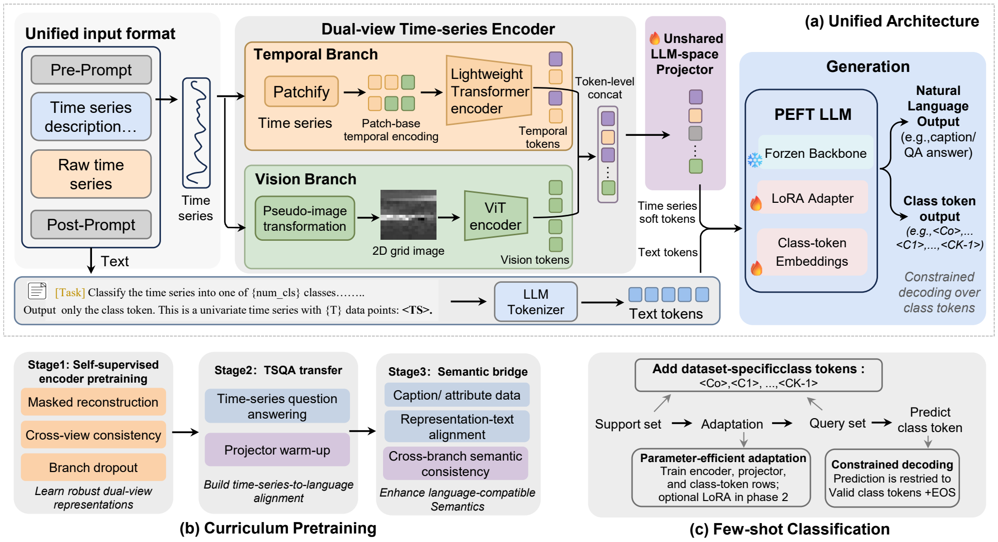

Figure 1: Overview of ChronoMorph. A temporal branch and a morphology-map visual branch produce dual-view time-series tokens, which are fused and projected into a pretrained causal LLM. Stage I learns dual-view representations, Stage II learns time-series-conditioned generation, and optional Stage III adds semantic enrichment; the main UCR benchmark uses the Stage II checkpoint. Downstream few-shot classification uses dataset-specific class tokens and one-pass scoring over the valid label set.

3.2 Dual-View Time-Series Tokens

ChronoMorph preserves two complementary evidence spaces before language conditioning. The temporal branch first extracts overlapping patches

$$R_i = x_{(i-1)s+1:(i-1)s+\ell}, \qquad i = 1, \dots, N_t, \qquad N_t = \left\lfloor \frac{T-\ell}{s} \right\rfloor + 1, \tag{3}$$

where ℓ and s denote patch length and stride. A dynamic PatchTST-style encoder with sinusoidal positions then maps the patch sequence to temporal tokens,

$$Z^t = G_t([W_t R_1 + \pi_1; \dots; W_t R_{N_t} + \pi_{N_t}]) , \tag{4}$$

using three Transformer blocks, hidden size 128, eight attention heads, and patch length/stride 16/8 in the main experiments. This branch targets ordered local relations, phase shifts, and long-range chronological dependencies.

The visual branch is designed to expose waveform morphology to a pretrained vision encoder without using dataset-level image statistics. We first apply a deterministic per-sequence transform. Each sequence is normalized by its own median and interquartile range,

$$\bar{x}_t = \frac{x_t - \mathrm{median}(x)}{\mathrm{IQR}(x) + \epsilon}, \tag{5}$$

where the statistics are computed only from the current sequence. To retain local waveform continuity while making non-adjacent windows jointly visible, we set the window length to $q = \lfloor \sqrt{T} \rfloor$ and the window stride to $\delta = \max(1, \lfloor \rho q \rfloor)$, with $\rho = 0.1$. The normalized signal is unfolded into an overlapping window grid, contrast-normalized, resized to the vision-backbone resolution, and replicated across RGB channels:

$$I(x) = \mathrm{RGB}(\mathrm{Resize}(\gamma(\mathrm{Unfold}_{q,\delta}(\bar{x})))) , \tag{6}$$

where $\gamma(\cdot)$ denotes per-sample image-grid normalization and contrast adjustment. The transform is fixed and the vision backbone remains frozen; adaptation is handled by the lightweight projectors introduced below. We feed $I(x)$ to `facebook/dinov2-base` and use the patch tokens from its fourth hidden layer, excluding the class token, as morphology-oriented tokens:

$$Z^v = G_v(I(x)). \tag{7}$$

The two streams are mapped to the same encoder dimension by lightweight MLP projectors and fused by token concatenation in the default configuration,

$$\tilde{Z}^t = Q_t(Z^t), \qquad \tilde{Z}^v = Q_v(Z^v), \qquad Z_x = \left[\tilde{Z}^t; \tilde{Z}^v\right]. \tag{8}$$

This late token-level fusion avoids prematurely collapsing chronological and morphological cues. Pooled temporal, visual, and fused representations are also used by auxiliary pretraining objectives.

3.3 Curriculum Bridging

The few-shot support set is too small to learn the time-series encoder, the time-series-to-LLM interface, and the decision rule from scratch. ChronoMorph therefore uses a staged transfer curriculum. Stage I trains the dual-view encoder with masked patch reconstruction, VICReg losses on two augmented views of the temporal, visual, and fused pooled representations, and branch dropout. For two augmentations $x^{(1)}$ and $x^{(2)}$, let $h_b^{(r)} = \mathrm{Pool}_b(Z_b(x^{(r)}))$ denote the pooled representation of branch $b \in \{t, v, f\}$, where f denotes the fused branch. We use

$$\mathcal{L}_{\mathrm{ssl}}^b = \alpha \left\| h_b^{(1)} - h_b^{(2)} \right\|_2^2 + \beta \sum_{r=1}^{2} \sum_j \max\Big(0, 1 - \sigma_j(h_b^{(r)})\Big) + \gamma \sum_{r=1}^{2} \left\| \mathrm{offdiag}\Big(\mathrm{Cov}(h_b^{(r)})\Big) \right\|_2^2, \tag{9}$$

which combines invariance, variance, and covariance terms. The masked reconstruction loss predicts temporal patches from a masked input,

$$\mathcal{L}_{\mathrm{rec}} = \frac{1}{2} \sum_{r=1}^{2} \left\| R_\omega\Big(Z_t(\mathcal{M}(x^{(r)}))\Big) - \mathcal{P}_{\ell,s}(x^{(r)}) \right\|_2^2, \tag{10}$$

where $\mathcal{M}$ masks a contiguous subsequence, $\mathcal{P}_{\ell,s}$ extracts temporal patches, and R_ω is the reconstruction head. The Stage I objective is

$$\mathcal{L}_{\mathrm{I}} = \lambda_{\mathrm{rec}} \mathcal{L}_{\mathrm{rec}} + \lambda_t \mathcal{L}_{\mathrm{ssl}}^t + \lambda_v \mathcal{L}_{\mathrm{ssl}}^v + \lambda_f \mathcal{L}_{\mathrm{ssl}}^f. \tag{11}$$

This stage uses raw TSQA-train series, raw M4-train series, and unlabeled official TRAIN-split sequences from a fixed 98-dataset UCR pool; no UCR labels or official TEST split are used. Because this pool is drawn from the same archive as the UCR benchmark, the main results should be interpreted as unlabeled-source transfer rather than strict leave-dataset-out pretraining.

Stage II attaches the projector and frozen causal LLM and trains time-series question answering,

$$\mathcal{L}_{\mathrm{II}} = \mathbb{E}_{(x,p,a)\sim\mathcal{D}_{\mathrm{TSQA}}} \left[\mathcal{L}_{\mathrm{LM}}(a \mid x, p)\right], \tag{12}$$

first warming up the projector and then jointly optimizing the encoder and projector. The Stage II checkpoint is the default initialization for UCR few-shot classification. Stage III is optional and mixes TSQA, M4 captioning, and synthetic attribute supervision with representation-text alignment and branch-consistency losses; it is used only for semantic enrichment, not for the default UCR results. Appendix B.1 gives the stage-wise data details.

3.4 Class-Token Decision Interface

For a dataset D with C classes, we introduce class tokens $\mathcal{C}_D = \{\langle c_0 \rangle, \ldots, \langle c_{C-1} \rangle\}$ with vocabulary indices $\mathcal{V}_D = \{v_0, \ldots, v_{C-1}\}$. Let $h_\Theta(x, p)$ be the final hidden state at the last prompt position after conditioning on H_x, and let W_{out} be the LLM output embedding matrix. The one-pass score for class c is

$$s_c(x) = \left[W_{\mathrm{out}} h_\Theta(x, p)\right]_{v_c}, \qquad \hat{y} = \arg\max_{c \in \{0,\ldots,C-1\}} s_c(x). \tag{13}$$

The supervised target answer is the single class token $\langle c_y \rangle$, giving

$$\mathcal{L}_{\text{cls-token}} = -\log p_\Theta(v_y \mid H_x). \tag{14}$$

This interface differs from attaching a conventional classifier head to the final representation. The decision is still made through the autoregressive output distribution of the LLM, but the candidate space is restricted to supervised class tokens that are valid for the current dataset. It also differs from phrase-likelihood classification with natural-language labels: each class is represented by exactly one learned token, so the score is not affected by label length, shared subword prefixes, or uncontrolled continuations.

Only the input embeddings and output rows of the new class tokens are trainable; the rest of the vocabulary remains fixed. This design turns generative classification into supervised decoding over a compact, dataset-specific label space: during training, the target answer is the class token for the labeled support example, and during inference the model is allowed to score only valid class tokens. For anonymous-label datasets such as UCR, these tokens are closed-set identifiers. For semantic-label datasets, each class has a label card with a canonical name and template-consistent verbalizers $\mathcal{S}_c$. We average their token embeddings to form a prototype,

$$p_c = \frac{1}{|\mathcal{S}_c|} \sum_{r \in \mathcal{S}_c} \frac{1}{|\tau(r)|} \sum_{u \in \tau(r)} E_{\mathrm{lm}}(u), \tag{15}$$

rescale it to the average vocabulary-row norm, and initialize the corresponding class-token rows from this prototype. During adaptation, we regularize the input row e_{v_c} and output row w_{v_c} toward the detached prototype $\tilde{p}_c$:

$$\mathcal{L}_{\mathrm{sem}} = \frac{1}{C} \sum_{c=0}^{C-1} \left[1 - \cos(e_{v_c}, \tilde{p}_c) + 1 - \cos(w_{v_c}, \tilde{p}_c)\right], \qquad \mathcal{L}_{\mathrm{adapt}} = \mathcal{L}_{\text{cls-token}} + \lambda_{\mathrm{row}} \mathcal{L}_{\mathrm{sem}}. \tag{16}$$

This injects label semantics as a prior while preserving a stable single-token decision space.

3.5 Downstream Adaptation

Few-shot adaptation uses a two-phase schedule. In the warm-up phase, we optimize the time-series encoder, projector, and dataset-specific class-token rows,

$$\Theta_{\mathrm{warm}} = \{\theta_{\mathrm{enc}}, \psi, e_{\mathcal{V}_D}, w_{\mathcal{V}_D}\}, \qquad \min_{\Theta_{\mathrm{warm}}} \mathcal{L}_{\mathrm{adapt}}. \tag{17}$$

In the joint phase, we additionally enable LoRA adapters on the language side,

$$\Theta_{\mathrm{joint}} = \Theta_{\mathrm{warm}} \cup \{\Delta\Theta_{\mathrm{LoRA}}\}, \qquad \Delta W_l = B_l A_l, \quad \mathrm{rank}(\Delta W_l) \leq r. \tag{18}$$

The main benchmark uses LoRA rank 16 and alpha 32, applied to the LLM adaptation modules exposed by the implementation. This schedule keeps the early optimization problem compact while still allowing limited language-side adaptation after the time-series interface has stabilized.

4 Experiments

4.1 Experimental Setup

Protocol. We evaluate on all 128 datasets in the UCR Archive 2018 with a fixed full-label-space few-shot protocol. For each dataset D and support size $S \in \{1, 2, 5, 10\}$, all classes are included; we sample up to S labeled examples per class from the official training split and evaluate on all examples in the official test split. All methods use the same sampled support sets within each of five independent runs.

Pretraining data and default initialization. The main benchmark uses the Stage II checkpoint: Stage I is trained on raw TSQA-train series, raw M4-train series, and unlabeled official TRAIN-split sequences from a fixed 98-dataset UCR pool; Stage II is trained on the TSQA training split. Stage III is not used for Table 1. No pretraining stage uses any official UCR TEST split or UCR class label. For UCR datasets included in the 98-dataset pool, ChronoMorph may have seen unlabeled official TRAIN sequences before downstream adaptation; after this pretraining, each target run uses only the sampled support labels for supervised adaptation.

Baselines and metrics. We compare against CNN/prototype baselines (RESNET, TAPNET, COSCO-RESNET), Transformer-style or decomposition-based baselines (PATCHTST, DLINEAR, TIMESNET, AUTOFORMER, CROSSFORMER, FEDFORMER, INFORMER), and the LLM/foundation-style GPT4TS baseline. Accuracy is averaged over five runs and then macro-averaged over datasets. `Avg` averages the four shot settings, and `AvgRank` averages dataset-level ranks across shots; lower rank is better. Standard deviations and full per-dataset tables are in Appendix A.

Table 1: Few-shot classification accuracy (%) on 128 UCR datasets, grouped by model family. We report the mean over five runs. `Avg` and `AvgRank` average the four shot-specific mean accuracies and average ranks, respectively. Best is bold and second-best is underlined.

Family	Model	1-shot	2-shot	5-shot	10-shot	Avg	AvgRank
CNN/prototype	ResNet	47.79	53.98	61.27	66.24	57.32	4.89
CNN/prototype	TapNet	45.40	50.79	59.47	64.93	55.15	5.45
CNN/prototype	COSCO-ResNet	<u>47.88</u>	<u>54.95</u>	<u>63.55</u>	<u>67.18</u>	<u>58.39</u>	<u>4.70</u>
Transformer/TS	DLinear	44.21	48.01	52.30	55.43	49.99	7.24
Transformer/TS	PatchTST	42.39	48.43	56.94	62.72	52.62	6.54
Transformer/TS	TimesNet	45.16	49.95	57.86	62.79	53.94	5.90
Transformer/TS	Autoformer	36.78	39.45	43.20	46.04	41.37	9.47
Transformer/TS	Crossformer	44.31	49.45	57.20	61.94	53.23	6.04
Transformer/TS	FEDformer	37.71	41.71	47.30	51.50	44.56	8.68
Transformer/TS	Informer	46.42	51.14	58.35	63.13	54.76	5.59
LLM/foundation	GPT4TS	35.61	38.30	42.94	46.88	40.93	9.26
Ours	ChronoMorph	**48.31**	**55.55**	**64.14**	**69.16**	**59.29**	**4.24**

Fairness and training details. All methods share the same official TRAIN support pool, TEST query split, and sampled support indices. We do not add a validation split by default; each method follows its runner's checkpoint-selection rule, and some baselines select checkpoints using support-set performance. Appendix B.8 lists the training recipes and hyperparameters.

Implementation details. ChronoMorph uses `meta-llama/Llama-3.2-1B`, `facebook/dinov2-base`, temporal patch length 16, stride 8, hidden size 128, and the Stage II checkpoint. Downstream adaptation uses 5 warm-up epochs, 55 joint-training epochs, and LoRA with rank 16 and alpha 32. Additional details are in Appendix B.

4.2 Main Results

Benchmark-level comparison. Table 1 shows that ChronoMorph obtains the highest mean accuracy under all four support sizes: 48.31, 55.55, 64.14, and 69.16 for 1-, 2-, 5-, and 10-shot. The strongest non-ChronoMorph baseline is COSCO-RESNET, with 47.88, 54.95, 63.55, and 67.18. Thus the improvement over the strongest specialized CNN/prototype baseline is modest in accuracy, especially in the lowest-shot regimes, but it is consistent across support sizes and is also reflected in the best `AvgRank` (4.24 vs. 4.70 for COSCO-RESNET). The gap is more substantial against Transformer-style and LLM/foundation baselines: ChronoMorph improves `Avg` by 4.53 points over INFORMER and 18.36 points over GPT4TS. These results suggest that the proposed interface is most clearly beneficial relative to generic sequence and LLM-style baselines, while remaining competitive with strong specialized CNN/prototype methods.

Dataset-level robustness. Table 2 reports Win/Tie/Loss counts against representative baselines. ChronoMorph wins substantially more often than it loses against PATCHTST, TIMESNET, and GPT4TS. Against the strongest CNN/prototype baselines, the picture is closer: ChronoMorph is nearly balanced with COSCO-RESNET and RESNET at several shot levels, which is consistent with the modest average-accuracy margins in Table 1.

Additional external datasets. We also evaluate MIT-BIH, SleepEDF, and CinC2017 under full-label-space 10-, 20-, and 30-shot protocols, using 10 runs for each external setting. Table 3 reports the average over these shot settings, with the full table in Appendix B.5. ChronoMorph obtains the best average accuracy on MIT-BIH and CinC2017 and the second-best overall external average, but it is clearly behind RESNET and TAPNET on SleepEDF. We therefore treat these experiments as evidence of transfer beyond UCR, not as a claim of uniform dominance across external domains.

Table 2: Win/Tie/Loss of ChronoMorph against selected baselines on 128 UCR datasets.

Baseline	1-shot	2-shot	5-shot	10-shot
COSCO-ResNet	68 / 0 / 60	68 / 1 / 59	58 / 0 / 70	64 / 0 / 64
PatchTST	97 / 0 / 31	97 / 0 / 31	99 / 0 / 29	94 / 2 / 32
TimesNet	88 / 0 / 40	91 / 1 / 36	86 / 0 / 42	93 / 1 / 34
GPT4TS	106 / 2 / 20	110 / 0 / 18	114 / 1 / 13	121 / 1 / 6
ResNet	64 / 1 / 63	66 / 1 / 61	67 / 0 / 61	69 / 0 / 59
TapNet	75 / 0 / 53	75 / 3 / 50	76 / 1 / 51	72 / 3 / 53

Table 3: Average few-shot accuracy (%) on three external datasets, computed over 10-, 20-, and 30-shot settings with 10 runs per setting. Best is bold and second-best is underlined.

Model	MIT-BIH	SleepEDF	CinC2017	Avg.
ChronoMorph	**61.04**	47.16	**34.63**	47.61
ResNet	50.23	**66.84**	33.63	**50.23**
TapNet	38.73	66.70	31.29	45.57
COSCO-ResNet	36.19	62.74	32.48	43.80
PatchTST	50.46	28.92	29.21	36.20
OneFitsAll	43.42	27.05	29.19	33.22
TimesNet	53.79	23.23	28.36	35.13
DLinear	57.29	19.81	24.87	33.99
Autoformer	41.64	26.48	27.45	31.86
Crossformer	56.76	24.93	25.62	35.77
FEDformer	56.93	21.79	25.05	34.59
Informer	53.25	22.45	28.39	34.70

4.3 Computational Efficiency and Adaptation Cost

Because ChronoMorph reuses a pretrained causal LLM, total parameters and adapted parameters must be reported separately. Table 4 shows the main trade-off: ChronoMorph updates only 12.3M parameters per target dataset, but the frozen LLM backbone dominates memory and inference cost. ChronoMorph is therefore most appropriate when transfer accuracy and reusable adaptation matter more than compact deployment. The complete efficiency table, including throughput and additional baselines, is in Appendix B.7.

4.4 Ablation Studies

Table 5 summarizes the main top-level UCR ablations on all 128 datasets. Each ablation follows the same five-run full-label-space few-shot protocol and modifies only the component under study. Focused compact views with the corresponding drops are provided in Tables 20–22. Because UCR labels are anonymous, the decision ablation tests closed-set class-token scoring rather than semantic label priors. The no-LLM ablation has a different completed coverage set and is therefore reported separately in Appendix Table 23.

Effect of curriculum pretraining. Removing curriculum pretraining drops `Avg` from 59.29 to 55.94, with shot-level accuracies decreasing from 48.31/55.55/64.14/69.16 to 44.70/51.81/60.65/66.60 for 1-/2-/5-/10-shot, respectively. The focused view in Appendix Table 20 reports a 3.35-point average drop, suggesting that the aggregate Stage I+II initialization helps when downstream supervision is too limited to learn the representation and LLM interface from scratch. This ablation removes the full pretraining pipeline at once and should not be read as isolating each substage.

Effect of dual-view time-series-to-language modeling. Temporal-only and visual-only variants reach `Avg` scores of 54.21 and 55.09, below the full model's 59.29. Appendix Table 21 shows that removing the visual branch yields a 5.08-point average drop, while removing the temporal branch yields a 4.20-point drop. The temporal-only variant is weaker at 1-, 2-, and 5-shot, whereas the

Table 4: Compact efficiency comparison under the 10-shot UCR protocol. Runtime and memory are measured on the representative UCR runtime subset. Total parameters count the deployed model; updated parameters count downstream-adapted parameters only.

Model	Total params	Updated params	Peak GB	Infer ms/ex.
ChronoMorph	1.28B	12.3M	13.75	156.85
COSCO-ResNet	662.1K	662.1K	0.58	0.25
ResNet	629.1K	629.1K	0.33	0.26
PatchTST	665.4K	598.3K	1.54	0.75
GPT4TS	82.2M	1.1M	0.56	0.86

Table 5: Summary of top-level ablations on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. Each value is mean accuracy (%) over five runs. Best is bold, and the best ablated variant in each column is underlined. Since UCR labels are anonymous, the label-space ablation isolates closed-set class-token scoring rather than semantic label priors.

Variant	1-shot	2-shot	5-shot	10-shot	Avg
ChronoMorph	**48.31**	**55.55**	**64.14**	**69.16**	**59.29**
w/o Curriculum Pretraining	44.70	<u>51.81</u>	60.65	<u>66.60</u>	<u>55.94</u>
Temporal only	43.51	50.55	58.40	64.39	54.21
Visual only	<u>45.88</u>	50.02	<u>60.86</u>	63.58	55.09
w/o Class-Token Label-Space Scoring	44.46	50.44	60.08	65.59	55.14

visual-only variant is weaker at 2- and 10-shot. Neither single view dominates across all shots, which supports the intended complementarity between chronological and morphological evidence.

Effect of class-token label-space scoring. Disabling class-token label-space scoring reduces `Avg` from 59.29 to 55.14. As detailed in Appendix Table 22, the corresponding shot-level accuracies decrease from 48.31/55.55/64.14/69.16 to 44.46/50.44/60.08/65.59, giving a 4.15-point average drop. This indicates that, under the same adaptation budget, unconstrained generative adaptation is less stable for closed-set few-shot classification than directly scoring the dataset-specific label tokens.

Effect of semantic label priors. For external datasets with meaningful label names, Table 6 compares anonymous class-token rows with rows initialized and regularized from label-verbalizer prototypes. Semantic priors improve the 10/20/30-shot average on MIT-BIH, SleepEDF, and CinC2017, increasing the overall average from 47.61 to 50.43. The full appendix table shows that the benefit is not uniform at every shot, with a decrease on CinC2017 20-shot, but the averaged result supports the semantic-prior class-token space as a useful extension of supervised closed-set decoding.

Effect of the LLM backbone. The no-LLM variant is evaluated on the 120 UCR datasets covered by both the full and no-LLM runs, so we report it separately in Appendix D, with the compact summary in Table 23. On this shared subset, replacing the language pathway with a lightweight linear classifier reduces `Avg` from 58.67 to 52.31, a 6.36-point drop. The decrease is consistent across support sizes: 47.93 to 42.95 at 1-shot, 54.98 to 48.20 at 2-shot, 63.27 to 57.06 at 5-shot, and 68.48 to 61.01 at 10-shot. This suggests that, under our implementation and adaptation schedule, the language pathway is beneficial; however, this ablation does not isolate whether the gain comes from pretrained knowledge, additional capacity, or the token-scoring interface.

Overall, the ablations support the central design: curriculum pretraining improves transferability, dual-view modeling preserves more complete evidence, closed-set class-token scoring stabilizes the final decision, and semantic-prior row initialization and regularization further improve semantic-label transfer.

Table 6: Semantic-prior class-token ablation on external datasets. Values are average accuracy (%) over 10-, 20-, and 30-shot settings with 10 runs per setting. Δ denotes Semantic prior minus Anonymous labels.

Variant	MIT-BIH	SleepEDF	CinC2017	Avg.
Anonymous class tokens	61.04	47.16	34.63	47.61
Semantic-prior class tokens	**64.18**	**50.83**	**36.27**	**50.43**
Δ	+3.14	+3.67	+1.64	+2.82

5 Limitations

ChronoMorph improves average accuracy and average rank, but its gains over the strongest CNN/prototype baselines are modest in extremely low-shot settings. The main empirical message is therefore not that LLM-based TSC now universally dominates specialized CNN-style few-shot classifiers, but that a carefully designed time-series-to-language interface can substantially outperform previous Transformer-style and LLM/foundation-style baselines while remaining competitive with strong task-specific methods. ChronoMorph also has a larger inference footprint than compact CNN baselines because it reuses a pretrained causal LLM, even though only a small fraction of parameters is updated during downstream adaptation. In our efficiency probe, this makes ChronoMorph two to three orders of magnitude slower at inference than compact CNN/prototype baselines. The current UCR pretraining setting also uses unlabeled official TRAIN sequences from a fixed subset of UCR datasets, so it does not establish strict leave-dataset-out transfer for every UCR target dataset. The external semantic-prior results improve the averaged performance on all three semantic-label datasets, but individual shot-level effects still vary, so label semantics should be viewed as a prior over the class-token rows rather than as a replacement for supervised adaptation. ChronoMorph may benefit applications where labeled time-series data are scarce, such as physiological monitoring or industrial inspection, but the present study is not a deployment-ready clinical or safety-critical system. Any deployment in such domains would require domain-specific validation, privacy review, and failure analysis beyond the scope of this paper.

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A Full UCR Benchmark Results

This appendix reports the complete per-dataset few-shot benchmark results for all 128 UCR datasets. Each cell is shown as mean $\pm$ standard deviation over five runs, and the summary rows at the end of each table report average accuracy, average rank, and the number of datasets attaining the best mean accuracy (counting ties).

Table 7: Per-dataset 1-shot classification results on 128 shared UCR datasets. Each value is reported as mean ± standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
ACSF1	**53.80 ± 4.97**	42.00 ± 4.95	48.40 ± 7.30	24.00 ± 4.24	35.60 ± 3.21	27.67 ± 7.02	24.33 ± 9.71	28.33 ± 2.52	27.80 ± 7.09	31.67 ± 9.81	18.60 ± 4.51	44.00 ± 5.20
Adiac	33.81 ± 1.23	12.84 ± 3.09	33.81 ± 1.23	8.85 ± 2.36	11.97 ± 1.69	21.40 ± 4.24	34.53 ± 1.77	8.10 ± 1.92	40.82 ± 3.61	24.21 ± 3.24	3.02 ± 1.05	**45.17 ± 5.28**
AllGestureWiimoteX	25.71 ± 4.01	**26.86 ± 2.52**	24.43 ± 2.55	15.57 ± 2.95	21.86 ± 4.59	20.71 ± 2.87	11.33 ± 1.19	16.76 ± 2.52	13.43 ± 1.74	18.90 ± 1.08	21.86 ± 4.59	25.11 ± 4.65
AllGestureWiimoteY	**26.37 ± 2.89**	26.17 ± 2.88	22.71 ± 4.62	17.49 ± 2.02	25.17 ± 5.50	21.81 ± 2.78	12.29 ± 1.27	20.29 ± 1.98	13.06 ± 1.34	21.33 ± 2.38	25.17 ± 5.50	25.54 ± 4.42
AllGestureWiimoteZ	**27.11 ± 1.95**	22.66 ± 4.13	25.49 ± 2.57	18.26 ± 2.01	19.71 ± 2.87	19.24 ± 1.57	13.29 ± 0.43	18.76 ± 2.79	14.14 ± 1.72	19.71 ± 1.79	19.71 ± 2.87	22.26 ± 3.96
ArrowHead	47.66 ± 8.94	46.63 ± 9.45	48.00 ± 7.74	52.00 ± 6.38	49.71 ± 1.40	47.62 ± 9.46	38.29 ± 4.00	50.86 ± 1.71	35.89 ± 2.60	50.86 ± 6.05	41.49 ± 12.34	**52.57 ± 3.43**
BME	49.60 ± 2.85	50.80 ± 2.13	49.73 ± 3.42	56.67 ± 3.43	62.93 ± 5.35	55.33 ± 7.86	51.56 ± 3.15	61.33 ± 4.06	55.87 ± 7.71	58.89 ± 6.19	35.73 ± 3.22	**66.40 ± 3.88**
Beef	36.00 ± 6.41	31.33 ± 5.06	27.33 ± 7.96	38.67 ± 9.89	32.00 ± 7.67	40.00 ± 8.82	34.44 ± 6.94	**41.11 ± 1.92**	30.67 ± 6.83	40.00 ± 6.67	31.33 ± 9.31	34.00 ± 2.79
BeetleFly	55.00 ± 24.24	60.00 ± 25.00	61.00 ± 24.85	67.00 ± 4.47	61.00 ± 9.62	60.00 ± 13.23	55.00 ± 5.00	61.67 ± 11.55	48.00 ± 2.74	60.00 ± 13.23	**70.00 ± 11.18**	**70.00 ± 11.73**
BirdChicken	62.00 ± 8.37	**71.00 ± 9.62**	70.00 ± 17.68	56.00 ± 4.18	56.00 ± 4.18	51.67 ± 2.89	51.67 ± 12.58	53.33 ± 5.77	55.00 ± 5.00	51.67 ± 7.64	59.00 ± 9.62	68.00 ± 14.83
CBF	81.67 ± 6.47	**87.62 ± 2.73**	71.71 ± 7.91	54.02 ± 3.44	66.38 ± 7.40	55.85 ± 2.16	33.63 ± 1.20	58.00 ± 5.61	36.89 ± 2.37	54.78 ± 1.68	35.20 ± 3.75	72.48 ± 4.85
Car	37.00 ± 4.62	39.00 ± 4.80	41.67 ± 9.28	38.67 ± 11.63	27.33 ± 9.90	42.22 ± 5.09	24.44 ± 3.47	43.89 ± 5.85	27.00 ± 4.15	40.00 ± 7.26	30.00 ± 9.65	**55.67 ± 8.94**
Chinatown	66.53 ± 19.14	63.73 ± 26.98	62.51 ± 27.20	67.41 ± 23.86	67.35 ± 11.32	77.07 ± 16.93	81.83 ± 2.19	77.94 ± 9.20	77.61 ± 8.88	75.41 ± 7.92	56.68 ± 20.52	**93.59 ± 3.92**
ChlorineConcentration	34.61 ± 2.26	35.41 ± 4.36	34.04 ± 2.01	33.26 ± 2.96	35.54 ± 3.20	33.64 ± 3.96	32.92 ± 4.26	33.42 ± 5.09	33.78 ± 2.74	34.11 ± 3.50	**39.93 ± 15.64**	32.61 ± 3.30
CinCECGTorso	37.20 ± 9.73	36.03 ± 6.56	36.51 ± 8.82	40.06 ± 3.86	35.72 ± 7.09	44.86 ± 6.03	35.36 ± 5.77	**50.53 ± 4.66**	34.97 ± 4.68	45.68 ± 5.50	42.49 ± 6.76	48.74 ± 6.47
Coffee	78.57 ± 9.78	**92.86 ± 8.38**	84.29 ± 13.02	72.86 ± 14.42	74.29 ± 18.80	63.10 ± 10.91	65.48 ± 7.43	71.43 ± 16.37	76.43 ± 4.07	64.29 ± 7.14	51.43 ± 4.07	75.71 ± 13.69
Computers	57.52 ± 7.64	**60.40 ± 6.13**	59.12 ± 4.47	47.12 ± 5.87	52.64 ± 10.26	50.93 ± 1.29	48.80 ± 0.69	51.87 ± 2.57	48.96 ± 2.79	52.27 ± 3.23	45.20 ± 5.79	56.40 ± 5.77
CricketX	26.15 ± 3.29	26.10 ± 3.67	20.56 ± 4.67	17.28 ± 1.96	18.15 ± 4.50	20.34 ± 0.39	8.89 ± 0.82	17.52 ± 1.26	11.38 ± 2.35	20.68 ± 0.74	7.28 ± 1.05	**26.26 ± 5.29**
CricketY	**26.97 ± 3.03**	24.46 ± 4.25	22.72 ± 1.74	18.56 ± 4.13	17.90 ± 3.61	22.82 ± 1.56	10.34 ± 1.89	20.94 ± 3.72	11.23 ± 0.80	21.20 ± 3.98	9.54 ± 2.33	23.33 ± 5.19
CricketZ	**32.51 ± 2.32**	28.72 ± 2.15	26.56 ± 4.19	19.64 ± 2.73	22.26 ± 3.91	20.34 ± 4.11	11.03 ± 1.54	20.00 ± 2.31	12.26 ± 1.25	20.34 ± 2.57	7.64 ± 1.05	29.28 ± 4.82
Crop	25.10 ± 4.94	28.14 ± 2.90	25.10 ± 4.94	10.23 ± 2.66	17.27 ± 1.43	27.73 ± 2.89	24.10 ± 4.30	26.19 ± 2.96	27.27 ± 3.58	30.31 ± 5.00	3.80 ± 0.58	**36.48 ± 3.38**
DiatomSizeReduction	79.67 ± 5.51	55.36 ± 19.14	84.12 ± 4.82	**89.22 ± 8.91**	49.61 ± 24.86	75.71 ± 7.74	55.01 ± 11.77	56.43 ± 20.40	50.07 ± 12.66	76.47 ± 11.70	41.44 ± 13.25	85.82 ± 8.55
DistalPhalanxOutlineAgeGroup	43.02 ± 17.16	43.02 ± 18.34	43.60 ± 14.92	42.73 ± 19.01	46.76 ± 16.49	45.80 ± 19.81	46.28 ± 21.27	**56.59 ± 9.36**	50.94 ± 19.66	46.28 ± 20.78	35.68 ± 14.61	42.01 ± 16.44
DistalPhalanxOutlineCorrect	62.10 ± 1.32	57.32 ± 9.22	**63.04 ± 2.44**	57.54 ± 11.13	53.55 ± 11.63	53.99 ± 14.78	53.26 ± 12.27	53.62 ± 11.97	55.14 ± 10.52	53.26 ± 12.97	56.74 ± 7.30	60.58 ± 2.98
DistalPhalanxTW	51.22 ± 5.15	44.03 ± 11.70	53.38 ± 4.02	43.74 ± 5.17	51.22 ± 6.58	45.32 ± 9.89	42.69 ± 4.40	38.13 ± 7.51	48.20 ± 5.84	47.48 ± 5.62	18.56 ± 14.51	**54.53 ± 1.72**
DodgerLoopDay	32.50 ± 7.34	25.00 ± 4.05	29.75 ± 7.68	**41.50 ± 2.05**	26.00 ± 6.58	40.42 ± 4.73	20.42 ± 1.91	32.92 ± 3.15	21.50 ± 4.45	40.42 ± 7.94	26.00 ± 6.58	34.50 ± 4.38
DodgerLoopGame	52.17 ± 6.30	55.07 ± 17.38	49.13 ± 8.90	**60.87 ± 12.35**	59.86 ± 22.05	51.45 ± 9.96	51.21 ± 5.14	56.52 ± 10.82	50.43 ± 2.44	51.45 ± 9.59	59.86 ± 22.05	56.96 ± 11.69
DodgerLoopWeekend	84.49 ± 8.37	74.78 ± 12.27	83.19 ± 10.24	93.62 ± 3.81	**96.23 ± 1.19**	95.65 ± 1.26	63.77 ± 4.41	94.44 ± 0.42	68.41 ± 4.57	93.72 ± 5.86	**96.23 ± 1.19**	85.36 ± 4.30
ECG200	63.00 ± 12.61	**72.00 ± 6.75**	66.20 ± 8.32	71.80 ± 4.44	61.60 ± 12.82	61.33 ± 10.97	65.00 ± 11.79	63.00 ± 20.66	64.20 ± 14.29	68.67 ± 7.09	51.40 ± 12.93	60.60 ± 6.91
ECG5000	54.22 ± 21.00	50.05 ± 20.59	61.53 ± 13.66	62.80 ± 19.19	53.35 ± 20.14	**69.07 ± 12.52**	54.44 ± 9.79	57.20 ± 16.09	62.35 ± 12.99	68.84 ± 13.99	19.70 ± 25.30	67.18 ± 10.57
ECGFiveDays	58.65 ± 7.69	57.24 ± 6.51	54.80 ± 13.00	57.63 ± 2.63	**62.21 ± 8.35**	57.99 ± 4.84	54.86 ± 3.17	55.87 ± 1.56	54.08 ± 6.73	58.46 ± 5.08	51.54 ± 6.38	61.77 ± 6.61
EOGHorizontalSignal	20.11 ± 2.01	22.38 ± 1.86	22.60 ± 2.77	22.43 ± 4.01	21.22 ± 3.15	20.63 ± 4.63	15.29 ± 2.63	23.57 ± 5.45	10.39 ± 2.26	22.01 ± 4.37	27.24 ± 2.99	**29.06 ± 4.16**
EOGVerticalSignal	11.27 ± 3.67	14.09 ± 3.23	13.26 ± 3.09	24.20 ± 3.27	23.76 ± 3.73	23.39 ± 2.79	11.23 ± 0.70	24.86 ± 4.17	11.27 ± 3.83	23.11 ± 4.07	19.06 ± 6.08	**27.35 ± 1.90**
Earthquakes	50.07 ± 8.55	51.08 ± 6.33	50.94 ± 6.46	49.35 ± 5.05	45.61 ± 20.94	48.68 ± 19.16	49.40 ± 1.10	49.40 ± 14.03	47.05 ± 5.90	47.00 ± 24.65	46.04 ± 13.71	**55.54 ± 11.57**
ElectricDevices	28.83 ± 9.01	32.82 ± 12.31	28.35 ± 12.24	17.59 ± 1.67	27.21 ± 7.38	21.93 ± 4.54	17.11 ± 1.87	15.12 ± 2.81	17.04 ± 2.05	15.61 ± 7.10	17.06 ± 3.10	**33.14 ± 10.96**
EthanolLevel	24.44 ± 1.30	25.40 ± 0.85	25.28 ± 1.57	24.92 ± 1.50	23.80 ± 1.65	26.13 ± 0.58	26.53 ± 1.55	26.67 ± 1.29	25.32 ± 1.50	25.80 ± 1.39	24.76 ± 1.40	**26.68 ± 1.86**
FaceAll	**42.17 ± 4.79**	32.36 ± 9.47	**42.17 ± 4.79**	28.66 ± 9.09	19.36 ± 3.07	29.11 ± 4.19	33.12 ± 5.46	25.94 ± 1.75	34.05 ± 3.40	32.92 ± 2.10	8.64 ± 3.46	32.84 ± 6.86
FaceFour	68.64 ± 10.56	47.50 ± 11.12	56.36 ± 4.30	64.32 ± 4.73	54.09 ± 8.34	70.08 ± 4.30	52.65 ± 3.99	67.05 ± 7.10	51.36 ± 4.28	**73.11 ± 3.99**	42.95 ± 6.60	54.77 ± 8.25
FacesUCR	**54.83 ± 5.21**	37.63 ± 5.89	43.65 ± 3.88	35.69 ± 7.47	20.81 ± 1.95	36.24 ± 3.26	41.98 ± 2.95	40.05 ± 2.80	41.87 ± 3.09	42.63 ± 2.58	16.92 ± 7.54	50.37 ± 2.98
FiftyWords	13.27 ± 1.89	12.79 ± 3.45	13.27 ± 1.89	32.70 ± 4.33	19.60 ± 4.90	35.90 ± 3.31	3.44 ± 1.59	**38.97 ± 6.44**	11.30 ± 2.86	36.70 ± 4.41	3.74 ± 2.40	36.48 ± 2.66
Fish	**49.03 ± 5.96**	21.26 ± 3.60	45.71 ± 1.71	36.00 ± 3.98	23.20 ± 3.80	32.57 ± 4.57	16.95 ± 2.93	30.29 ± 2.62	21.71 ± 2.77	31.81 ± 4.76	20.80 ± 5.10	34.74 ± 4.91
FordA	59.23 ± 9.68	57.68 ± 10.00	**59.83 ± 8.76**	50.73 ± 1.01	56.70 ± 8.04	50.30 ± 0.33	49.57 ± 1.42	49.72 ± 0.63	49.68 ± 1.11	49.92 ± 0.80	50.61 ± 1.68	50.71 ± 0.98
FordB	52.54 ± 6.44	52.64 ± 6.88	52.40 ± 4.53	50.12 ± 1.83	**53.60 ± 4.09**	49.96 ± 0.94	50.53 ± 0.72	50.21 ± 0.80	50.81 ± 0.65	49.88 ± 0.98	50.05 ± 1.16	49.75 ± 2.56
FreezerRegularTrain	61.33 ± 21.55	58.22 ± 20.21	56.76 ± 19.80	61.31 ± 21.84	57.07 ± 18.69	59.91 ± 27.91	48.78 ± 1.68	53.81 ± 23.62	49.55 ± 8.37	60.27 ± 28.68	61.45 ± 8.09	**61.90 ± 22.25**
FreezerSmallTrain	67.85 ± 12.31	66.48 ± 12.45	67.42 ± 12.38	67.47 ± 12.29	63.19 ± 10.91	**73.32 ± 3.65**	55.33 ± 0.97	69.60 ± 5.96	57.00 ± 2.79	73.03 ± 4.44	65.92 ± 7.33	66.53 ± 10.64
Fungi	93.23 ± 4.00	35.81 ± 4.97	64.62 ± 2.56	68.39 ± 5.78	67.20 ± 19.03	**95.70 ± 0.93**	73.66 ± 2.69	86.74 ± 1.89	75.16 ± 1.03	**95.70 ± 0.93**	14.09 ± 7.48	79.35 ± 5.30
GestureMidAirD1	17.08 ± 4.26	19.38 ± 4.43	17.08 ± 4.26	34.31 ± 3.87	19.08 ± 6.45	31.54 ± 2.04	17.44 ± 0.89	34.62 ± 4.07	21.85 ± 4.63	**34.87 ± 2.47**	19.08 ± 6.45	32.00 ± 3.94
GestureMidAirD2	13.23 ± 5.14	17.08 ± 3.90	13.23 ± 5.14	29.08 ± 4.19	20.46 ± 5.77	29.74 ± 3.20	11.79 ± 2.22	29.23 ± 5.38	17.38 ± 5.95	**36.41 ± 0.89**	20.46 ± 5.77	33.85 ± 3.03
GestureMidAirD3	8.62 ± 1.84	9.08 ± 2.52	8.62 ± 1.84	20.00 ± 3.17	10.92 ± 1.26	**24.87 ± 4.24**	9.49 ± 1.60	23.08 ± 0.77	14.77 ± 4.47	22.56 ± 2.91	10.92 ± 1.26	18.46 ± 2.61

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Table 7: Per-dataset 1-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
GesturePebbleZ1	49.07 ± 5.03	38.37 ± 6.03	**53.26 ± 6.97**	46.16 ± 13.08	23.60 ± 6.95	40.50 ± 15.60	20.93 ± 3.54	42.44 ± 7.05	18.95 ± 2.08	43.99 ± 8.75	23.60 ± 6.95	35.81 ± 4.48
GesturePebbleZ2	55.19 ± 11.38	42.91 ± 15.92	**57.47 ± 8.89**	48.35 ± 13.33	21.01 ± 3.48	47.68 ± 13.93	18.35 ± 4.94	43.88 ± 16.17	16.20 ± 5.17	55.06 ± 14.60	21.01 ± 3.48	33.29 ± 5.85
GunPoint	55.47 ± 4.79	58.13 ± 9.13	55.87 ± 5.32	53.20 ± 9.11	56.40 ± 7.58	**64.00 ± 8.08**	58.89 ± 10.59	63.33 ± 4.81	58.93 ± 8.45	62.89 ± 3.01	52.40 ± 11.44	58.00 ± 7.79
GunPointAgeSpan	49.94 ± 8.02	48.67 ± 3.53	49.11 ± 3.99	60.82 ± 22.09	58.73 ± 19.72	59.70 ± 21.56	52.11 ± 7.60	59.07 ± 20.81	52.41 ± 6.32	**61.92 ± 19.97**	48.10 ± 2.61	60.51 ± 10.32
GunPointMaleVersusFemale	63.99 ± 11.97	63.29 ± 11.59	64.30 ± 12.20	54.56 ± 4.47	56.65 ± 3.27	55.91 ± 6.97	50.95 ± 7.17	55.59 ± 9.92	50.19 ± 6.98	56.75 ± 8.76	58.04 ± 11.38	**81.01 ± 10.12**
GunPointOldVersusYoung	**92.38 ± 10.44**	89.97 ± 13.74	90.35 ± 13.23	53.90 ± 25.42	53.33 ± 25.91	58.31 ± 30.15	54.81 ± 4.33	57.46 ± 29.14	54.79 ± 8.98	58.10 ± 28.87	78.41 ± 20.62	57.97 ± 10.78
Ham	50.10 ± 9.32	51.24 ± 7.01	47.81 ± 11.28	55.24 ± 11.13	49.33 ± 2.81	48.89 ± 5.25	53.65 ± 3.06	51.43 ± 7.44	**55.81 ± 6.79**	50.16 ± 4.89	51.81 ± 10.55	52.76 ± 9.30
HandOutlines	46.81 ± 15.76	58.86 ± 12.82	55.84 ± 9.47	70.59 ± 7.20	75.68 ± 8.02	71.17 ± 7.55	51.08 ± 2.34	**76.76 ± 10.41**	52.81 ± 15.40	74.86 ± 11.22	61.19 ± 14.73	61.73 ± 6.31
Haptics	23.96 ± 5.22	22.66 ± 2.12	22.86 ± 4.94	**26.49 ± 6.85**	22.66 ± 4.38	25.11 ± 9.19	19.16 ± 1.81	23.16 ± 7.32	23.44 ± 3.46	25.43 ± 9.51	25.65 ± 5.26	25.65 ± 4.99
Herring	49.69 ± 6.39	49.69 ± 10.50	50.94 ± 11.51	48.12 ± 8.44	49.69 ± 7.44	47.40 ± 10.40	**51.04 ± 2.39**	49.48 ± 9.55	48.75 ± 8.44	48.44 ± 9.50	48.12 ± 10.39	50.00 ± 10.00
HouseTwenty	**80.84 ± 7.67**	74.62 ± 5.52	68.40 ± 16.13	50.25 ± 3.68	57.65 ± 8.25	68.63 ± 3.50	49.30 ± 7.15	61.62 ± 2.57	46.05 ± 6.81	66.67 ± 1.28	53.61 ± 8.52	62.18 ± 8.04
InlineSkate	16.40 ± 1.60	15.64 ± 0.76	18.00 ± 0.73	15.89 ± 3.32	14.69 ± 2.92	18.06 ± 2.11	15.15 ± 0.73	18.00 ± 2.36	15.02 ± 3.34	**18.67 ± 3.56**	14.87 ± 3.22	17.13 ± 2.24
InsectEPGRegularTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	43.53 ± 11.83	50.44 ± 13.06	48.19 ± 5.92	42.03 ± 6.13	45.25 ± 3.22	46.27 ± 8.59	49.26 ± 6.69	87.15 ± 20.64	55.34 ± 14.47
InsectEPGSmallTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	47.95 ± 6.69	55.10 ± 5.49	51.14 ± 8.16	39.36 ± 1.75	52.88 ± 6.50	37.67 ± 14.87	52.48 ± 11.47	85.78 ± 19.79	50.76 ± 7.86
InsectWingbeatSound	16.18 ± 3.00	14.23 ± 3.87	14.35 ± 2.82	39.88 ± 4.03	22.79 ± 2.65	43.42 ± 0.26	17.56 ± 4.92	43.16 ± 3.28	21.14 ± 2.99	**45.61 ± 5.13**	12.52 ± 3.37	25.27 ± 2.20
ItalyPowerDemand	61.83 ± 17.15	62.29 ± 17.84	64.74 ± 16.58	55.32 ± 7.10	60.35 ± 16.10	69.81 ± 17.24	68.35 ± 9.65	**78.33 ± 24.61**	68.98 ± 15.67	71.62 ± 21.13	47.31 ± 2.37	58.52 ± 15.27
LargeKitchenAppliances	48.48 ± 12.88	52.85 ± 11.74	46.72 ± 15.63	33.39 ± 3.38	**54.88 ± 8.28**	40.09 ± 11.55	34.93 ± 5.70	32.53 ± 3.23	31.47 ± 3.69	41.69 ± 12.56	33.81 ± 2.87	45.44 ± 5.60
Lightning2	50.49 ± 6.80	55.08 ± 10.15	51.80 ± 4.27	56.72 ± 11.39	51.80 ± 12.52	58.47 ± 11.16	48.63 ± 3.41	57.38 ± 11.36	51.48 ± 5.26	59.02 ± 8.67	55.08 ± 13.25	**59.67 ± 8.87**
Lightning7	44.93 ± 4.48	44.38 ± 4.60	40.82 ± 7.47	43.01 ± 4.18	42.19 ± 7.34	53.42 ± 10.70	15.07 ± 6.28	45.66 ± 6.76	18.63 ± 4.40	**56.16 ± 8.33**	24.11 ± 7.72	41.64 ± 7.60
Mallat	64.03 ± 14.52	50.42 ± 9.84	64.49 ± 9.43	80.37 ± 4.76	56.96 ± 10.09	**82.32 ± 3.73**	54.67 ± 2.15	75.96 ± 4.82	64.55 ± 4.53	75.72 ± 2.60	37.30 ± 5.46	68.24 ± 7.54
Meat	42.00 ± 9.96	44.00 ± 10.18	58.33 ± 10.99	**77.67 ± 17.26**	52.67 ± 11.94	46.11 ± 9.18	38.33 ± 8.66	33.33 ± 0.00	36.00 ± 2.53	66.11 ± 20.97	33.67 ± 0.75	55.67 ± 9.62
MedicalImages	27.92 ± 5.84	28.45 ± 4.75	24.95 ± 5.95	26.34 ± 8.67	24.16 ± 4.26	22.32 ± 5.35	26.84 ± 12.59	23.86 ± 6.12	**28.53 ± 8.23**	23.16 ± 6.10	4.21 ± 2.27	27.66 ± 4.41
MelbournePedestrian	**51.29 ± 5.58**	44.27 ± 4.81	47.47 ± 6.50	9.85 ± 0.23	31.99 ± 4.95	21.74 ± 7.37	46.44 ± 8.26	40.62 ± 4.65	43.40 ± 7.11	43.73 ± 5.33	31.99 ± 4.95	39.16 ± 5.22
MiddlePhalanxOutlineAgeGroup	38.83 ± 7.74	37.53 ± 10.57	38.05 ± 7.83	34.03 ± 9.03	35.97 ± 8.21	37.45 ± 11.78	29.87 ± 4.55	38.53 ± 15.64	33.90 ± 7.76	32.90 ± 7.86	**41.69 ± 18.94**	39.48 ± 7.40
MiddlePhalanxOutlineCorrect	53.68 ± 13.63	52.10 ± 14.21	54.43 ± 13.33	53.75 ± 17.89	56.49 ± 9.58	**58.42 ± 17.26**	55.56 ± 15.16	54.07 ± 23.81	54.50 ± 10.22	57.50 ± 21.09	52.85 ± 12.08	51.68 ± 5.00
MiddlePhalanxTW	34.03 ± 16.27	44.68 ± 7.76	32.47 ± 17.97	30.52 ± 10.75	32.34 ± 10.78	29.00 ± 15.96	38.96 ± 3.37	**46.10 ± 3.62**	35.45 ± 13.57	42.21 ± 5.07	15.19 ± 11.64	37.14 ± 16.84
MixedShapesRegularTrain	45.36 ± 5.29	40.59 ± 8.80	43.58 ± 10.39	47.60 ± 10.74	39.59 ± 5.62	43.05 ± 15.67	17.31 ± 1.15	45.53 ± 11.76	25.71 ± 3.77	42.76 ± 16.81	45.03 ± 12.47	**56.06 ± 5.34**
MixedShapesSmallTrain	52.78 ± 8.02	39.55 ± 6.53	48.84 ± 9.40	53.90 ± 7.66	45.57 ± 7.91	52.81 ± 6.78	17.32 ± 0.61	53.84 ± 4.25	24.17 ± 3.24	53.31 ± 5.69	48.73 ± 8.44	**63.30 ± 3.34**
MoteStrain	62.99 ± 13.96	60.67 ± 12.58	64.35 ± 11.70	63.40 ± 15.50	63.27 ± 13.76	68.45 ± 11.94	56.26 ± 12.55	63.82 ± 16.98	61.65 ± 9.57	**68.66 ± 11.89**	57.99 ± 9.98	54.36 ± 12.77
NonInvasiveFetalECGThorax1	19.05 ± 4.42	8.04 ± 2.52	19.05 ± 4.42	38.23 ± 1.81	12.07 ± 7.41	36.76 ± 3.30	12.57 ± 2.38	28.62 ± 3.97	14.44 ± 2.07	**42.58 ± 4.62**	5.58 ± 2.48	39.16 ± 2.28
NonInvasiveFetalECGThorax2	20.69 ± 5.30	9.71 ± 2.65	20.69 ± 5.30	47.48 ± 2.72	14.81 ± 5.33	46.67 ± 0.18	20.97 ± 3.22	34.59 ± 7.78	23.69 ± 1.73	**49.89 ± 4.96**	8.30 ± 1.72	47.48 ± 2.06
OSULeaf	**45.62 ± 7.45**	30.41 ± 8.35	38.35 ± 12.00	24.05 ± 2.14	22.15 ± 5.63	27.82 ± 3.13	17.77 ± 1.80	26.03 ± 1.89	20.00 ± 5.09	26.17 ± 1.56	17.60 ± 2.44	29.83 ± 5.82
OliveOil	28.00 ± 23.64	26.67 ± 11.06	47.33 ± 13.21	**65.33 ± 15.20**	36.00 ± 22.66	32.22 ± 16.44	44.44 ± 10.72	28.89 ± 11.71	32.00 ± 9.60	61.11 ± 15.40	27.33 ± 11.88	28.00 ± 14.26
PLAID	15.46 ± 6.16	13.97 ± 4.45	16.95 ± 4.17	17.77 ± 3.34	21.79 ± 0.97	21.54 ± 5.77	13.78 ± 3.08	19.93 ± 5.41	14.41 ± 2.63	20.30 ± 4.75	21.79 ± 0.97	**32.10 ± 5.72**
PhalangesOutlinesCorrect	**54.90 ± 6.98**	51.28 ± 10.93	54.41 ± 7.38	53.40 ± 8.33	52.54 ± 10.38	48.87 ± 4.48	51.01 ± 10.34	53.38 ± 8.94	52.73 ± 9.70	52.49 ± 9.30	54.83 ± 6.32	51.93 ± 9.54
Phoneme	**10.43 ± 1.42**	8.46 ± 1.00	**10.43 ± 1.42**	4.21 ± 0.54	5.61 ± 2.38	6.14 ± 1.81	2.39 ± 1.01	5.87 ± 1.54	3.20 ± 2.06	4.98 ± 3.20	3.42 ± 0.41	6.87 ± 1.15
PickupGestureWiimoteZ	35.60 ± 7.80	44.80 ± 4.82	42.00 ± 5.10	40.80 ± 5.40	36.00 ± 3.16	46.67 ± 9.87	17.33 ± 5.03	39.33 ± 6.43	20.40 ± 4.98	**48.67 ± 8.33**	36.00 ± 3.16	47.20 ± 7.29
PigAirwayPressure	23.94 ± 3.76	9.13 ± 2.66	**38.65 ± 2.92**	3.65 ± 1.47	3.75 ± 1.33	5.29 ± 0.83	4.01 ± 0.73	4.01 ± 1.11	3.27 ± 0.92	4.81 ± 0.96	6.15 ± 1.64	5.48 ± 2.19
PigArtPressure	40.58 ± 5.14	10.10 ± 2.36	**60.10 ± 2.28**	9.90 ± 2.42	2.21 ± 0.80	10.58 ± 2.10	8.01 ± 2.00	10.58 ± 1.73	4.81 ± 0.76	9.94 ± 2.27	15.58 ± 3.05	15.38 ± 4.07
PigCVP	22.50 ± 3.47	14.33 ± 2.79	**50.29 ± 2.90**	5.00 ± 1.25	3.27 ± 0.71	5.93 ± 0.73	3.53 ± 1.39	6.89 ± 0.56	2.02 ± 0.22	4.33 ± 0.83	8.94 ± 1.94	11.44 ± 2.08
Plane	**98.86 ± 2.56**	91.62 ± 6.26	95.43 ± 3.65	87.24 ± 8.67	85.14 ± 9.49	82.86 ± 4.15	78.73 ± 5.25	88.57 ± 9.09	86.48 ± 4.39	88.57 ± 6.67	26.10 ± 13.02	93.90 ± 6.92
PowerCons	76.89 ± 9.68	**80.22 ± 9.41**	78.11 ± 13.88	60.22 ± 11.82	61.00 ± 7.74	62.96 ± 3.70	49.44 ± 7.47	58.33 ± 2.78	51.33 ± 4.56	62.78 ± 6.41	55.22 ± 21.33	60.89 ± 11.09
ProximalPhalanxOutlineAgeGroup	48.49 ± 34.51	48.20 ± 31.13	48.68 ± 30.82	39.61 ± 19.82	39.12 ± 18.60	43.74 ± 32.89	49.59 ± 25.02	**60.81 ± 40.85**	38.15 ± 22.87	46.50 ± 27.14	32.68 ± 16.57	48.98 ± 23.27
ProximalPhalanxOutlineCorrect	54.43 ± 13.05	53.40 ± 15.85	51.20 ± 16.57	56.22 ± 5.47	53.26 ± 10.68	44.67 ± 13.09	46.96 ± 12.63	**58.76 ± 6.63**	52.78 ± 12.27	43.99 ± 6.30	57.59 ± 13.02	52.23 ± 12.03
ProximalPhalanxTW	54.63 ± 10.31	57.56 ± 9.33	58.15 ± 6.64	54.34 ± 11.16	49.76 ± 11.67	**60.81 ± 11.71**	55.93 ± 9.23	38.37 ± 0.75	55.80 ± 7.24	57.07 ± 10.70	17.37 ± 16.23	55.22 ± 6.63
RefrigerationDevices	36.05 ± 6.49	35.68 ± 5.16	37.44 ± 5.37	34.99 ± 2.77	33.97 ± 4.76	**41.16 ± 5.10**	34.58 ± 1.47	34.93 ± 0.92	34.72 ± 0.99	38.84 ± 5.04	32.75 ± 2.24	29.76 ± 11.02
Rock	29.20 ± 8.07	34.40 ± 7.92	28.40 ± 6.54	41.60 ± 16.52	26.40 ± 8.53	38.67 ± 25.32	**57.33 ± 5.03**	44.67 ± 18.90	36.80 ± 8.07	40.00 ± 24.25	36.40 ± 14.93	39.20 ± 13.75
ScreenType	33.92 ± 4.73	34.19 ± 7.39	**34.29 ± 6.61**	33.87 ± 2.72	33.87 ± 4.61	32.44 ± 5.27	32.18 ± 3.99	28.00 ± 3.40	30.99 ± 3.01	33.60 ± 4.19	30.61 ± 2.66	32.53 ± 4.84
SemgHandGenderCh2	50.87 ± 2.35	55.37 ± 5.38	53.17 ± 4.35	50.17 ± 11.45	**61.00 ± 15.57**	42.78 ± 13.47	52.78 ± 1.84	39.89 ± 8.47	44.67 ± 12.14	35.61 ± 1.06	55.73 ± 4.74	51.90 ± 14.92
SemgHandMovementCh2	28.89 ± 4.22	**29.87 ± 4.86**	29.60 ± 5.24	25.29 ± 1.13	24.89 ± 3.57	20.37 ± 0.13	21.70 ± 2.31	22.52 ± 2.76	18.71 ± 1.91	20.07 ± 3.46	27.02 ± 5.10	28.13 ± 4.95

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Table 7: Per-dataset 1-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
SemgHandSubjectCh2	28.89 ± 5.77	29.07 ± 4.60	28.00 ± 2.65	35.60 ± 4.19	29.91 ± 10.77	26.37 ± 1.89	23.11 ± 2.89	26.67 ± 4.87	21.56 ± 1.94	26.96 ± 3.68	27.91 ± 3.11	**37.42 ± 8.82**
ShakeGestureWiimoteZ	63.20 ± 7.69	58.00 ± 11.83	**67.60 ± 11.61**	40.40 ± 11.26	43.20 ± 9.55	45.33 ± 3.06	16.00 ± 7.21	40.00 ± 3.46	24.40 ± 4.98	46.00 ± 6.00	43.20 ± 9.55	60.40 ± 7.92
ShapeletSim	66.33 ± 12.25	**87.00 ± 16.91**	73.89 ± 23.85	50.11 ± 2.27	75.22 ± 8.30	53.89 ± 1.67	52.96 ± 1.40	48.89 ± 5.64	50.44 ± 1.82	51.11 ± 0.96	47.78 ± 3.07	62.44 ± 7.44
ShapesAll	23.63 ± 4.60	15.60 ± 1.98	23.63 ± 4.60	45.70 ± 2.59	19.27 ± 4.58	44.72 ± 2.58	6.00 ± 0.58	43.94 ± 2.14	8.07 ± 3.90	44.06 ± 1.11	8.13 ± 2.75	**50.20 ± 3.03**
SmallKitchenAppliances	40.37 ± 15.51	**46.24 ± 15.42**	44.32 ± 15.50	42.19 ± 3.16	38.13 ± 22.42	32.98 ± 5.23	32.80 ± 3.67	35.82 ± 4.27	35.41 ± 3.13	36.89 ± 7.74	38.88 ± 7.80	42.88 ± 15.27
SmoothSubspace	**74.93 ± 5.59**	74.40 ± 5.13	74.80 ± 8.48	31.73 ± 3.96	50.00 ± 10.52	46.67 ± 1.76	53.33 ± 9.45	50.22 ± 10.96	62.27 ± 6.56	60.67 ± 10.35	56.67 ± 5.19	68.53 ± 8.95
SonyAIBORobotSurface1	77.57 ± 10.30	**80.80 ± 8.99**	80.37 ± 5.89	59.80 ± 6.73	64.53 ± 18.83	52.86 ± 10.38	56.96 ± 5.82	55.35 ± 9.37	56.07 ± 4.43	57.68 ± 9.32	55.61 ± 9.09	57.20 ± 8.62
SonyAIBORobotSurface2	80.29 ± 6.43	**84.09 ± 5.18**	80.90 ± 6.99	67.39 ± 19.08	73.03 ± 13.78	66.18 ± 11.61	60.37 ± 27.05	61.35 ± 27.10	69.49 ± 15.13	68.07 ± 16.06	57.31 ± 15.03	66.09 ± 10.78
StarLightCurves	63.75 ± 3.45	62.30 ± 9.88	57.59 ± 9.08	57.39 ± 13.40	57.55 ± 15.28	**67.54 ± 17.11**	37.78 ± 3.53	57.92 ± 15.55	27.99 ± 0.00	67.36 ± 17.40	64.81 ± 14.61	63.03 ± 14.23
Strawberry	52.27 ± 4.26	51.84 ± 3.86	54.43 ± 4.33	54.86 ± 4.34	53.62 ± 3.45	58.29 ± 0.95	54.86 ± 13.21	**60.18 ± 0.68**	56.11 ± 9.26	57.93 ± 0.83	45.89 ± 11.15	51.46 ± 5.21
SwedishLeaf	**65.95 ± 6.01**	46.08 ± 6.36	**65.95 ± 6.01**	27.42 ± 3.44	40.83 ± 4.40	36.69 ± 1.99	27.20 ± 2.42	24.53 ± 4.08	31.65 ± 2.48	38.19 ± 0.81	8.10 ± 1.23	57.98 ± 2.72
Symbols	**85.77 ± 4.83**	70.17 ± 4.91	76.68 ± 7.53	79.60 ± 7.65	72.50 ± 6.12	81.54 ± 3.92	37.15 ± 0.97	82.18 ± 0.42	46.13 ± 9.06	80.57 ± 0.95	51.54 ± 6.72	80.94 ± 7.54
SyntheticControl	72.53 ± 6.61	76.40 ± 5.24	68.67 ± 10.59	38.47 ± 6.28	80.33 ± 9.12	47.78 ± 5.87	44.89 ± 3.20	48.22 ± 0.38	42.13 ± 2.87	50.00 ± 4.48	24.53 ± 7.68	**84.20 ± 6.54**
ToeSegmentation1	**69.82 ± 7.39**	69.65 ± 4.27	65.09 ± 8.05	51.67 ± 1.47	55.79 ± 6.85	49.27 ± 1.10	49.71 ± 3.29	46.64 ± 4.73	52.46 ± 2.82	48.39 ± 3.23	49.47 ± 2.47	58.16 ± 3.70
ToeSegmentation2	68.92 ± 8.63	70.31 ± 3.38	70.92 ± 10.76	50.31 ± 5.37	65.54 ± 9.84	50.26 ± 2.91	35.90 ± 10.81	50.51 ± 4.24	33.23 ± 16.47	52.05 ± 6.54	41.54 ± 14.95	**75.23 ± 12.13**
Trace	61.00 ± 4.64	65.20 ± 9.09	65.80 ± 6.61	49.60 ± 3.65	74.40 ± 6.15	51.33 ± 5.51	42.33 ± 4.51	47.67 ± 0.58	44.80 ± 7.69	51.33 ± 4.93	37.80 ± 6.42	**99.20 ± 1.79**
TwoLeadECG	**69.48 ± 12.74**	64.97 ± 10.26	66.51 ± 13.23	54.03 ± 2.33	57.82 ± 5.04	59.32 ± 3.05	53.32 ± 7.30	52.62 ± 3.32	54.47 ± 5.41	60.32 ± 1.06	50.73 ± 1.02	64.88 ± 11.28
TwoPatterns	48.77 ± 2.41	**61.51 ± 6.85**	46.17 ± 11.65	31.86 ± 3.24	36.86 ± 6.10	29.76 ± 2.56	25.52 ± 1.93	28.23 ± 0.74	26.06 ± 1.27	29.87 ± 1.68	26.32 ± 2.53	45.74 ± 2.64
UMD	43.06 ± 4.02	**54.58 ± 9.53**	45.56 ± 6.14	46.81 ± 7.70	50.14 ± 6.28	46.30 ± 5.02	45.60 ± 3.21	47.69 ± 4.19	48.61 ± 2.55	47.45 ± 6.56	31.39 ± 4.35	45.47 ± 8.73
UWaveGestureLibraryAll	29.36 ± 4.20	24.46 ± 5.19	28.34 ± 3.82	66.77 ± 5.89	41.63 ± 5.40	69.19 ± 5.01	25.78 ± 1.68	67.23 ± 2.91	26.67 ± 3.53	**69.21 ± 6.24**	50.70 ± 5.42	33.37 ± 0.26
UWaveGestureLibraryX	26.45 ± 4.19	26.60 ± 3.44	24.71 ± 1.83	47.27 ± 6.01	35.36 ± 5.04	**49.30 ± 7.17**	22.37 ± 0.14	47.22 ± 8.02	24.34 ± 5.30	49.17 ± 7.61	17.28 ± 8.20	30.33 ± 0.23
UWaveGestureLibraryY	25.37 ± 4.63	26.13 ± 4.45	24.62 ± 4.93	43.95 ± 4.51	38.35 ± 7.26	47.05 ± 4.02	24.58 ± 1.89	46.37 ± 4.93	26.81 ± 2.91	**47.18 ± 5.20**	20.37 ± 4.71	40.36 ± 0.25
UWaveGestureLibraryZ	24.77 ± 4.74	27.03 ± 3.18	24.45 ± 3.15	42.90 ± 1.96	36.37 ± 3.72	44.61 ± 1.78	22.47 ± 2.82	44.70 ± 2.64	25.10 ± 2.80	44.75 ± 1.63	22.77 ± 3.47	**47.44 ± 2.61**
Wafer	50.39 ± 4.56	44.94 ± 10.31	49.91 ± 13.57	**67.87 ± 4.81**	51.56 ± 17.59	63.90 ± 3.73	64.15 ± 3.76	63.99 ± 9.60	63.82 ± 9.82	65.48 ± 1.03	43.02 ± 26.21	46.27 ± 7.11
Wine	**57.41 ± 14.70**	54.07 ± 4.61	54.81 ± 11.31	52.96 ± 8.34	50.74 ± 6.23	48.77 ± 3.85	50.00 ± 0.00	50.00 ± 0.00	52.22 ± 4.97	37.65 ± 19.80	51.48 ± 3.31	51.11 ± 7.48
WordSynonyms	14.92 ± 3.03	13.70 ± 5.79	14.92 ± 3.03	19.78 ± 2.70	16.96 ± 4.46	19.75 ± 1.90	6.79 ± 1.46	**21.16 ± 1.50**	10.06 ± 3.54	20.53 ± 2.41	4.98 ± 1.34	20.91 ± 2.73
Worms	**36.10 ± 9.87**	33.25 ± 6.66	35.06 ± 11.43	17.14 ± 4.04	23.90 ± 9.39	19.91 ± 6.54	23.38 ± 4.50	18.61 ± 5.86	17.66 ± 7.03	19.05 ± 5.25	17.66 ± 2.36	24.68 ± 8.95
WormsTwoClass	44.79 ± 3.48	49.09 ± 6.96	52.99 ± 6.71	49.09 ± 3.72	50.39 ± 3.23	49.35 ± 3.44	48.92 ± 10.50	52.38 ± 1.98	53.51 ± 4.81	**53.68 ± 3.75**	52.99 ± 2.82	47.53 ± 10.69
Yoga	**55.60 ± 4.57**	50.68 ± 5.18	53.54 ± 4.28	49.99 ± 1.44	49.75 ± 4.01	49.48 ± 0.91	51.11 ± 3.62	49.62 ± 0.72	51.12 ± 3.30	48.62 ± 0.52	49.35 ± 2.85	50.42 ± 4.51
Avg. Acc.	47.79	45.40	47.88	44.21	42.39	45.16	36.78	44.31	37.71	46.42	35.61	**48.31**
Avg. Rank	5.11	5.80	5.19	6.45	6.92	6.04	9.11	6.17	8.51	5.30	8.82	**4.58**
#Best	25	16	14	7	5	11	2	11	2	15	4	**26**

Table 8: Per-dataset 2-shot classification results on 128 shared UCR datasets. Each value is reported as mean ± standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
ACSF1	**61.60 ± 7.86**	47.20 ± 6.30	59.20 ± 5.26	25.80 ± 4.38	42.00 ± 3.16	40.33 ± 2.31	30.67 ± 2.52	44.00 ± 6.24	35.80 ± 3.19	40.67 ± 5.69	24.80 ± 4.66	54.33 ± 4.16
Adiac	35.45 ± 2.60	14.27 ± 3.74	35.45 ± 2.60	18.87 ± 2.86	17.80 ± 5.08	27.79 ± 2.14	44.67 ± 0.39	16.62 ± 1.77	55.24 ± 4.67	32.91 ± 3.49	3.48 ± 1.27	**56.01 ± 1.25**
AllGestureWiimoteX	**32.43 ± 2.94**	27.69 ± 3.48	32.09 ± 3.58	19.46 ± 2.47	23.77 ± 3.63	24.43 ± 4.71	11.86 ± 1.00	22.67 ± 2.29	14.09 ± 1.28	21.10 ± 5.46	23.77 ± 3.63	26.51 ± 4.36
AllGestureWiimoteY	**35.11 ± 5.43**	28.57 ± 4.31	28.91 ± 5.81	20.54 ± 2.69	33.31 ± 2.65	24.33 ± 2.50	12.00 ± 1.38	21.90 ± 1.33	16.14 ± 1.67	22.57 ± 1.00	33.31 ± 2.65	29.43 ± 1.69
AllGestureWiimoteZ	**29.80 ± 2.53**	26.94 ± 2.84	29.31 ± 4.56	16.77 ± 2.02	22.97 ± 5.35	19.86 ± 4.12	12.76 ± 0.36	19.57 ± 2.48	14.97 ± 0.98	20.29 ± 3.48	22.97 ± 5.35	27.91 ± 5.21
ArrowHead	50.17 ± 5.13	44.80 ± 3.82	45.14 ± 5.25	**57.03 ± 4.92**	49.14 ± 13.02	53.14 ± 6.02	34.10 ± 6.27	50.67 ± 2.01	42.29 ± 4.92	55.43 ± 6.44	43.77 ± 8.76	**57.03 ± 2.81**
BME	48.40 ± 4.91	55.33 ± 7.02	54.40 ± 8.12	50.80 ± 5.11	67.33 ± 7.90	48.89 ± 2.14	64.00 ± 5.21	58.89 ± 3.91	67.33 ± 7.44	54.89 ± 9.20	50.80 ± 8.05	**79.47 ± 11.40**
Beef	42.00 ± 2.98	36.67 ± 8.50	36.70 ± 9.60	49.33 ± 7.23	39.33 ± 6.41	48.89 ± 5.09	40.00 ± 3.33	34.44 ± 5.09	42.67 ± 4.94	**51.11 ± 7.70**	32.67 ± 7.23	43.33 ± 5.27
BeetleFly	50.00 ± 9.35	67.00 ± 18.23	55.00 ± 12.75	52.00 ± 11.51	57.00 ± 16.43	45.00 ± 10.00	48.33 ± 12.58	38.33 ± 7.64	45.00 ± 9.35	56.67 ± 7.64	57.00 ± 4.47	**75.00 ± 18.03**
BirdChicken	71.00 ± 6.52	72.00 ± 4.47	**76.00 ± 7.42**	52.00 ± 14.40	65.00 ± 10.61	45.00 ± 8.66	51.67 ± 7.64	46.67 ± 5.77	53.00 ± 2.74	50.00 ± 13.23	49.00 ± 2.24	69.00 ± 22.19
CBF	85.76 ± 5.83	**90.62 ± 7.16**	82.18 ± 10.67	64.78 ± 6.67	82.69 ± 10.89	68.07 ± 6.00	34.89 ± 2.41	73.78 ± 6.88	36.78 ± 1.75	70.00 ± 6.65	42.84 ± 7.09	85.91 ± 4.23
Car	42.00 ± 15.43	41.67 ± 13.49	52.67 ± 7.87	55.33 ± 10.63	43.00 ± 11.45	54.44 ± 6.74	32.78 ± 0.96	52.78 ± 10.84	37.00 ± 7.76	55.56 ± 4.19	31.00 ± 10.71	**63.33 ± 7.45**
Chinatown	95.74 ± 1.26	**96.73 ± 1.19**	**96.73 ± 0.81**	54.52 ± 24.75	91.08 ± 5.39	85.33 ± 8.50	86.10 ± 2.71	84.55 ± 3.36	80.70 ± 5.41	82.80 ± 4.37	45.01 ± 24.65	91.84 ± 4.85
ChlorineConcentration	34.61 ± 4.95	35.24 ± 5.79	35.19 ± 5.79	34.54 ± 2.93	34.28 ± 2.22	31.22 ± 6.21	33.24 ± 2.95	33.84 ± 2.69	33.91 ± 2.07	33.69 ± 2.59	**39.08 ± 9.47**	32.47 ± 3.03
CinCECGTorso	46.52 ± 12.75	48.35 ± 9.58	51.06 ± 10.06	39.77 ± 4.46	54.91 ± 12.73	48.14 ± 2.63	36.18 ± 3.15	53.94 ± 2.51	50.30 ± 12.26	48.89 ± 2.81	49.00 ± 5.40	**62.95 ± 6.66**
Coffee	**94.29 ± 7.41**	88.57 ± 12.73	**94.29 ± 3.19**	92.14 ± 3.91	90.71 ± 10.89	73.81 ± 8.99	63.10 ± 7.43	85.71 ± 3.57	84.29 ± 10.29	77.38 ± 4.12	65.71 ± 15.07	88.57 ± 4.66
Computers	52.24 ± 10.15	54.40 ± 10.81	52.72 ± 12.14	48.08 ± 7.70	53.28 ± 11.25	48.00 ± 10.51	48.67 ± 4.60	48.00 ± 11.45	50.40 ± 1.72	46.67 ± 11.22	50.16 ± 7.53	**57.73 ± 17.44**
CricketX	**36.51 ± 3.27**	34.10 ± 3.98	33.03 ± 3.86	20.62 ± 2.07	30.56 ± 3.87	27.35 ± 2.06	8.12 ± 1.41	26.15 ± 1.85	12.15 ± 1.24	27.44 ± 2.96	11.54 ± 1.86	35.98 ± 3.13
CricketY	**37.54 ± 5.48**	30.62 ± 2.69	33.28 ± 3.86	22.56 ± 4.44	24.21 ± 2.69	26.50 ± 7.27	9.06 ± 0.39	25.90 ± 6.29	12.67 ± 0.94	27.52 ± 6.80	10.00 ± 3.24	30.43 ± 5.23
CricketZ	**40.41 ± 4.15**	35.49 ± 2.72	33.69 ± 5.21	20.36 ± 1.13	24.92 ± 3.09	26.75 ± 2.72	8.12 ± 1.50	24.44 ± 0.78	12.97 ± 2.47	26.50 ± 2.74	11.18 ± 2.42	33.33 ± 3.02
Crop	31.31 ± 3.87	37.82 ± 2.09	31.31 ± 3.87	14.67 ± 0.98	24.66 ± 2.54	36.33 ± 2.63	35.06 ± 3.15	37.89 ± 1.28	35.83 ± 2.39	38.66 ± 2.99	4.63 ± 0.75	**43.94 ± 2.11**
DiatomSizeReduction	78.43 ± 9.98	84.44 ± 4.38	89.74 ± 8.40	83.73 ± 5.41	63.01 ± 13.25	84.42 ± 6.89	59.91 ± 2.78	84.97 ± 5.38	59.28 ± 18.44	82.35 ± 7.47	35.10 ± 12.79	**90.39 ± 5.62**
DistalPhalanxOutlineAgeGroup	66.91 ± 6.08	**67.77 ± 3.31**	65.76 ± 6.53	55.54 ± 12.17	58.42 ± 11.15	57.07 ± 19.57	50.84 ± 14.56	56.12 ± 13.32	56.26 ± 12.92	57.07 ± 14.47	38.99 ± 8.80	**67.77 ± 3.39**
DistalPhalanxOutlineCorrect	52.32 ± 12.15	51.59 ± 11.49	52.54 ± 12.44	49.57 ± 11.83	51.45 ± 15.36	**57.49 ± 13.09**	56.76 ± 11.56	55.80 ± 11.93	51.38 ± 12.34	56.64 ± 12.97	50.65 ± 9.31	55.80 ± 14.95
DistalPhalanxTW	49.50 ± 8.83	**55.54 ± 4.21**	54.10 ± 4.92	39.28 ± 8.61	53.96 ± 7.56	41.97 ± 11.56	42.45 ± 10.74	39.81 ± 18.49	47.05 ± 8.79	45.80 ± 16.17	22.45 ± 8.85	51.37 ± 8.97
DodgerLoopDay	28.75 ± 2.34	23.75 ± 2.50	28.00 ± 5.70	**44.00 ± 4.09**	38.50 ± 7.42	39.58 ± 4.02	23.75 ± 2.50	42.50 ± 7.60	19.75 ± 4.09	42.50 ± 3.31	38.50 ± 7.42	35.50 ± 8.78
DodgerLoopGame	54.93 ± 6.45	63.04 ± 8.02	58.41 ± 11.09	53.62 ± 8.51	65.65 ± 18.08	67.87 ± 7.26	48.55 ± 4.53	**71.50 ± 0.84**	50.14 ± 3.81	68.36 ± 3.42	65.65 ± 18.08	61.01 ± 13.03
DodgerLoopWeekend	92.75 ± 2.40	87.68 ± 5.59	92.03 ± 3.69	94.64 ± 2.74	96.09 ± 1.41	96.38 ± 1.26	64.98 ± 4.71	**97.34 ± 0.42**	65.22 ± 8.45	96.14 ± 0.84	96.09 ± 1.41	84.64 ± 2.73
ECG200	69.40 ± 7.77	75.20 ± 7.05	71.80 ± 5.54	76.80 ± 3.56	71.80 ± 4.38	**78.33 ± 1.53**	74.33 ± 5.69	75.67 ± 3.21	73.60 ± 3.65	75.33 ± 0.58	52.20 ± 18.20	73.00 ± 3.46
ECG5000	61.90 ± 13.09	69.05 ± 11.38	61.86 ± 10.65	**74.23 ± 10.96**	58.61 ± 14.82	72.42 ± 4.03	65.61 ± 11.76	69.40 ± 4.84	67.93 ± 11.62	66.36 ± 9.81	30.14 ± 19.40	67.68 ± 9.76
ECGFiveDays	68.78 ± 9.46	56.79 ± 4.91	71.73 ± 8.76	57.12 ± 2.23	67.29 ± 3.58	66.55 ± 6.20	74.64 ± 13.99	**75.18 ± 2.37**	68.76 ± 10.24	68.52 ± 6.54	57.79 ± 7.24	69.12 ± 9.82
EOGHorizontalSignal	20.66 ± 1.70	23.43 ± 5.48	21.38 ± 4.60	25.86 ± 4.54	28.07 ± 2.90	30.66 ± 2.91	12.06 ± 1.96	31.68 ± 2.61	9.56 ± 1.50	29.01 ± 2.49	29.83 ± 3.23	**37.35 ± 4.07**
EOGVerticalSignal	13.37 ± 1.32	16.57 ± 3.25	14.86 ± 2.77	26.80 ± 1.98	29.56 ± 4.11	26.98 ± 4.28	11.88 ± 1.54	26.34 ± 4.16	10.83 ± 1.45	26.61 ± 6.09	20.88 ± 2.94	**31.82 ± 3.47**
Earthquakes	52.95 ± 7.17	57.84 ± 11.11	49.35 ± 10.38	53.24 ± 4.93	**63.88 ± 6.42**	63.31 ± 10.45	49.16 ± 1.10	54.92 ± 17.17	56.55 ± 8.34	60.43 ± 21.88	57.12 ± 15.34	55.25 ± 14.33
ElectricDevices	37.63 ± 9.44	41.80 ± 2.08	34.70 ± 5.95	21.27 ± 5.73	37.69 ± 10.00	28.63 ± 3.21	22.56 ± 2.12	20.63 ± 2.87	22.76 ± 4.51	29.88 ± 3.14	14.36 ± 4.97	**47.14 ± 3.99**
EthanolLevel	25.32 ± 0.64	25.52 ± 1.56	25.12 ± 2.09	25.76 ± 1.71	26.96 ± 0.82	26.87 ± 2.61	**29.20 ± 1.91**	25.67 ± 3.18	25.24 ± 0.98	26.47 ± 2.08	25.12 ± 0.81	28.00 ± 3.11
FaceAll	**55.98 ± 6.19**	45.22 ± 8.18	**55.98 ± 6.19**	40.27 ± 5.76	24.12 ± 4.85	38.56 ± 3.11	38.84 ± 3.81	36.71 ± 2.55	38.89 ± 3.64	41.40 ± 3.79	11.35 ± 5.01	39.09 ± 1.28
FaceFour	77.73 ± 4.99	52.05 ± 10.37	81.82 ± 7.14	74.09 ± 5.98	59.32 ± 5.76	79.55 ± 9.02	57.58 ± 7.31	80.68 ± 3.41	62.50 ± 7.14	**84.85 ± 2.37**	40.45 ± 14.16	68.55 ± 4.91
FacesUCR	**67.20 ± 6.42**	51.90 ± 6.85	62.04 ± 8.47	41.67 ± 3.36	25.42 ± 6.93	44.55 ± 2.52	45.95 ± 4.27	49.07 ± 5.45	48.56 ± 3.79	49.27 ± 6.47	10.96 ± 5.26	55.08 ± 2.32
FiftyWords	19.03 ± 1.51	13.49 ± 1.95	19.03 ± 1.51	40.09 ± 3.22	28.35 ± 3.05	44.69 ± 2.97	5.05 ± 0.44	46.74 ± 2.43	16.53 ± 1.75	44.18 ± 3.70	15.16 ± 3.75	**47.30 ± 2.89**
Fish	**50.97 ± 10.77**	26.97 ± 9.88	50.51 ± 16.65	50.17 ± 6.36	31.54 ± 5.18	45.90 ± 8.88	20.00 ± 4.12	39.05 ± 5.64	22.97 ± 3.62	45.71 ± 10.47	22.86 ± 2.83	49.91 ± 10.25
FordA	**64.98 ± 8.59**	62.27 ± 9.82	62.82 ± 9.33	49.92 ± 1.84	60.48 ± 7.29	50.30 ± 1.31	50.66 ± 0.90	50.15 ± 0.79	50.27 ± 1.01	50.58 ± 1.43	50.76 ± 2.11	53.35 ± 2.84
FordB	**61.19 ± 3.67**	58.10 ± 1.40	58.67 ± 2.63	49.09 ± 1.35	56.89 ± 2.99	49.84 ± 0.63	50.99 ± 2.05	50.25 ± 1.29	49.95 ± 1.77	50.58 ± 0.68	50.77 ± 0.87	50.42 ± 2.72
FreezerRegularTrain	68.01 ± 9.57	67.73 ± 22.81	61.99 ± 17.00	67.56 ± 20.68	71.73 ± 5.42	59.79 ± 26.51	53.13 ± 0.84	60.39 ± 25.70	58.85 ± 2.14	60.74 ± 25.80	72.39 ± 6.35	**79.14 ± 7.94**
FreezerSmallTrain	66.47 ± 12.08	65.70 ± 14.41	66.00 ± 12.69	66.60 ± 13.71	64.67 ± 11.73	**71.47 ± 6.03**	56.34 ± 1.10	68.49 ± 7.01	56.88 ± 4.22	71.17 ± 6.80	65.57 ± 16.06	64.30 ± 13.28
Fungi	97.10 ± 2.45	34.84 ± 7.53	66.24 ± 2.06	68.17 ± 5.25	68.17 ± 13.75	**97.13 ± 0.82**	75.99 ± 1.64	82.62 ± 2.42	74.73 ± 0.85	96.95 ± 0.31	8.06 ± 3.93	81.97 ± 3.24
GestureMidAirD1	24.62 ± 1.44	29.08 ± 3.94	20.62 ± 1.44	38.31 ± 3.66	29.38 ± 2.74	37.69 ± 3.35	17.95 ± 2.70	36.67 ± 1.78	35.69 ± 2.53	36.92 ± 2.77	29.38 ± 2.74	**44.92 ± 6.31**
GestureMidAirD2	16.46 ± 2.22	21.85 ± 4.16	16.46 ± 2.22	35.69 ± 3.47	33.69 ± 3.66	42.05 ± 0.44	12.05 ± 1.60	44.36 ± 2.47	30.92 ± 4.26	**45.64 ± 3.20**	33.69 ± 3.66	43.85 ± 4.03
GestureMidAirD3	10.15 ± 1.99	11.85 ± 3.91	10.15 ± 1.99	25.08 ± 3.59	18.00 ± 2.64	26.41 ± 1.18	14.62 ± 4.07	**26.92 ± 1.54**	22.00 ± 3.63	**26.92 ± 2.77**	18.00 ± 2.64	21.08 ± 1.85

Continued on next page

Table 8: Per-dataset 2-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
GesturePebbleZ1	62.44 ± 5.24	41.40 ± 5.40	**64.88 ± 2.42**	53.26 ± 4.18	29.53 ± 10.46	58.91 ± 3.41	16.86 ± 3.54	57.17 ± 0.89	24.19 ± 4.00	61.82 ± 3.41	29.53 ± 10.46	37.91 ± 6.35
GesturePebbleZ2	**66.58 ± 5.17**	44.05 ± 14.70	61.70 ± 6.13	51.90 ± 9.19	35.32 ± 9.94	55.06 ± 7.46	15.82 ± 0.63	46.84 ± 8.79	20.76 ± 3.15	56.75 ± 4.12	35.32 ± 9.94	41.39 ± 4.24
GunPoint	67.07 ± 8.15	65.20 ± 5.95	66.00 ± 5.89	60.13 ± 7.74	65.47 ± 7.68	55.33 ± 6.57	64.44 ± 5.18	58.44 ± 9.46	68.13 ± 5.80	57.11 ± 7.95	54.00 ± 6.53	**79.33 ± 16.25**
GunPointAgeSpan	65.38 ± 2.37	63.73 ± 4.38	**68.61 ± 1.37**	67.41 ± 13.19	61.08 ± 13.41	61.08 ± 8.21	52.22 ± 8.74	55.49 ± 10.60	52.41 ± 8.13	58.76 ± 12.22	47.41 ± 14.13	64.49 ± 7.90
GunPointMaleVersusFemale	79.30 ± 6.81	84.62 ± 8.74	81.65 ± 8.32	57.15 ± 7.06	65.95 ± 9.57	60.65 ± 4.86	50.11 ± 5.57	60.76 ± 10.27	52.59 ± 5.35	65.82 ± 3.12	65.00 ± 5.79	**87.66 ± 5.66**
GunPointOldVersusYoung	**91.94 ± 11.58**	89.97 ± 13.74	91.37 ± 12.08	72.89 ± 4.20	73.14 ± 5.24	73.54 ± 7.07	56.51 ± 2.29	72.49 ± 6.74	59.30 ± 3.02	75.87 ± 9.07	78.03 ± 17.26	70.92 ± 5.05
Ham	45.71 ± 9.69	50.86 ± 8.18	48.00 ± 10.27	54.86 ± 11.40	53.33 ± 7.28	54.92 ± 1.98	**58.73 ± 5.25**	50.48 ± 13.83	55.24 ± 4.62	57.78 ± 9.77	52.57 ± 10.93	50.86 ± 8.07
HandOutlines	57.19 ± 12.29	63.30 ± 2.05	60.65 ± 3.52	69.19 ± 14.14	63.57 ± 17.16	67.21 ± 16.82	52.16 ± 12.39	68.11 ± 20.52	41.57 ± 12.57	67.48 ± 15.69	**69.89 ± 13.52**	64.59 ± 10.14
Haptics	29.16 ± 3.65	24.35 ± 4.42	25.32 ± 2.75	28.31 ± 7.72	28.25 ± 7.08	**29.33 ± 7.26**	23.81 ± 0.68	29.22 ± 2.66	20.97 ± 3.07	29.00 ± 7.31	26.95 ± 8.77	26.62 ± 4.64
Herring	52.81 ± 11.18	**56.88 ± 7.13**	52.81 ± 11.23	51.25 ± 5.89	52.19 ± 12.13	47.92 ± 6.51	46.35 ± 3.61	45.83 ± 3.25	50.00 ± 4.69	48.96 ± 4.51	45.00 ± 9.84	55.00 ± 4.34
HouseTwenty	**76.30 ± 9.04**	69.24 ± 11.83	67.39 ± 13.31	60.00 ± 7.10	63.03 ± 13.56	68.91 ± 7.47	47.34 ± 6.31	59.94 ± 1.75	52.10 ± 8.21	71.15 ± 7.81	60.34 ± 11.20	71.93 ± 10.30
InlineSkate	18.58 ± 2.99	19.56 ± 1.81	**19.93 ± 2.98**	18.29 ± 2.68	17.05 ± 3.64	16.85 ± 1.00	15.76 ± 0.93	15.82 ± 1.62	16.29 ± 1.31	17.03 ± 0.46	16.36 ± 2.46	18.98 ± 2.54
InsectEPGRegularTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	51.97 ± 7.58	58.39 ± 6.76	58.37 ± 4.44	37.08 ± 10.70	54.89 ± 2.35	37.99 ± 13.89	59.71 ± 4.91	81.69 ± 17.66	68.76 ± 9.60
InsectEPGSmallTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	51.33 ± 4.40	58.47 ± 6.25	57.56 ± 4.71	41.37 ± 7.24	58.23 ± 4.08	42.65 ± 10.05	58.10 ± 4.04	80.40 ± 17.17	67.79 ± 2.45
InsectWingbeatSound	20.38 ± 3.08	16.62 ± 4.40	18.70 ± 3.47	47.87 ± 2.61	33.18 ± 5.98	50.44 ± 4.17	20.69 ± 3.83	49.41 ± 1.90	27.56 ± 2.55	**51.03 ± 4.83**	16.68 ± 7.51	31.49 ± 3.72
ItalyPowerDemand	68.88 ± 12.96	75.63 ± 13.47	73.61 ± 12.06	50.03 ± 0.16	79.18 ± 15.47	54.49 ± 7.77	75.87 ± 16.00	76.35 ± 22.01	**83.15 ± 15.39**	73.57 ± 21.06	50.07 ± 0.34	65.03 ± 13.70
LargeKitchenAppliances	45.65 ± 8.51	**54.19 ± 10.41**	50.93 ± 13.42	33.60 ± 7.89	44.85 ± 7.66	37.96 ± 12.60	32.00 ± 3.23	34.40 ± 7.77	32.96 ± 7.01	36.62 ± 11.05	32.00 ± 6.67	42.56 ± 5.00
Lightning2	57.05 ± 8.63	53.77 ± 7.90	59.34 ± 7.64	54.10 ± 10.50	51.80 ± 7.11	57.38 ± 4.34	49.73 ± 9.88	51.91 ± 4.13	52.13 ± 11.90	55.19 ± 6.21	50.16 ± 8.80	**63.28 ± 8.25**
Lightning7	45.48 ± 6.31	43.01 ± 5.94	42.74 ± 6.38	38.90 ± 8.91	47.95 ± 8.16	**59.36 ± 11.07**	19.63 ± 3.45	50.23 ± 12.43	27.40 ± 4.94	56.16 ± 10.34	20.82 ± 11.60	46.03 ± 6.17
Mallat	73.35 ± 15.31	47.53 ± 14.01	80.84 ± 4.13	86.79 ± 1.54	65.16 ± 19.15	**91.78 ± 0.99**	63.31 ± 2.66	78.79 ± 4.89	70.90 ± 5.59	87.65 ± 2.43	49.21 ± 6.11	80.67 ± 4.06
Meat	54.33 ± 7.42	42.00 ± 7.76	73.00 ± 16.39	**89.67 ± 7.85**	52.00 ± 12.16	53.33 ± 6.01	43.33 ± 8.33	41.11 ± 7.52	51.33 ± 18.61	57.22 ± 7.88	36.33 ± 5.82	57.67 ± 17.30
MedicalImages	33.45 ± 8.28	25.74 ± 5.92	33.61 ± 5.62	24.45 ± 6.71	27.45 ± 4.51	28.46 ± 3.05	27.85 ± 6.61	28.55 ± 2.80	**34.53 ± 1.25**	30.00 ± 1.71	29.82 ± 16.29	31.63 ± 5.59
MelbournePedestrian	**63.62 ± 4.54**	57.50 ± 8.49	63.00 ± 2.07	14.05 ± 5.17	42.42 ± 4.22	40.29 ± 3.90	51.77 ± 5.29	50.24 ± 2.74	54.32 ± 5.40	46.37 ± 2.70	42.42 ± 4.22	49.78 ± 4.68
MiddlePhalanxOutlineAgeGroup	47.01 ± 9.52	44.55 ± 12.64	45.45 ± 9.45	**50.39 ± 7.58**	44.29 ± 10.12	37.45 ± 10.11	46.75 ± 13.26	45.24 ± 16.76	48.57 ± 12.10	48.92 ± 14.26	22.86 ± 4.12	42.99 ± 12.95
MiddlePhalanxOutlineCorrect	47.01 ± 7.12	51.13 ± 7.83	49.42 ± 8.07	53.75 ± 12.47	44.54 ± 7.47	52.92 ± 7.99	47.54 ± 10.52	**54.07 ± 11.30**	48.59 ± 8.66	53.26 ± 15.07	50.72 ± 11.13	48.87 ± 3.56
MiddlePhalanxTW	37.92 ± 11.48	44.68 ± 3.99	42.73 ± 9.27	38.96 ± 14.59	39.22 ± 13.51	45.45 ± 5.95	43.07 ± 3.27	36.36 ± 13.61	**46.36 ± 2.32**	45.24 ± 3.58	16.36 ± 7.24	44.81 ± 7.78
MixedShapesRegularTrain	55.18 ± 4.21	33.76 ± 13.30	55.82 ± 8.68	58.38 ± 6.27	54.48 ± 12.34	62.60 ± 3.23	19.42 ± 0.39	63.00 ± 3.88	22.36 ± 6.85	63.64 ± 2.51	54.05 ± 4.44	**67.98 ± 4.06**
MixedShapesSmallTrain	45.73 ± 9.51	37.76 ± 12.10	51.77 ± 7.21	61.69 ± 3.51	54.28 ± 10.55	63.00 ± 2.46	19.05 ± 0.77	64.04 ± 1.32	24.01 ± 6.26	62.64 ± 3.39	57.53 ± 4.77	**68.35 ± 3.26**
MoteStrain	81.95 ± 4.07	75.34 ± 9.74	78.95 ± 3.25	70.97 ± 10.57	75.59 ± 8.13	78.41 ± 9.87	68.53 ± 3.31	82.48 ± 1.96	68.63 ± 10.07	**83.65 ± 4.68**	55.75 ± 11.79	58.56 ± 15.12
NonInvasiveFetalECGThorax1	25.55 ± 2.02	14.44 ± 3.90	25.55 ± 2.02	**55.96 ± 3.30**	18.44 ± 9.38	50.77 ± 4.84	18.37 ± 2.16	51.65 ± 2.21	21.42 ± 2.01	53.89 ± 1.99	13.55 ± 0.51	50.58 ± 2.13
NonInvasiveFetalECGThorax2	27.07 ± 5.56	22.30 ± 6.32	27.07 ± 5.56	**62.09 ± 2.37**	28.27 ± 8.71	55.69 ± 4.28	27.23 ± 2.03	56.05 ± 0.94	30.73 ± 3.23	58.63 ± 2.42	18.29 ± 1.75	59.72 ± 2.76
OSULeaf	55.70 ± 9.28	38.35 ± 6.31	**56.03 ± 5.89**	25.79 ± 2.77	25.29 ± 5.19	29.89 ± 3.34	18.18 ± 1.43	28.10 ± 2.07	21.90 ± 3.33	28.51 ± 4.37	23.31 ± 3.85	36.28 ± 6.97
OliveOil	64.67 ± 10.70	32.67 ± 5.48	71.33 ± 10.70	**80.67 ± 3.65**	17.33 ± 10.90	32.22 ± 11.71	38.89 ± 7.70	25.56 ± 12.62	24.00 ± 10.90	58.89 ± 22.69	22.00 ± 13.04	23.33 ± 9.72
PLAID	19.29 ± 3.99	16.65 ± 2.54	20.89 ± 4.30	22.98 ± 3.80	35.53 ± 7.89	27.56 ± 4.82	17.44 ± 2.97	25.88 ± 4.40	19.81 ± 3.24	28.68 ± 3.91	35.53 ± 7.89	**40.19 ± 5.06**
PhalangesOutlinesCorrect	52.91 ± 9.24	53.03 ± 10.79	52.61 ± 9.16	56.64 ± 11.15	54.13 ± 11.04	**60.64 ± 3.81**	58.70 ± 2.16	60.37 ± 4.20	52.26 ± 8.88	60.06 ± 3.62	49.04 ± 14.25	50.84 ± 2.93
Phoneme	**12.22 ± 1.88**	10.75 ± 2.12	11.22 ± 1.88	5.41 ± 0.40	7.31 ± 0.69	7.77 ± 0.84	4.13 ± 1.74	7.00 ± 0.45	6.19 ± 2.34	5.29 ± 0.61	5.20 ± 0.75	10.02 ± 0.59
PickupGestureWiimoteZ	43.20 ± 11.37	53.20 ± 9.55	**62.40 ± 1.67**	43.20 ± 6.72	40.40 ± 8.29	52.67 ± 10.07	21.33 ± 1.15	53.33 ± 4.16	27.60 ± 3.58	51.33 ± 6.43	40.40 ± 8.29	58.80 ± 3.03
PigAirwayPressure	28.37 ± 3.10	13.08 ± 1.90	**49.52 ± 0.76**	5.00 ± 0.26	3.65 ± 0.87	4.49 ± 0.28	3.21 ± 1.21	7.05 ± 0.73	4.62 ± 1.05	6.09 ± 0.73	5.10 ± 0.80	9.81 ± 0.43
PigArtPressure	50.58 ± 2.62	23.65 ± 3.56	**69.42 ± 3.86**	7.88 ± 0.26	5.29 ± 1.63	10.10 ± 0.48	8.01 ± 1.39	9.78 ± 1.11	7.12 ± 1.24	8.33 ± 0.56	19.13 ± 1.04	30.67 ± 1.38
PigCVP	31.54 ± 0.94	18.08 ± 2.27	**61.92 ± 1.78**	7.02 ± 0.43	3.94 ± 1.10	6.73 ± 0.00	4.01 ± 0.56	7.85 ± 1.00	4.13 ± 0.80	5.29 ± 0.83	10.10 ± 1.40	19.33 ± 0.99
Plane	**100.00 ± 0.00**	94.29 ± 3.75	99.05 ± 1.17	86.10 ± 6.69	91.62 ± 3.39	86.98 ± 7.02	84.44 ± 7.02	81.90 ± 1.90	87.24 ± 4.98	86.98 ± 6.34	21.71 ± 8.89	99.24 ± 0.80
PowerCons	82.11 ± 6.61	**83.78 ± 3.63**	82.33 ± 7.91	72.11 ± 2.98	70.78 ± 14.65	69.81 ± 4.10	52.22 ± 1.47	70.93 ± 7.46	50.67 ± 4.38	70.00 ± 8.73	65.89 ± 16.73	66.78 ± 15.52
ProximalPhalanxOutlineAgeGroup	**68.10 ± 13.07**	64.29 ± 28.01	64.78 ± 22.07	58.44 ± 28.63	68.00 ± 16.32	53.17 ± 31.21	56.26 ± 31.37	56.91 ± 37.90	59.22 ± 23.54	51.38 ± 35.67	33.95 ± 22.63	66.24 ± 17.89
ProximalPhalanxOutlineCorrect	**65.43 ± 5.55**	60.07 ± 16.64	63.85 ± 6.55	63.37 ± 8.02	62.96 ± 6.50	54.75 ± 14.29	54.30 ± 16.96	60.94 ± 6.65	59.86 ± 17.09	55.21 ± 16.86	44.88 ± 18.17	59.93 ± 12.82
ProximalPhalanxTW	50.83 ± 4.70	56.00 ± 9.92	**67.12 ± 8.12**	49.76 ± 11.43	53.37 ± 7.43	58.70 ± 5.35	53.98 ± 16.76	42.28 ± 24.98	56.00 ± 8.06	48.29 ± 19.01	16.98 ± 12.37	57.76 ± 7.59
RefrigerationDevices	**41.39 ± 9.49**	40.37 ± 10.57	41.23 ± 9.97	33.92 ± 0.58	35.31 ± 3.72	36.71 ± 1.61	33.87 ± 4.03	35.64 ± 1.78	33.81 ± 1.50	33.60 ± 3.03	33.07 ± 2.39	33.39 ± 7.13
Rock	32.00 ± 5.48	38.40 ± 8.65	37.80 ± 13.31	67.60 ± 10.53	44.80 ± 18.25	58.67 ± 13.61	**70.00 ± 11.14**	66.67 ± 8.08	47.60 ± 12.28	60.00 ± 14.00	44.40 ± 14.24	64.00 ± 7.07
ScreenType	36.91 ± 4.95	**40.21 ± 6.72**	38.40 ± 6.31	35.25 ± 3.07	36.75 ± 2.52	39.56 ± 4.47	33.78 ± 3.74	38.67 ± 2.63	33.71 ± 2.21	39.11 ± 4.89	33.81 ± 3.79	35.63 ± 4.40
SemgHandGenderCh2	55.13 ± 5.29	58.23 ± 11.97	54.27 ± 6.20	53.27 ± 13.91	**62.87 ± 8.65**	49.22 ± 10.20	55.56 ± 1.55	62.44 ± 3.53	43.87 ± 10.45	56.22 ± 13.04	53.70 ± 4.90	59.73 ± 10.87
SemgHandMovementCh2	33.51 ± 7.12	31.29 ± 5.18	**36.00 ± 6.56**	29.78 ± 3.16	23.56 ± 5.26	24.44 ± 2.52	18.89 ± 0.59	24.44 ± 1.39	18.71 ± 1.13	24.07 ± 5.35	27.91 ± 3.59	31.16 ± 4.70

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Table 8: Per-dataset 2-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
SemgHandSubjectCh2	32.00 ± 3.88	30.71 ± 4.13	33.56 ± 5.04	**45.91 ± 5.66**	40.67 ± 4.39	29.56 ± 4.02	23.56 ± 2.47	32.52 ± 7.62	24.09 ± 3.07	30.81 ± 0.84	29.47 ± 4.28	$\underline{43.20 \pm 6.72}$
ShakeGestureWiimoteZ	76.00 ± 6.63	$\underline{76.80 \pm 7.43}$	**81.60 ± 0.89**	47.20 ± 7.95	58.40 ± 5.18	54.00 ± 6.00	26.00 ± 4.00	54.67 ± 4.62	32.40 ± 2.61	56.00 ± 5.29	58.40 ± 5.18	**81.60 ± 2.61**
ShapeletSim	78.89 ± 14.16	**98.33 ± 2.08**	$\underline{92.11 \pm 5.26}$	50.11 ± 2.87	80.22 ± 16.23	47.59 ± 3.16	50.74 ± 3.06	48.15 ± 1.79	49.22 ± 2.93	46.48 ± 3.06	49.67 ± 1.50	67.44 ± 10.92
ShapesAll	36.80 ± 3.60	28.17 ± 3.75	$\underline{36.80 \pm 3.60}$	50.73 ± 1.71	32.23 ± 5.06	$\underline{55.78 \pm 1.50}$	6.11 ± 0.63	53.17 ± 1.00	18.33 ± 5.95	53.33 ± 2.62	27.63 ± 3.10	**63.03 ± 2.22**
SmallKitchenAppliances	48.85 ± 7.91	$\underline{50.40 \pm 8.87}$	50.19 ± 7.71	41.49 ± 3.33	47.15 ± 6.04	$\underline{49.60 \pm 6.54}$	35.20 ± 2.93	40.09 ± 2.14	42.03 ± 4.49	44.09 ± 8.36	34.61 ± 4.49	**55.68 ± 8.74**
SmoothSubspace	78.93 ± 10.17	$\underline{77.87 \pm 5.55}$	80.13 ± 5.80	33.33 ± 0.00	51.20 ± 5.65	58.22 ± 4.73	68.89 ± 7.31	59.11 ± 4.23	70.67 ± 4.88	$\underline{81.33 \pm 6.36}$	48.27 ± 9.05	**85.07 ± 5.20**
SonyAIBORobotSurface1	$\underline{90.42 \pm 4.34}$	**94.08 ± 1.77**	89.38 ± 5.35	64.83 ± 12.29	70.05 ± 3.77	62.90 ± 13.91	61.90 ± 7.89	66.61 ± 10.89	59.10 ± 5.18	$\underline{62.73 \pm 9.45}$	52.65 ± 8.42	71.85 ± 8.28
SonyAIBORobotSurface2	**88.00 ± 2.82**	$\underline{84.87 \pm 6.58}$	83.67 ± 9.41	74.59 ± 3.28	78.70 ± 2.68	69.60 ± 7.04	77.26 ± 1.79	78.49 ± 0.86	78.30 ± 1.49	79.29 ± 1.17	56.81 ± 17.39	68.21 ± 5.28
StarLightCurves	**76.49 ± 11.97**	$\underline{66.10 \pm 9.91}$	66.22 ± 10.70	67.92 ± 13.52	71.57 ± 12.97	63.90 ± 14.70	35.37 ± 11.93	65.31 ± 14.87	34.69 ± 14.98	63.78 ± 14.15	68.45 ± 12.08	$\underline{74.47 \pm 15.08}$
Strawberry	$\underline{62.81 \pm 6.83}$	57.14 ± 6.89	60.22 ± 9.39	51.84 ± 8.28	56.59 ± 2.41	53.33 ± 7.10	**66.58 ± 2.89**	52.97 ± 8.18	59.62 ± 11.05	54.59 ± 4.47	51.03 ± 11.13	$\underline{59.41 \pm 6.38}$
SwedishLeaf	**72.45 ± 3.37**	54.30 ± 6.89	**72.45 ± 3.37**	34.02 ± 3.00	43.26 ± 7.26	48.91 ± 2.97	33.28 ± 3.94	46.24 ± 1.25	38.85 ± 4.16	47.52 ± 2.78	7.94 ± 0.74	$\underline{64.99 \pm 3.83}$
Symbols	**90.49 ± 1.42**	75.88 ± 7.72	$\underline{88.48 \pm 4.41}$	81.23 ± 0.99	76.18 ± 6.87	81.37 ± 1.84	37.42 ± 3.12	82.04 ± 0.58	58.95 ± 6.86	80.00 ± 1.74	51.50 ± 7.93	$\underline{82.69 \pm 3.43}$
SyntheticControl	79.67 ± 4.05	$\underline{85.07 \pm 4.63}$	$\underline{82.60 \pm 6.01}$	41.40 ± 6.37	83.80 ± 5.24	55.00 ± 3.38	44.11 ± 1.39	59.89 ± 3.15	47.93 ± 4.58	62.89 ± 3.15	20.80 ± 4.95	**87.93 ± 3.95**
ToeSegmentation1	**68.42 ± 7.02**	$\underline{60.79 \pm 8.14}$	$\underline{61.58 \pm 7.54}$	53.95 ± 2.19	58.60 ± 10.97	51.90 ± 4.30	53.80 ± 2.21	53.95 ± 0.44	52.37 ± 4.44	51.61 ± 3.08	50.70 ± 2.53	60.26 ± 8.54
ToeSegmentation2	$\underline{76.15 \pm 9.44}$	**79.69 ± 9.24**	72.15 ± 7.68	48.92 ± 5.64	68.62 ± 12.68	66.41 ± 15.71	51.54 ± 5.38	63.08 ± 15.78	47.08 ± 8.33	63.59 ± 15.79	54.31 ± 14.96	$\underline{76.15 \pm 12.44}$
Trace	$\underline{76.00 \pm 3.00}$	76.40 ± 3.05	73.80 ± 2.95	52.40 ± 4.04	$\underline{76.60 \pm 4.39}$	47.33 ± 3.21	51.00 ± 4.00	49.00 ± 3.00	55.60 ± 4.62	49.00 ± 5.00	46.00 ± 6.78	**98.40 ± 3.05**
TwoLeadECG	**82.53 ± 6.47**	$\underline{79.19 \pm 7.45}$	70.63 ± 4.92	55.68 ± 3.84	$\underline{59.95 \pm 3.74}$	62.36 ± 5.86	56.60 ± 1.27	55.90 ± 1.72	55.49 ± 2.80	60.14 ± 7.03	48.48 ± 5.21	74.63 ± 8.09
TwoPatterns	52.71 ± 4.44	$\underline{71.75 \pm 5.17}$	65.30 ± 11.45	32.62 ± 1.84	44.61 ± 9.45	31.66 ± 1.59	24.88 ± 1.30	28.57 ± 1.14	25.14 ± 0.61	29.88 ± 2.34	27.30 ± 2.15	**78.23 ± 0.41**
UMD	54.86 ± 7.95	$\underline{71.67 \pm 9.36}$	68.75 ± 16.74	54.17 ± 4.55	62.50 ± 12.00	52.31 ± 6.22	47.69 ± 9.93	60.19 ± 1.45	53.06 ± 4.83	56.25 ± 6.05	47.50 ± 4.16	**78.86 ± 14.11**
UWaveGestureLibraryAll	32.50 ± 3.38	$\underline{28.30 \pm 5.01}$	31.51 ± 4.14	**69.44 ± 6.55**	43.65 ± 6.39	66.54 ± 6.96	26.72 ± 0.50	66.58 ± 7.48	30.82 ± 2.10	$\underline{66.61 \pm 7.59}$	54.22 ± 7.86	42.54 ± 0.28
UWaveGestureLibraryX	31.94 ± 2.72	28.97 ± 3.44	30.99 ± 3.77	50.42 ± 3.39	42.01 ± 2.19	**51.69 ± 1.72**	21.35 ± 1.20	$\underline{51.05 \pm 1.76}$	28.72 ± 2.91	$\underline{50.93 \pm 1.76}$	22.53 ± 5.80	42.54 ± 0.28
UWaveGestureLibraryY	28.00 ± 4.17	26.42 ± 2.97	27.96 ± 5.26	$\underline{43.36 \pm 3.42}$	39.43 ± 5.20	42.54 ± 7.86	19.88 ± 2.66	**43.61 ± 6.37**	28.02 ± 2.78	41.75 ± 7.40	28.22 ± 7.42	42.54 ± 0.28
UWaveGestureLibraryZ	31.19 ± 2.60	30.19 ± 2.10	28.42 ± 4.12	$\underline{44.60 \pm 3.60}$	41.59 ± 3.78	42.17 ± 5.87	23.93 ± 2.40	43.23 ± 7.30	29.59 ± 1.71	40.53 ± 6.44	23.38 ± 7.02	**59.88 ± 2.03**
Wafer	47.08 ± 18.17	57.95 ± 22.06	52.16 ± 16.40	$\underline{56.77 \pm 13.38}$	49.59 ± 17.01	$\underline{67.22 \pm 1.00}$	55.02 ± 6.93	66.32 ± 1.28	56.21 ± 9.73	**67.64 ± 1.29**	54.99 ± 20.43	55.01 ± 15.62
Wine	**60.74 ± 8.43**	50.74 ± 7.92	$\underline{60.48 \pm 12.52}$	53.70 ± 8.88	47.41 ± 11.54	$\underline{53.70 \pm 3.21}$	53.09 ± 5.35	50.00 ± 0.00	50.74 ± 1.66	54.94 ± 26.28	50.74 ± 1.66	48.89 ± 5.34
WordSynonyms	20.63 ± 4.03	16.05 ± 1.67	$\underline{20.63 \pm 4.03}$	21.35 ± 4.96	20.06 ± 3.25	29.05 ± 0.77	6.11 ± 1.90	$\underline{29.73 \pm 1.73}$	13.92 ± 2.48	29.57 ± 1.41	6.43 ± 4.20	**34.92 ± 2.64**
Worms	**54.03 ± 11.78**	44.94 ± 11.42	$\underline{52.73 \pm 7.32}$	22.60 ± 2.99	28.57 ± 6.43	19.48 ± 1.30	17.32 ± 5.41	$\underline{17.75 \pm 5.25}$	15.58 ± 4.00	20.35 ± 3.75	18.70 ± 5.00	38.70 ± 4.63
WormsTwoClass	$\underline{55.32 \pm 9.03}$	52.47 ± 8.50	**55.84 ± 3.31**	48.57 ± 2.69	49.09 ± 4.63	49.78 ± 2.70	51.52 ± 8.84	47.19 ± 3.97	47.27 ± 4.37	46.32 ± 1.50	48.83 ± 7.82	54.03 ± 7.72
Yoga	**52.99 ± 4.56**	50.68 ± 2.33	50.48 ± 3.16	50.86 ± 0.91	51.49 ± 3.75	50.12 ± 2.91	51.48 ± 2.93	50.70 ± 2.90	51.46 ± 3.58	50.58 ± 2.12	$\underline{51.69 \pm 4.15}$	50.64 ± 4.90
Avg. Acc.	53.98	50.79	$\underline{54.95}$	48.01	48.43	49.95	39.45	49.45	41.71	51.14	38.30	**55.55**
Avg. Rank	4.84	5.63	$\underline{4.83}$	6.78	6.42	6.13	9.36	6.29	8.63	5.64	9.17	**4.28**
#Best	**33**	13	19	10	2	9	4	6	3	7	2	$\underline{32}$

Table 9: Per-dataset 5-shot classification results on 128 shared UCR datasets. Each value is reported as mean ± standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
ACSF1	**76.00 ± 2.92**	64.60 ± 5.18	72.80 ± 1.64	36.20 ± 3.49	50.40 ± 2.61	60.00 ± 2.00	40.00 ± 1.00	62.67 ± 2.52	46.80 ± 3.70	58.67 ± 1.53	30.60 ± 4.83	68.33 ± 2.89
Adiac	56.21 ± 4.18	42.81 ± 4.46	56.21 ± 4.18	35.29 ± 1.32	34.58 ± 2.33	43.82 ± 1.54	46.97 ± 3.07	38.62 ± 1.68	64.50 ± 5.47	43.99 ± 3.86	7.11 ± 1.15	**69.51 ± 2.42**
AllGestureWiimoteX	**40.46 ± 4.50**	34.14 ± 3.58	38.71 ± 5.08	23.63 ± 2.19	35.40 ± 4.31	32.24 ± 3.43	13.57 ± 1.57	29.95 ± 4.86	18.17 ± 3.28	29.38 ± 3.89	35.40 ± 4.31	39.40 ± 2.86
AllGestureWiimoteY	43.20 ± 1.70	40.60 ± 1.51	36.20 ± 3.42	27.89 ± 2.07	40.97 ± 3.06	36.76 ± 2.07	15.38 ± 0.81	34.43 ± 2.02	19.49 ± 2.37	35.43 ± 1.55	40.97 ± 3.06	**48.11 ± 2.89**
AllGestureWiimoteZ	**41.46 ± 4.33**	35.71 ± 5.12	36.46 ± 5.30	20.20 ± 1.22	27.11 ± 2.51	26.57 ± 0.25	13.86 ± 1.71	26.67 ± 2.65	17.37 ± 2.23	26.86 ± 0.99	27.11 ± 2.51	36.14 ± 0.97
ArrowHead	66.40 ± 5.44	61.03 ± 5.46	64.23 ± 7.62	59.43 ± 5.27	58.86 ± 12.66	60.19 ± 7.46	42.67 ± 0.87	60.76 ± 1.19	48.11 ± 2.87	63.43 ± 8.42	54.06 ± 9.79	**73.60 ± 3.71**
BME	55.60 ± 7.80	56.93 ± 3.90	68.00 ± 9.90	58.53 ± 7.20	89.60 ± 10.65	58.89 ± 5.55	73.33 ± 8.11	76.89 ± 7.34	80.00 ± 6.70	73.33 ± 5.93	53.07 ± 5.63	**97.73 ± 1.53**
Beef	54.67 ± 4.47	47.33 ± 2.79	55.33 ± 2.98	52.67 ± 7.60	43.33 ± 4.71	64.44 ± 3.85	50.00 ± 3.33	52.22 ± 8.39	62.00 ± 8.03	**84.44 ± 1.92**	35.33 ± 4.47	51.33 ± 6.91
BeetleFly	72.00 ± 7.58	60.00 ± 11.73	79.00 ± 10.84	71.00 ± 14.32	57.00 ± 8.37	70.00 ± 8.66	51.67 ± 2.89	66.67 ± 16.07	49.00 ± 6.52	70.00 ± 10.00	59.00 ± 18.51	**91.67 ± 2.89**
BirdChicken	69.00 ± 12.94	77.00 ± 9.75	**90.00 ± 3.54**	57.00 ± 5.70	61.00 ± 16.73	63.33 ± 7.64	36.67 ± 2.89	65.00 ± 10.00	54.00 ± 8.94	68.33 ± 15.28	54.00 ± 4.18	85.00 ± 11.73
CBF	95.87 ± 2.59	95.58 ± 3.67	**95.89 ± 3.27**	61.47 ± 7.78	95.67 ± 4.48	74.52 ± 5.66	39.04 ± 0.80	80.11 ± 4.97	42.29 ± 1.09	80.59 ± 4.09	41.20 ± 4.62	93.76 ± 3.90
Car	59.33 ± 9.02	51.00 ± 7.96	65.67 ± 5.73	52.33 ± 6.41	45.67 ± 2.79	68.89 ± 6.31	31.67 ± 5.00	48.89 ± 4.19	41.00 ± 12.56	57.78 ± 5.85	36.00 ± 4.50	**69.67 ± 3.98**
Chinatown	96.15 ± 2.16	96.97 ± 0.49	**97.43 ± 0.38**	50.96 ± 33.10	96.03 ± 2.24	84.94 ± 15.12	91.06 ± 1.94	93.00 ± 5.85	89.56 ± 2.00	87.95 ± 8.75	73.00 ± 0.76	93.46 ± 10.56
ChlorineConcentration	35.97 ± 2.09	35.63 ± 3.63	36.15 ± 2.53	35.57 ± 3.02	35.80 ± 3.16	35.02 ± 2.94	36.15 ± 1.24	35.49 ± 4.56	37.02 ± 1.05	35.96 ± 1.50	38.60 ± 5.65	**39.58 ± 3.20**
CinCECGTorso	49.41 ± 5.43	57.77 ± 5.88	56.17 ± 3.91	43.81 ± 1.84	69.26 ± 4.67	63.41 ± 2.73	37.80 ± 1.42	64.95 ± 2.40	64.83 ± 11.83	68.41 ± 5.18	54.58 ± 3.46	**82.56 ± 7.22**
Coffee	86.43 ± 14.81	97.86 ± 1.96	**100.00 ± 0.00**	92.86 ± 3.57	97.86 ± 1.96	86.90 ± 2.06	76.19 ± 2.06	82.14 ± 9.45	92.86 ± 10.41	85.71 ± 0.00	50.00 ± 6.19	94.29 ± 1.96
Computers	58.80 ± 7.81	61.60 ± 5.76	60.96 ± 6.58	46.64 ± 4.37	54.48 ± 7.54	49.87 ± 5.33	50.53 ± 1.29	48.00 ± 8.43	49.04 ± 4.09	48.67 ± 4.31	46.56 ± 5.23	**62.67 ± 4.74**
CricketX	45.90 ± 4.41	43.08 ± 3.97	43.08 ± 2.15	25.23 ± 1.80	36.10 ± 5.19	34.87 ± 4.89	11.79 ± 1.43	30.77 ± 0.51	16.92 ± 1.32	33.68 ± 3.44	11.33 ± 1.41	**49.06 ± 3.21**
CricketY	42.67 ± 4.49	43.08 ± 3.40	43.28 ± 6.01	30.41 ± 3.78	36.92 ± 6.92	37.86 ± 2.42	9.06 ± 1.21	36.58 ± 2.06	15.23 ± 1.85	37.26 ± 2.44	12.05 ± 2.18	**49.66 ± 1.41**
CricketZ	51.59 ± 3.02	49.64 ± 3.11	47.79 ± 6.26	26.31 ± 2.67	36.31 ± 4.35	37.18 ± 1.18	10.77 ± 0.51	34.87 ± 1.43	15.44 ± 1.54	37.18 ± 0.77	10.92 ± 1.59	**53.68 ± 2.06**
Crop	49.93 ± 1.45	46.93 ± 2.27	49.93 ± 1.45	31.91 ± 2.21	38.13 ± 2.76	46.92 ± 2.69	44.47 ± 0.97	47.82 ± 2.55	47.76 ± 1.51	48.69 ± 2.69	8.15 ± 3.22	**52.18 ± 2.24**
DiatomSizeReduction	85.75 ± 6.98	70.26 ± 16.84	**95.95 ± 1.26**	61.57 ± 2.74	65.49 ± 10.17	87.69 ± 0.50	68.63 ± 2.55	66.34 ± 8.52	73.53 ± 9.37	87.25 ± 0.00	41.83 ± 13.83	93.66 ± 1.97
DistalPhalanxOutlineAgeGroup	65.76 ± 4.54	68.63 ± 3.96	67.19 ± 5.75	58.99 ± 14.81	63.31 ± 4.46	64.75 ± 3.74	67.63 ± 2.16	69.30 ± 3.24	68.20 ± 2.35	68.59 ± 2.31	53.67 ± 12.57	**69.35 ± 1.20**
DistalPhalanxOutlineCorrect	66.81 ± 3.92	61.96 ± 7.85	**68.62 ± 6.57**	56.23 ± 10.81	60.07 ± 6.07	62.44 ± 3.65	62.68 ± 3.32	58.94 ± 3.37	63.41 ± 7.40	59.30 ± 3.08	55.87 ± 8.58	63.77 ± 11.63
DistalPhalanxTW	58.99 ± 2.69	57.12 ± 1.40	58.56 ± 3.40	49.64 ± 7.01	**60.58 ± 2.61**	57.31 ± 0.42	51.32 ± 3.62	59.47 ± 4.40	54.10 ± 4.24	56.59 ± 5.05	31.94 ± 15.91	56.40 ± 4.87
DodgerLoopDay	37.75 ± 3.35	40.50 ± 4.11	37.00 ± 3.26	46.75 ± 4.01	44.25 ± 2.09	**52.50 ± 5.45**	22.08 ± 0.72	45.42 ± 2.60	24.25 ± 3.49	50.83 ± 4.73	44.25 ± 2.09	45.75 ± 3.26
DodgerLoopGame	66.52 ± 4.02	82.32 ± 3.38	72.17 ± 3.64	76.09 ± 5.15	**84.49 ± 1.89**	75.36 ± 5.02	50.00 ± 2.61	78.99 ± 7.63	53.33 ± 2.09	77.29 ± 4.82	**84.49 ± 1.89**	72.61 ± 10.25
DodgerLoopWeekend	95.51 ± 1.65	92.46 ± 3.46	94.78 ± 1.88	97.39 ± 0.40	97.25 ± 0.61	96.62 ± 0.84	66.43 ± 4.25	**97.83 ± 0.00**	72.90 ± 1.82	96.86 ± 0.42	97.25 ± 0.61	93.62 ± 2.37
ECG200	76.40 ± 3.58	77.80 ± 6.26	76.20 ± 4.60	75.00 ± 1.58	62.80 ± 16.36	77.67 ± 3.06	76.67 ± 3.79	78.00 ± 2.00	77.60 ± 3.21	**79.33 ± 4.93**	69.60 ± 4.51	67.40 ± 8.91
ECG5000	78.85 ± 9.03	75.51 ± 9.29	75.56 ± 6.88	**84.88 ± 3.51**	71.69 ± 7.99	82.01 ± 5.12	76.33 ± 7.90	83.48 ± 5.73	74.58 ± 8.80	82.23 ± 5.84	18.64 ± 16.84	73.89 ± 8.21
ECGFiveDays	71.54 ± 13.30	70.27 ± 8.41	70.22 ± 9.52	69.38 ± 6.83	77.54 ± 3.65	72.94 ± 9.66	**87.65 ± 8.42**	75.22 ± 2.60	83.30 ± 11.51	75.61 ± 4.26	60.49 ± 6.59	70.96 ± 3.91
EOGHorizontalSignal	24.97 ± 1.88	32.71 ± 1.62	27.35 ± 3.43	29.56 ± 3.95	37.73 ± 6.23	36.92 ± 1.62	10.96 ± 1.15	38.40 ± 2.19	9.94 ± 1.90	35.17 ± 4.76	31.99 ± 1.89	**46.35 ± 2.27**
EOGVerticalSignal	18.18 ± 3.62	22.10 ± 3.22	18.78 ± 4.22	29.01 ± 2.45	**38.23 ± 4.43**	31.77 ± 5.00	8.47 ± 1.80	31.95 ± 3.43	10.39 ± 1.59	31.22 ± 4.29	22.82 ± 2.72	36.74 ± 2.24
Earthquakes	50.94 ± 12.41	50.22 ± 8.85	52.23 ± 11.61	**56.83 ± 2.92**	52.09 ± 5.05	56.35 ± 11.87	51.32 ± 9.11	56.12 ± 3.14	53.38 ± 4.56	54.68 ± 7.61	52.81 ± 9.99	54.68 ± 9.31
ElectricDevices	36.69 ± 5.52	36.63 ± 6.97	36.08 ± 5.74	23.65 ± 1.86	38.77 ± 8.33	38.73 ± 7.16	22.89 ± 3.05	25.09 ± 5.42	31.29 ± 6.08	38.61 ± 5.70	11.47 ± 4.02	**47.06 ± 4.18**
EthanolLevel	27.52 ± 1.92	26.36 ± 0.86	26.60 ± 1.66	27.88 ± 1.25	24.88 ± 0.18	28.40 ± 1.73	28.60 ± 2.78	26.53 ± 0.12	25.40 ± 0.75	**29.53 ± 0.64**	26.32 ± 1.07	25.92 ± 1.40
FaceAll	67.63 ± 4.66	**67.75 ± 7.28**	67.63 ± 4.66	53.96 ± 2.14	41.22 ± 3.29	55.76 ± 2.36	53.79 ± 5.19	55.44 ± 2.51	51.51 ± 5.08	54.30 ± 2.41	17.64 ± 1.73	54.54 ± 1.26
FaceFour	88.18 ± 2.85	61.82 ± 13.74	**89.77 ± 4.62**	78.86 ± 3.17	57.50 ± 8.82	86.36 ± 1.14	64.77 ± 2.27	87.88 ± 1.74	71.82 ± 1.24	85.61 ± 1.74	41.14 ± 6.88	75.68 ± 4.37
FacesUCR	**78.67 ± 2.69**	73.00 ± 4.70	76.25 ± 3.16	50.94 ± 2.11	43.00 ± 7.53	68.33 ± 3.31	58.46 ± 2.45	68.99 ± 1.59	60.62 ± 1.49	67.61 ± 2.78	27.20 ± 3.34	68.58 ± 2.57
FiftyWords	30.07 ± 3.14	27.91 ± 3.50	30.07 ± 3.14	47.03 ± 2.26	37.71 ± 3.50	52.23 ± 1.46	9.45 ± 3.91	56.19 ± 3.63	25.10 ± 2.19	49.38 ± 1.41	40.84 ± 2.35	**61.58 ± 2.55**
Fish	61.26 ± 15.63	48.69 ± 6.91	67.66 ± 6.55	56.80 ± 6.22	42.97 ± 5.32	**68.76 ± 3.49**	25.52 ± 2.93	56.57 ± 5.51	40.80 ± 3.78	67.05 ± 1.44	26.40 ± 8.24	64.57 ± 5.92
FordA	71.41 ± 10.22	**71.94 ± 9.09**	70.55 ± 10.29	50.33 ± 1.09	63.62 ± 6.82	50.28 ± 2.11	51.67 ± 1.61	49.67 ± 1.72	51.11 ± 2.04	49.82 ± 1.42	50.91 ± 1.25	54.00 ± 2.22
FordB	**61.26 ± 2.75**	58.96 ± 1.47	59.04 ± 3.16	50.00 ± 1.41	55.88 ± 4.39	50.86 ± 0.77	49.51 ± 0.81	48.81 ± 1.26	49.56 ± 0.80	50.21 ± 1.36	50.27 ± 2.02	52.32 ± 2.78
FreezerRegularTrain	71.93 ± 7.16	**86.24 ± 2.60**	79.17 ± 4.44	76.42 ± 0.69	77.88 ± 9.36	76.43 ± 0.19	57.27 ± 4.31	76.76 ± 0.69	62.60 ± 6.21	71.04 ± 6.76	59.41 ± 11.73	85.11 ± 3.83
FreezerSmallTrain	**77.10 ± 3.40**	75.64 ± 1.31	75.24 ± 0.61	75.24 ± 1.24	71.21 ± 6.86	75.59 ± 1.06	60.63 ± 1.62	76.02 ± 0.68	66.32 ± 5.59	73.06 ± 2.85	63.24 ± 10.73	75.05 ± 3.10
Fungi	96.34 ± 4.07	34.19 ± 4.54	66.45 ± 5.80	74.09 ± 4.22	74.73 ± 9.52	**97.13 ± 0.31**	75.63 ± 1.35	90.14 ± 2.24	75.27 ± 0.54	96.06 ± 0.82	10.75 ± 6.51	81.18 ± 3.16
GestureMidAirD1	37.54 ± 4.63	52.92 ± 3.41	37.54 ± 4.63	46.00 ± 0.64	43.38 ± 3.51	53.85 ± 0.77	21.03 ± 3.11	52.05 ± 3.11	40.46 ± 4.85	52.82 ± 1.94	43.38 ± 3.51	**60.31 ± 2.35**
GestureMidAirD2	28.15 ± 3.51	42.15 ± 7.00	28.15 ± 3.51	48.31 ± 1.58	46.46 ± 3.63	50.00 ± 1.54	12.82 ± 3.55	49.49 ± 1.60	33.08 ± 9.04	50.00 ± 0.77	46.46 ± 3.63	**54.62 ± 1.96**
GestureMidAirD3	14.00 ± 2.69	19.85 ± 2.33	14.00 ± 2.69	30.92 ± 2.20	26.92 ± 3.31	25.13 ± 1.18	14.10 ± 3.11	31.79 ± 1.60	26.31 ± 4.82	27.18 ± 1.78	26.92 ± 3.31	**32.15 ± 3.33**

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Table 9: Per-dataset 5-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
GesturePebbleZ1	71.74 ± 3.41	64.53 ± 3.05	73.60 ± 2.12	65.35 ± 3.15	63.95 ± 8.94	65.89 ± 3.50	23.45 ± 1.21	73.64 ± 2.20	28.95 ± 4.06	**76.16 ± 6.07**	63.95 ± 8.94	63.84 ± 8.99
GesturePebbleZ2	67.34 ± 3.97	52.53 ± 17.40	**73.04 ± 4.07**	58.35 ± 6.18	48.10 ± 7.12	62.03 ± 7.30	19.41 ± 4.45	64.98 ± 9.05	24.81 ± 3.02	60.34 ± 8.06	48.10 ± 7.12	60.89 ± 8.06
GunPoint	82.53 ± 12.46	77.33 ± 12.19	79.87 ± 13.86	59.60 ± 4.87	73.07 ± 2.03	66.00 ± 6.36	55.11 ± 6.19	70.22 ± 2.52	65.87 ± 7.58	66.00 ± 2.91	62.40 ± 5.79	**90.13 ± 10.49**
GunPointAgeSpan	75.70 ± 7.16	69.81 ± 6.17	76.46 ± 8.05	75.51 ± 2.81	73.67 ± 12.74	76.79 ± 5.43	49.26 ± 6.34	75.74 ± 4.37	60.38 ± 3.16	69.20 ± 7.84	65.70 ± 10.14	**81.52 ± 9.45**
GunPointMaleVersusFemale	91.20 ± 8.64	**96.65 ± 4.17**	90.32 ± 9.07	61.14 ± 7.95	87.53 ± 12.91	74.37 ± 13.69	55.17 ± 0.48	75.00 ± 13.22	61.20 ± 4.73	77.95 ± 14.78	63.92 ± 13.57	94.62 ± 6.24
GunPointOldVersusYoung	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	75.87 ± 2.49	70.29 ± 8.33	70.79 ± 4.48	56.93 ± 3.97	73.23 ± 4.31	58.98 ± 5.43	70.48 ± 8.59	92.38 ± 8.67	80.44 ± 8.63
Ham	60.95 ± 6.21	60.00 ± 6.77	62.10 ± 4.59	62.86 ± 11.45	58.86 ± 7.48	60.00 ± 11.70	63.49 ± 0.55	56.51 ± 8.59	63.62 ± 3.26	**66.35 ± 16.28**	56.76 ± 9.56	60.57 ± 9.91
HandOutlines	46.86 ± 11.04	52.86 ± 13.07	67.24 ± 3.85	80.76 ± 4.40	74.49 ± 7.28	**83.06 ± 3.92**	47.84 ± 8.25	81.62 ± 4.95	35.95 ± 0.00	82.34 ± 4.91	75.57 ± 5.75	68.54 ± 6.21
Haptics	32.14 ± 2.79	28.25 ± 2.44	31.49 ± 1.92	35.39 ± 2.49	28.18 ± 3.44	36.58 ± 0.19	21.32 ± 3.65	**37.12 ± 1.14**	24.68 ± 3.46	**37.12 ± 0.94**	31.82 ± 7.39	26.95 ± 4.88
Herring	50.94 ± 5.70	55.00 ± 8.93	52.50 ± 7.46	51.88 ± 6.76	50.94 ± 6.68	48.96 ± 5.02	**57.29 ± 4.51**	51.56 ± 7.16	46.88 ± 13.58	49.48 ± 2.39	49.06 ± 7.13	45.00 ± 4.87
HouseTwenty	83.70 ± 6.31	81.18 ± 9.08	81.85 ± 9.95	58.99 ± 5.13	76.81 ± 7.55	73.67 ± 7.34	54.90 ± 2.57	66.95 ± 4.78	59.66 ± 6.85	72.83 ± 4.23	63.03 ± 4.28	**84.37 ± 7.51**
InlineSkate	20.15 ± 4.70	22.95 ± 2.93	19.75 ± 2.66	21.05 ± 2.12	18.98 ± 2.35	24.85 ± 1.07	16.00 ± 1.31	24.06 ± 0.93	16.51 ± 0.15	24.06 ± 0.90	19.20 ± 1.41	**28.07 ± 1.19**
InsectEPGRegularTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	56.47 ± 1.83	78.96 ± 9.81	59.57 ± 2.67	38.96 ± 2.13	57.30 ± 1.41	53.17 ± 6.32	61.31 ± 3.42	89.88 ± 9.24	80.24 ± 3.80
InsectEPGSmallTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	53.98 ± 1.90	77.03 ± 8.38	59.44 ± 2.44	52.61 ± 2.78	60.51 ± 1.98	55.66 ± 1.65	62.12 ± 1.62	83.13 ± 0.00	83.05 ± 2.80
InsectWingbeatSound	25.39 ± 1.91	24.63 ± 1.27	26.10 ± 5.93	53.74 ± 0.54	42.18 ± 1.58	56.25 ± 3.65	25.84 ± 1.75	**58.96 ± 0.98**	38.75 ± 2.42	57.51 ± 1.69	20.02 ± 5.68	41.44 ± 2.25
ItalyPowerDemand	87.66 ± 4.88	89.29 ± 4.25	88.36 ± 4.24	62.66 ± 11.58	93.39 ± 3.05	58.50 ± 7.42	89.57 ± 5.31	93.78 ± 2.95	**95.72 ± 1.14**	85.46 ± 2.44	56.64 ± 14.68	84.26 ± 5.94
LargeKitchenAppliances	60.00 ± 4.27	58.51 ± 9.39	**62.67 ± 10.64**	37.55 ± 2.23	60.48 ± 4.88	44.09 ± 3.07	36.98 ± 2.46	40.44 ± 3.60	37.71 ± 3.74	44.09 ± 6.61	37.17 ± 1.99	50.13 ± 3.07
Lightning2	64.92 ± 6.82	62.30 ± 7.42	67.87 ± 5.51	60.33 ± 3.91	64.92 ± 4.86	**69.95 ± 2.50**	50.82 ± 5.68	68.85 ± 5.68	61.64 ± 7.91	68.31 ± 3.41	59.02 ± 6.86	69.51 ± 10.47
Lightning7	58.90 ± 7.12	54.25 ± 7.09	60.55 ± 3.80	51.23 ± 2.84	50.96 ± 8.07	67.58 ± 3.45	18.26 ± 3.95	61.64 ± 3.62	30.41 ± 0.61	**70.32 ± 2.85**	22.19 ± 2.63	66.30 ± 3.83
Mallat	71.03 ± 13.07	70.33 ± 5.20	93.72 ± 2.08	88.67 ± 2.40	77.56 ± 9.50	**94.66 ± 0.33**	65.67 ± 5.01	87.51 ± 2.78	79.72 ± 5.50	90.06 ± 1.99	56.21 ± 5.78	90.88 ± 1.15
Meat	64.33 ± 14.22	50.00 ± 9.93	81.00 ± 8.47	85.33 ± 10.70	49.33 ± 15.48	76.67 ± 10.41	47.78 ± 15.03	37.22 ± 5.36	50.00 ± 11.61	**88.89 ± 2.55**	34.00 ± 1.49	64.00 ± 11.82
MedicalImages	**45.61 ± 3.94**	43.95 ± 4.74	42.63 ± 4.22	27.26 ± 7.75	32.26 ± 2.93	37.41 ± 0.59	36.84 ± 3.23	38.03 ± 3.53	40.16 ± 6.60	38.20 ± 1.38	5.24 ± 3.29	42.03 ± 2.75
MelbournePedestrian	76.79 ± 2.31	74.96 ± 2.28	**77.84 ± 3.11**	11.65 ± 1.29	58.25 ± 3.82	62.66 ± 1.83	70.73 ± 4.35	71.76 ± 3.72	70.23 ± 2.95	67.90 ± 5.27	58.25 ± 3.82	65.81 ± 3.77
MiddlePhalanxOutlineAgeGroup	47.01 ± 8.99	54.03 ± 8.97	50.78 ± 7.01	52.08 ± 8.63	49.09 ± 6.85	50.22 ± 4.61	51.73 ± 7.75	53.90 ± 7.87	**55.06 ± 4.76**	54.11 ± 9.46	31.82 ± 16.75	49.74 ± 4.72
MiddlePhalanxOutlineCorrect	53.20 ± 7.83	52.71 ± 15.62	58.14 ± 12.45	46.94 ± 12.76	48.93 ± 10.44	49.60 ± 3.67	**58.65 ± 6.95**	55.44 ± 15.07	51.68 ± 10.94	49.60 ± 9.23	50.93 ± 8.48	48.45 ± 7.93
MiddlePhalanxTW	**45.84 ± 9.64**	**45.84 ± 3.45**	45.71 ± 4.74	**45.84 ± 4.10**	43.12 ± 4.67	44.37 ± 0.37	40.04 ± 2.62	43.72 ± 4.56	44.16 ± 4.95	44.59 ± 2.28	22.86 ± 6.13	43.64 ± 5.71
MixedShapesRegularTrain	63.93 ± 9.77	61.94 ± 4.99	68.21 ± 8.35	68.04 ± 1.34	68.63 ± 3.58	67.15 ± 2.55	20.78 ± 0.21	70.05 ± 0.70	30.67 ± 6.49	67.15 ± 1.60	64.11 ± 1.37	**76.68 ± 5.11**
MixedShapesSmallTrain	66.24 ± 12.23	57.21 ± 4.26	72.37 ± 7.91	66.37 ± 2.62	66.13 ± 3.66	68.30 ± 1.42	22.61 ± 0.88	67.60 ± 3.67	25.23 ± 4.04	66.78 ± 1.26	60.00 ± 5.54	**78.71 ± 1.70**
MoteStrain	84.74 ± 4.04	83.85 ± 4.63	85.10 ± 2.03	80.19 ± 8.46	77.96 ± 3.58	**85.73 ± 4.20**	72.68 ± 2.25	80.09 ± 3.68	77.11 ± 6.04	85.36 ± 1.87	60.30 ± 15.64	80.37 ± 4.50
NonInvasiveFetalECGThorax1	40.80 ± 1.72	31.63 ± 3.55	40.80 ± 1.72	**72.10 ± 1.96**	51.79 ± 9.05	70.09 ± 3.02	22.04 ± 0.40	70.72 ± 1.13	32.89 ± 2.08	69.19 ± 0.16	28.56 ± 4.30	67.42 ± 0.56
NonInvasiveFetalECGThorax2	41.15 ± 5.46	43.13 ± 8.61	41.15 ± 5.46	**78.55 ± 0.96**	60.71 ± 3.55	75.84 ± 1.88	37.61 ± 1.37	75.40 ± 1.50	50.66 ± 2.46	74.93 ± 2.49	37.30 ± 2.25	75.31 ± 0.82
OSULeaf	57.85 ± 5.97	50.74 ± 10.99	**72.23 ± 4.84**	32.40 ± 1.86	35.37 ± 3.13	38.84 ± 1.09	16.80 ± 1.04	32.64 ± 2.71	24.05 ± 2.60	36.91 ± 1.04	23.06 ± 4.07	49.92 ± 6.46
OliveOil	59.33 ± 13.21	28.67 ± 13.66	**74.00 ± 6.83**	31.33 ± 9.60	26.67 ± 10.00	42.22 ± 25.02	40.00 ± 10.00	32.22 ± 13.47	40.00 ± 21.73	32.22 ± 22.19	26.67 ± 10.00	28.67 ± 11.69
PLAID	24.02 ± 4.96	24.51 ± 3.44	22.87 ± 2.95	30.06 ± 4.34	47.93 ± 3.93	36.56 ± 2.69	26.88 ± 2.01	40.16 ± 2.24	27.26 ± 3.48	37.93 ± 4.09	47.93 ± 3.93	**52.96 ± 5.39**
PhalangesOutlinesCorrect	56.22 ± 7.89	55.22 ± 9.82	**56.85 ± 6.96**	52.75 ± 11.57	53.15 ± 7.53	51.83 ± 11.55	50.35 ± 9.47	51.20 ± 13.21	53.05 ± 9.33	50.85 ± 11.66	53.61 ± 9.97	53.61 ± 8.69
Phoneme	**19.05 ± 2.97**	18.15 ± 2.47	**19.05 ± 2.97**	6.61 ± 0.25	9.00 ± 1.08	10.58 ± 0.95	6.87 ± 0.83	9.00 ± 0.29	9.61 ± 1.88	9.30 ± 1.34	8.21 ± 0.75	15.63 ± 1.56
PickupGestureWiimoteZ	56.00 ± 10.20	66.80 ± 5.93	**73.60 ± 4.56**	52.40 ± 1.67	56.80 ± 9.55	57.33 ± 3.06	27.33 ± 1.15	58.00 ± 3.46	42.00 ± 2.00	58.67 ± 1.15	56.80 ± 9.55	68.00 ± 2.45
PigAirwayPressure	25.00 ± 2.61	13.37 ± 2.05	**48.17 ± 0.40**	5.10 ± 0.26	4.62 ± 0.55	4.81 ± 0.00	4.17 ± 2.22	6.73 ± 0.48	4.04 ± 0.87	5.93 ± 0.56	4.42 ± 0.86	9.81 ± 0.73
PigArtPressure	50.96 ± 5.81	20.77 ± 3.29	**68.17 ± 4.63**	8.46 ± 0.65	5.10 ± 1.82	8.97 ± 1.21	6.57 ± 2.00	8.97 ± 0.73	6.92 ± 1.65	9.29 ± 1.11	18.17 ± 2.24	33.08 ± 1.33
PigCVP	29.33 ± 2.90	18.75 ± 2.40	**62.98 ± 1.73**	6.83 ± 0.22	3.37 ± 1.32	7.53 ± 0.56	4.65 ± 1.11	7.53 ± 0.28	3.75 ± 1.24	4.81 ± 0.48	10.38 ± 1.21	19.52 ± 0.55
Plane	**100.00 ± 0.00**	98.10 ± 2.43	99.62 ± 0.52	88.95 ± 7.93	95.24 ± 3.16	97.78 ± 0.55	96.51 ± 1.98	95.87 ± 1.45	97.14 ± 1.51	95.87 ± 0.55	23.05 ± 11.32	99.24 ± 1.24
PowerCons	80.56 ± 3.36	**84.44 ± 5.02**	84.00 ± 6.30	70.11 ± 5.26	81.11 ± 10.74	75.37 ± 5.45	54.26 ± 3.90	80.93 ± 10.72	52.22 ± 4.39	78.52 ± 10.32	68.89 ± 15.97	76.11 ± 6.55
ProximalPhalanxOutlineAgeGroup	73.37 ± 8.19	81.76 ± 3.12	79.90 ± 3.95	65.56 ± 9.78	75.41 ± 5.61	79.02 ± 7.19	79.02 ± 8.46	**84.07 ± 1.71**	74.34 ± 7.73	80.00 ± 3.41	43.61 ± 12.69	76.88 ± 5.79
ProximalPhalanxOutlineCorrect	**69.35 ± 11.27**	69.00 ± 5.91	69.28 ± 9.35	64.47 ± 0.52	63.71 ± 8.48	65.18 ± 3.44	64.60 ± 1.57	63.69 ± 1.73	63.16 ± 5.61	66.67 ± 4.47	55.26 ± 17.89	65.84 ± 4.70
ProximalPhalanxTW	62.34 ± 7.16	66.44 ± 4.62	**69.37 ± 2.52**	53.37 ± 11.45	65.27 ± 5.74	58.05 ± 7.67	58.21 ± 7.39	55.45 ± 4.15	60.00 ± 4.28	55.28 ± 3.66	27.32 ± 18.90	66.24 ± 3.19
RefrigerationDevices	42.83 ± 1.65	43.20 ± 6.13	**44.00 ± 6.18**	33.33 ± 2.73	37.92 ± 6.42	38.93 ± 3.24	31.20 ± 3.23	32.98 ± 2.22	32.32 ± 2.29	38.04 ± 3.74	34.61 ± 1.83	40.96 ± 5.59
Rock	42.00 ± 5.29	36.80 ± 7.16	38.40 ± 4.34	80.40 ± 2.97	60.80 ± 5.93	80.67 ± 1.15	81.33 ± 3.06	**88.67 ± 2.31**	59.20 ± 3.63	85.33 ± 3.06	33.60 ± 5.90	76.00 ± 3.16
ScreenType	**42.56 ± 7.61**	41.33 ± 8.02	40.64 ± 8.17	35.41 ± 2.78	36.16 ± 4.77	35.56 ± 2.39	32.18 ± 2.24	36.09 ± 2.30	32.53 ± 1.41	35.02 ± 0.41	34.08 ± 2.12	34.51 ± 4.14
SemgHandGenderCh2	58.13 ± 2.49	66.47 ± 7.67	56.07 ± 2.82	65.13 ± 3.61	**67.13 ± 2.65**	50.28 ± 5.31	55.83 ± 13.67	55.06 ± 2.04	54.63 ± 4.30	47.94 ± 16.07	59.97 ± 7.08	66.80 ± 7.25
SemgHandMovementCh2	30.89 ± 3.99	33.91 ± 3.59	34.40 ± 6.06	32.84 ± 5.27	26.49 ± 1.89	27.70 ± 2.36	20.59 ± 1.85	29.48 ± 2.44	21.64 ± 2.79	23.41 ± 3.03	33.16 ± 2.38	**36.31 ± 5.92**

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Table 9: Per-dataset 5-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
SemgHandSubjectCh2	37.11 $\pm$ 4.58	44.09 $\pm$ 1.20	35.60 $\pm$ 3.13	50.76 $\pm$ 1.90	49.02 $\pm$ 0.83	46.67 $\pm$ 2.12	27.78 $\pm$ 0.59	51.19 $\pm$ 2.74	30.62 $\pm$ 1.93	41.48 $\pm$ 6.03	36.62 $\pm$ 4.30	**54.31 $\pm$ 3.37**
ShakeGestureWiimoteZ	80.40 $\pm$ 4.34	91.60 $\pm$ 2.61	89.20 $\pm$ 1.10	56.40 $\pm$ 1.67	72.80 $\pm$ 5.22	62.00 $\pm$ 0.00	35.33 $\pm$ 1.15	60.67 $\pm$ 3.06	44.00 $\pm$ 2.45	55.33 $\pm$ 2.31	72.80 $\pm$ 5.22	**92.00 $\pm$ 0.00**
ShapeletSim	82.78 $\pm$ 10.96	**99.67 $\pm$ 0.75**	92.56 $\pm$ 6.55	49.44 $\pm$ 2.69	88.22 $\pm$ 10.06	55.93 $\pm$ 2.31	49.81 $\pm$ 2.80	49.63 $\pm$ 4.21	50.67 $\pm$ 1.82	49.44 $\pm$ 3.33	49.67 $\pm$ 3.98	92.00 $\pm$ 3.88
ShapesAll	57.57 $\pm$ 3.68	48.33 $\pm$ 5.58	57.57 $\pm$ 3.68	58.13 $\pm$ 1.68	51.50 $\pm$ 3.69	64.67 $\pm$ 1.64	9.22 $\pm$ 0.79	65.83 $\pm$ 3.88	13.83 $\pm$ 3.47	59.22 $\pm$ 2.28	56.47 $\pm$ 1.46	**76.20 $\pm$ 1.83**
SmallKitchenAppliances	59.47 $\pm$ 8.51	57.39 $\pm$ 6.38	**62.35 $\pm$ 6.85**	39.04 $\pm$ 4.11	57.55 $\pm$ 2.52	57.51 $\pm$ 6.62	38.67 $\pm$ 2.54	39.56 $\pm$ 5.26	46.67 $\pm$ 3.53	56.62 $\pm$ 3.96	44.00 $\pm$ 7.15	51.63 $\pm$ 12.25
SmoothSubspace	86.80 $\pm$ 2.13	82.00 $\pm$ 2.83	**88.67 $\pm$ 2.36**	33.87 $\pm$ 1.19	57.07 $\pm$ 4.58	65.78 $\pm$ 2.04	69.33 $\pm$ 3.71	61.11 $\pm$ 0.77	77.07 $\pm$ 5.51	84.00 $\pm$ 4.81	37.07 $\pm$ 6.74	88.00 $\pm$ 2.87
SonyAIBORobotSurface1	93.94 $\pm$ 2.22	**95.71 $\pm$ 1.41**	92.85 $\pm$ 2.46	61.83 $\pm$ 10.72	84.73 $\pm$ 2.02	71.21 $\pm$ 3.16	67.39 $\pm$ 1.20	71.60 $\pm$ 6.11	64.96 $\pm$ 2.40	64.89 $\pm$ 3.03	54.54 $\pm$ 12.34	84.86 $\pm$ 2.67
SonyAIBORobotSurface2	**92.51 $\pm$ 3.32**	91.14 $\pm$ 0.89	91.54 $\pm$ 4.19	78.15 $\pm$ 2.59	85.83 $\pm$ 4.10	79.96 $\pm$ 1.73	83.11 $\pm$ 2.73	81.78 $\pm$ 2.73	80.76 $\pm$ 4.57	83.21 $\pm$ 2.88	61.15 $\pm$ 14.84	77.21 $\pm$ 3.67
StarLightCurves	81.52 $\pm$ 8.85	60.67 $\pm$ 16.70	76.44 $\pm$ 1.22	74.70 $\pm$ 4.25	76.72 $\pm$ 3.78	75.63 $\pm$ 4.10	28.48 $\pm$ 9.45	75.33 $\pm$ 3.39	27.99 $\pm$ 0.00	75.69 $\pm$ 5.23	75.95 $\pm$ 2.54	**81.86 $\pm$ 7.93**
Strawberry	63.51 $\pm$ 2.99	64.22 $\pm$ 4.90	60.22 $\pm$ 9.83	58.49 $\pm$ 2.90	63.03 $\pm$ 7.09	61.89 $\pm$ 4.92	**67.21 $\pm$ 4.90**	59.82 $\pm$ 1.80	61.03 $\pm$ 3.64	65.77 $\pm$ 7.40	52.92 $\pm$ 6.34	58.05 $\pm$ 7.04
SwedishLeaf	**79.81 $\pm$ 1.70**	76.00 $\pm$ 4.97	**79.81 $\pm$ 1.70**	47.97 $\pm$ 1.68	65.66 $\pm$ 4.54	62.61 $\pm$ 4.53	49.23 $\pm$ 4.26	67.15 $\pm$ 4.00	56.93 $\pm$ 3.43	62.29 $\pm$ 1.68	15.90 $\pm$ 6.80	79.78 $\pm$ 2.56
Symbols	88.78 $\pm$ 7.99	88.74 $\pm$ 7.74	**91.24 $\pm$ 0.96**	83.44 $\pm$ 1.29	84.92 $\pm$ 3.31	84.29 $\pm$ 1.01	36.11 $\pm$ 0.23	86.40 $\pm$ 1.63	58.19 $\pm$ 7.40	83.75 $\pm$ 0.90	27.56 $\pm$ 4.43	88.04 $\pm$ 1.30
SyntheticControl	87.80 $\pm$ 2.38	**96.33 $\pm$ 0.33**	92.60 $\pm$ 1.01	48.53 $\pm$ 3.68	92.73 $\pm$ 2.94	73.22 $\pm$ 3.10	52.11 $\pm$ 3.20	66.22 $\pm$ 3.66	56.27 $\pm$ 3.45	72.78 $\pm$ 3.66	20.93 $\pm$ 4.87	95.07 $\pm$ 1.62
ToeSegmentation1	**85.96 $\pm$ 7.05**	77.37 $\pm$ 5.14	81.58 $\pm$ 5.14	54.74 $\pm$ 3.58	68.16 $\pm$ 3.79	55.56 $\pm$ 2.16	53.95 $\pm$ 2.74	51.61 $\pm$ 0.91	53.25 $\pm$ 3.38	54.82 $\pm$ 1.32	48.77 $\pm$ 4.31	70.53 $\pm$ 10.39
ToeSegmentation2	74.77 $\pm$ 12.53	77.85 $\pm$ 7.33	74.77 $\pm$ 8.79	50.77 $\pm$ 2.88	68.46 $\pm$ 6.25	58.97 $\pm$ 1.60	50.26 $\pm$ 0.44	59.49 $\pm$ 6.72	37.54 $\pm$ 3.14	64.10 $\pm$ 2.70	52.31 $\pm$ 9.12	**78.00 $\pm$ 6.36**
Trace	88.60 $\pm$ 7.37	99.00 $\pm$ 1.00	93.00 $\pm$ 8.00	53.00 $\pm$ 4.42	76.40 $\pm$ 5.77	51.67 $\pm$ 4.93	53.33 $\pm$ 2.08	49.67 $\pm$ 3.79	59.80 $\pm$ 2.59	55.67 $\pm$ 8.50	48.80 $\pm$ 10.71	**100.00 $\pm$ 0.00**
TwoLeadECG	**97.98 $\pm$ 4.17**	95.47 $\pm$ 5.89	75.84 $\pm$ 5.58	58.75 $\pm$ 2.38	82.55 $\pm$ 14.28	62.16 $\pm$ 1.52	60.81 $\pm$ 0.10	56.34 $\pm$ 2.59	61.88 $\pm$ 1.65	68.51 $\pm$ 9.61	52.10 $\pm$ 2.73	96.68 $\pm$ 5.58
TwoPatterns	64.62 $\pm$ 3.64	**74.98 $\pm$ 5.04**	74.94 $\pm$ 1.43	38.09 $\pm$ 3.07	51.91 $\pm$ 7.48	40.83 $\pm$ 3.89	26.00 $\pm$ 0.92	39.83 $\pm$ 3.80	26.50 $\pm$ 1.83	37.54 $\pm$ 1.63	27.50 $\pm$ 4.13	74.22 $\pm$ 2.81
UMD	87.08 $\pm$ 10.10	91.53 $\pm$ 5.28	**96.67 $\pm$ 2.05**	56.53 $\pm$ 3.69	74.44 $\pm$ 17.00	61.57 $\pm$ 1.45	62.96 $\pm$ 5.82	68.52 $\pm$ 10.79	71.81 $\pm$ 2.17	70.14 $\pm$ 10.69	48.47 $\pm$ 7.32	95.78 $\pm$ 2.33
UWaveGestureLibraryAll	42.53 $\pm$ 1.15	39.33 $\pm$ 5.11	44.53 $\pm$ 1.84	77.02 $\pm$ 5.21	57.17 $\pm$ 2.24	**78.18 $\pm$ 3.39**	26.42 $\pm$ 1.62	77.47 $\pm$ 3.61	35.67 $\pm$ 1.53	76.96 $\pm$ 4.32	65.94 $\pm$ 6.48	48.36 $\pm$ 0.30
UWaveGestureLibraryX	39.64 $\pm$ 3.12	46.31 $\pm$ 4.46	46.23 $\pm$ 4.40	56.88 $\pm$ 2.64	50.95 $\pm$ 2.22	58.42 $\pm$ 2.37	19.27 $\pm$ 3.18	**58.58 $\pm$ 2.34**	31.22 $\pm$ 3.55	57.86 $\pm$ 1.47	26.48 $\pm$ 5.59	47.36 $\pm$ 0.30
UWaveGestureLibraryY	37.00 $\pm$ 1.20	38.44 $\pm$ 2.57	40.23 $\pm$ 3.63	**52.69 $\pm$ 2.24**	48.41 $\pm$ 3.12	51.42 $\pm$ 3.10	21.22 $\pm$ 1.11	51.72 $\pm$ 2.98	30.92 $\pm$ 5.43	50.81 $\pm$ 4.53	19.79 $\pm$ 3.07	51.36 $\pm$ 0.31
UWaveGestureLibraryZ	38.74 $\pm$ 2.41	44.95 $\pm$ 4.51	40.85 $\pm$ 4.13	48.35 $\pm$ 3.41	47.76 $\pm$ 4.10	50.34 $\pm$ 2.37	20.23 $\pm$ 1.68	49.48 $\pm$ 1.66	31.23 $\pm$ 4.16	48.05 $\pm$ 3.36	25.20 $\pm$ 6.78	**55.54 $\pm$ 2.61**
Wafer	76.83 $\pm$ 9.15	69.99 $\pm$ 14.88	71.79 $\pm$ 15.46	55.60 $\pm$ 16.77	73.34 $\pm$ 19.66	80.42 $\pm$ 16.85	76.25 $\pm$ 9.34	**83.86 $\pm$ 16.37**	75.80 $\pm$ 12.54	83.26 $\pm$ 17.85	64.01 $\pm$ 22.29	71.65 $\pm$ 9.58
Wine	57.41 $\pm$ 13.03	48.15 $\pm$ 3.93	**61.11 $\pm$ 5.40**	55.93 $\pm$ 3.56	53.33 $\pm$ 5.62	47.53 $\pm$ 8.35	54.94 $\pm$ 8.55	50.00 $\pm$ 0.00	46.67 $\pm$ 6.47	37.65 $\pm$ 19.80	50.00 $\pm$ 0.00	51.48 $\pm$ 2.03
WordSynonyms	25.08 $\pm$ 1.54	21.22 $\pm$ 1.56	25.08 $\pm$ 1.54	29.56 $\pm$ 2.06	30.38 $\pm$ 2.74	45.87 $\pm$ 4.63	7.99 $\pm$ 1.24	44.67 $\pm$ 4.83	18.81 $\pm$ 2.07	41.33 $\pm$ 4.67	22.26 $\pm$ 2.64	**47.05 $\pm$ 3.87**
Worms	57.14 $\pm$ 9.76	55.58 $\pm$ 5.91	**62.86 $\pm$ 4.55**	24.68 $\pm$ 4.68	40.26 $\pm$ 5.11	22.94 $\pm$ 0.75	19.91 $\pm$ 3.27	25.97 $\pm$ 0.00	16.62 $\pm$ 2.32	22.08 $\pm$ 1.30	22.08 $\pm$ 5.11	44.42 $\pm$ 6.26
WormsTwoClass	**68.05 $\pm$ 7.21**	59.74 $\pm$ 6.69	62.08 $\pm$ 7.97	48.83 $\pm$ 3.13	51.17 $\pm$ 8.09	52.38 $\pm$ 2.70	42.86 $\pm$ 3.44	51.08 $\pm$ 5.25	48.83 $\pm$ 3.26	52.81 $\pm$ 3.27	48.83 $\pm$ 5.92	54.81 $\pm$ 7.59
Yoga	**55.71 $\pm$ 3.37**	53.57 $\pm$ 4.72	55.19 $\pm$ 2.57	50.13 $\pm$ 2.22	53.89 $\pm$ 3.78	49.79 $\pm$ 3.21	52.94 $\pm$ 1.57	47.50 $\pm$ 3.55	51.67 $\pm$ 1.99	49.68 $\pm$ 4.28	48.87 $\pm$ 2.05	51.31 $\pm$ 4.34
Avg. Acc.	61.27	59.47	63.55	52.30	56.94	57.86	43.20	57.20	47.30	58.35	42.94	**64.14**
Avg. Rank	4.77	5.34	4.23	7.73	6.60	5.69	9.55	5.94	8.70	5.79	9.48	**4.19**
#Best	21	13	30	6	4	8	4	7	2	8	1	**36**

Table 10: Per-dataset 10-shot classification results on 128 shared UCR datasets. Each value is reported as mean ± standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
ACSF1	**85.40 ± 2.19**	76.80 ± 3.35	80.80 ± 0.84	38.60 ± 0.89	58.80 ± 2.59	64.67 ± 0.58	42.67 ± 1.53	67.00 ± 2.00	53.60 ± 2.51	67.00 ± 2.00	36.80 ± 3.27	81.33 ± 1.53
Adiac	62.46 ± 4.53	56.98 ± 3.84	62.46 ± 4.53	45.32 ± 1.06	47.72 ± 5.48	56.01 ± 2.77	48.68 ± 0.78	47.91 ± 6.68	71.71 ± 1.94	48.00 ± 3.58	12.84 ± 1.42	**76.42 ± 1.68**
AllGestureWiimoteX	49.37 ± 3.15	43.11 ± 2.57	39.26 ± 2.50	23.91 ± 1.60	43.54 ± 2.85	37.14 ± 0.52	15.00 ± 0.87	35.19 ± 1.33	25.80 ± 2.24	31.00 ± 1.57	43.54 ± 2.85	**52.31 ± 2.26**
AllGestureWiimoteY	54.89 ± 2.09	50.74 ± 4.62	40.17 ± 2.24	27.26 ± 1.11	48.03 ± 1.58	42.57 ± 3.19	16.95 ± 2.22	42.57 ± 3.81	30.37 ± 4.65	41.24 ± 2.84	48.03 ± 1.58	**56.34 ± 3.54**
AllGestureWiimoteZ	**49.34 ± 1.38**	46.03 ± 2.76	39.80 ± 2.97	22.43 ± 1.80	34.54 ± 4.20	31.29 ± 1.13	16.67 ± 1.15	29.67 ± 3.32	21.71 ± 1.69	32.19 ± 0.59	34.54 ± 4.20	44.43 ± 2.91
ArrowHead	72.23 ± 5.22	63.89 ± 7.06	65.49 ± 9.09	64.34 ± 0.65	62.86 ± 7.57	60.38 ± 1.19	48.19 ± 2.58	59.62 ± 2.88	57.83 ± 5.80	63.43 ± 3.48	45.83 ± 13.23	**81.49 ± 3.22**
BME	64.27 ± 9.29	69.33 ± 1.56	78.67 ± 4.64	64.53 ± 0.30	97.60 ± 3.29	71.11 ± 2.34	78.89 ± 1.02	88.44 ± 1.02	91.33 ± 2.31	86.67 ± 0.00	46.40 ± 7.46	**99.20 ± 0.30**
Beef	51.33 ± 6.91	44.00 ± 6.83	52.00 ± 2.98	61.33 ± 4.47	46.00 ± 6.41	70.00 ± 3.33	51.11 ± 5.09	54.44 ± 3.85	64.00 ± 7.23	**80.00 ± 0.00**	37.33 ± 4.35	68.00 ± 1.83
BeetleFly	71.00 ± 8.94	74.00 ± 6.52	75.00 ± 6.12	86.00 ± 4.18	86.00 ± 2.24	81.67 ± 2.89	48.33 ± 2.89	83.33 ± 2.89	49.00 ± 11.94	75.00 ± 5.00	70.00 ± 12.25	**96.67 ± 2.89**
BirdChicken	84.00 ± 12.45	92.00 ± 5.70	**94.00 ± 4.18**	68.00 ± 2.74	81.00 ± 9.62	71.67 ± 7.64	41.67 ± 7.64	68.33 ± 16.07	56.00 ± 5.48	56.67 ± 7.64	56.00 ± 4.18	89.00 ± 5.48
CBF	98.96 ± 0.23	98.27 ± 0.96	**99.24 ± 0.32**	67.49 ± 2.09	95.98 ± 4.64	90.07 ± 1.09	39.37 ± 1.51	91.44 ± 1.93	51.24 ± 7.49	90.11 ± 0.88	49.31 ± 9.05	97.71 ± 0.69
Car	57.67 ± 10.84	58.67 ± 3.21	65.33 ± 2.17	73.00 ± 2.98	62.67 ± 5.22	70.00 ± 2.89	27.22 ± 5.85	65.56 ± 0.96	37.33 ± 3.84	72.22 ± 1.92	50.33 ± 8.28	**76.33 ± 5.45**
Chinatown	98.19 ± 0.13	98.13 ± 0.16	**98.54 ± 0.21**	54.58 ± 24.80	97.73 ± 0.75	94.75 ± 2.10	95.63 ± 1.01	97.28 ± 0.61	96.62 ± 0.60	96.60 ± 1.10	63.56 ± 20.21	94.56 ± 3.09
ChlorineConcentration	**41.18 ± 3.66**	34.41 ± 8.63	37.86 ± 5.20	34.42 ± 4.40	35.51 ± 6.32	35.06 ± 1.39	38.85 ± 1.62	32.99 ± 4.01	39.05 ± 4.08	39.50 ± 4.07	33.01 ± 11.68	34.77 ± 6.93
CinCECGTorso	54.03 ± 3.03	65.94 ± 5.62	64.64 ± 5.11	45.36 ± 1.09	80.65 ± 7.24	81.59 ± 1.76	35.82 ± 2.41	84.42 ± 2.94	63.93 ± 1.27	85.19 ± 1.24	52.77 ± 4.47	**94.71 ± 4.65**
Coffee	97.14 ± 6.39	97.14 ± 1.60	**100.00 ± 0.00**	96.43 ± 4.37	97.86 ± 1.96	91.67 ± 2.06	86.90 ± 16.50	92.86 ± 0.00	98.57 ± 3.19	90.48 ± 2.06	56.43 ± 7.74	95.71 ± 2.99
Computers	66.48 ± 3.03	63.20 ± 6.81	**68.08 ± 4.62**	48.88 ± 2.52	56.40 ± 10.98	52.13 ± 9.60	52.13 ± 6.02	50.93 ± 12.73	49.68 ± 5.04	53.87 ± 12.94	51.52 ± 3.69	65.60 ± 6.66
CricketX	60.72 ± 2.35	55.18 ± 4.36	51.28 ± 3.40	30.15 ± 2.84	42.92 ± 1.99	46.50 ± 1.21	14.27 ± 1.71	45.38 ± 2.45	20.87 ± 2.94	45.38 ± 1.94	19.33 ± 2.95	**66.92 ± 2.71**
CricketY	56.92 ± 2.81	49.44 ± 1.94	47.69 ± 2.43	32.82 ± 4.21	42.15 ± 5.17	44.62 ± 0.44	9.32 ± 1.04	44.02 ± 2.09	17.90 ± 2.44	41.11 ± 0.53	23.95 ± 3.97	**57.95 ± 1.33**
CricketZ	63.18 ± 1.66	57.38 ± 2.58	48.82 ± 4.98	30.56 ± 3.52	43.18 ± 5.27	46.32 ± 1.71	12.91 ± 2.15	47.78 ± 1.41	18.26 ± 2.08	45.30 ± 1.50	21.95 ± 3.02	**65.30 ± 1.16**
Crop	57.92 ± 1.22	54.52 ± 1.67	57.92 ± 1.22	42.46 ± 2.10	44.96 ± 1.35	54.14 ± 2.07	52.84 ± 0.15	54.82 ± 1.79	54.74 ± 0.79	54.60 ± 1.45	25.02 ± 2.10	**57.97 ± 1.25**
DiatomSizeReduction	87.32 ± 3.50	71.05 ± 13.58	**96.80 ± 0.91**	44.77 ± 1.16	62.68 ± 4.64	87.04 ± 1.80	66.88 ± 2.86	51.96 ± 5.03	48.89 ± 18.63	87.47 ± 0.19	30.13 ± 0.15	92.61 ± 2.00
DistalPhalanxOutlineAgeGroup	68.20 ± 2.98	68.78 ± 4.24	70.50 ± 1.61	69.50 ± 1.66	69.50 ± 2.81	66.67 ± 1.10	65.71 ± 3.62	70.50 ± 1.25	68.78 ± 2.72	66.91 ± 1.25	48.06 ± 11.76	**70.65 ± 2.18**
DistalPhalanxOutlineCorrect	64.78 ± 12.49	65.07 ± 11.29	63.62 ± 12.87	50.00 ± 13.55	60.14 ± 9.92	58.94 ± 4.04	**67.15 ± 2.41**	55.43 ± 14.84	62.39 ± 9.46	61.71 ± 6.72	47.61 ± 8.80	60.51 ± 11.85
DistalPhalanxTW	55.83 ± 3.65	59.14 ± 2.12	**60.58 ± 3.58**	47.34 ± 6.42	57.55 ± 6.08	53.96 ± 3.14	53.00 ± 2.91	54.92 ± 0.83	52.52 ± 4.35	54.92 ± 2.91	32.37 ± 14.50	54.10 ± 3.11
DodgerLoopDay	49.25 ± 3.71	47.25 ± 4.28	39.75 ± 3.11	41.50 ± 2.24	47.00 ± 4.11	**49.58 ± 5.64**	28.75 ± 2.50	45.42 ± 1.91	29.00 ± 2.40	48.75 ± 1.25	47.00 ± 4.11	48.75 ± 1.77
DodgerLoopGame	68.70 ± 6.09	83.33 ± 5.28	75.07 ± 2.64	79.28 ± 0.97	**88.84 ± 1.67**	76.81 ± 1.26	48.79 ± 3.42	81.40 ± 1.51	53.48 ± 2.83	78.99 ± 1.45	**88.84 ± 1.67**	83.62 ± 0.65
DodgerLoopWeekend	94.06 ± 1.39	94.20 ± 2.11	96.81 ± 0.83	**97.83 ± 0.00**	96.81 ± 0.65	96.86 ± 0.42	72.71 ± 1.82	97.58 ± 0.42	76.38 ± 1.82	97.10 ± 0.00	96.81 ± 0.65	**97.83 ± 0.00**
ECG200	80.00 ± 4.00	**83.00 ± 2.00**	80.40 ± 4.28	77.60 ± 4.34	79.80 ± 6.53	79.33 ± 3.21	78.00 ± 3.61	81.00 ± 5.00	82.00 ± 2.92	82.00 ± 3.61	63.00 ± 15.26	77.40 ± 4.88
ECG5000	83.56 ± 5.09	78.23 ± 2.20	80.72 ± 4.37	**88.37 ± 1.22**	76.31 ± 5.83	79.59 ± 2.75	73.64 ± 2.21	84.75 ± 1.16	79.28 ± 5.11	82.19 ± 1.79	24.56 ± 15.15	73.78 ± 6.18
ECGFiveDays	95.40 ± 1.88	88.99 ± 4.44	**97.56 ± 1.20**	74.87 ± 4.61	94.61 ± 2.00	83.82 ± 7.05	96.44 ± 1.42	91.64 ± 1.90	97.51 ± 1.12	86.18 ± 8.29	57.47 ± 4.38	75.28 ± 1.71
EOGHorizontalSignal	30.61 ± 3.99	43.15 ± 5.84	30.94 ± 3.48	29.89 ± 1.60	38.78 ± 6.14	39.59 ± 1.67	8.01 ± 1.27	41.62 ± 1.42	10.61 ± 3.44	40.06 ± 3.72	39.56 ± 4.17	**49.17 ± 1.62**
EOGVerticalSignal	21.71 ± 2.53	28.23 ± 2.47	19.83 ± 3.08	31.55 ± 2.19	40.06 ± 1.98	36.00 ± 1.12	8.47 ± 0.84	37.75 ± 1.67	11.99 ± 1.33	37.02 ± 1.99	30.55 ± 3.17	**42.21 ± 2.26**
Earthquakes	53.81 ± 8.62	53.96 ± 9.50	51.37 ± 3.47	52.52 ± 6.49	54.82 ± 2.85	60.91 ± 4.63	49.16 ± 8.34	59.71 ± 6.86	54.68 ± 5.26	**61.87 ± 3.60**	47.91 ± 6.66	54.68 ± 7.63
ElectricDevices	40.12 ± 4.42	38.65 ± 3.96	36.00 ± 4.64	28.04 ± 1.63	39.08 ± 8.26	48.59 ± 6.21	30.07 ± 3.26	33.74 ± 3.43	37.44 ± 1.49	48.14 ± 2.51	16.08 ± 5.57	**49.42 ± 5.02**
EthanolLevel	26.56 ± 1.96	25.76 ± 1.05	26.16 ± 0.71	29.28 ± 1.78	25.92 ± 1.70	28.07 ± 2.55	25.20 ± 3.34	26.80 ± 0.69	25.60 ± 1.13	**29.87 ± 2.47**	25.56 ± 1.05	28.24 ± 0.50
FaceAll	77.73 ± 3.48	**80.30 ± 1.67**	77.73 ± 3.48	67.09 ± 2.40	54.49 ± 5.41	67.46 ± 1.82	59.68 ± 1.83	67.32 ± 2.82	60.69 ± 2.63	64.06 ± 2.32	32.89 ± 2.71	63.01 ± 1.34
FaceFour	87.95 ± 3.07	59.32 ± 7.51	**92.95 ± 2.03**	74.77 ± 0.95	66.36 ± 10.77	85.98 ± 1.31	65.91 ± 2.27	88.26 ± 0.66	69.55 ± 1.69	83.71 ± 1.31	34.55 ± 3.82	79.09 ± 1.90
FacesUCR	**87.80 ± 1.60**	84.26 ± 2.06	82.41 ± 2.48	63.80 ± 1.79	54.40 ± 3.05	80.54 ± 1.15	70.11 ± 1.05	83.69 ± 1.13	72.08 ± 1.16	79.51 ± 1.53	34.67 ± 3.28	79.08 ± 1.41
FiftyWords	45.36 ± 2.39	36.70 ± 1.89	45.36 ± 2.39	53.67 ± 1.47	50.46 ± 2.83	64.32 ± 1.87	16.85 ± 0.67	67.99 ± 1.84	37.05 ± 3.62	57.36 ± 1.54	56.40 ± 1.98	**71.34 ± 2.10**
Fish	**83.31 ± 3.78**	56.00 ± 4.28	71.54 ± 6.28	65.60 ± 3.49	62.74 ± 2.15	68.95 ± 4.29	24.38 ± 3.80	71.05 ± 1.44	57.14 ± 3.45	75.05 ± 2.31	38.17 ± 5.05	78.29 ± 3.92
FordA	78.26 ± 5.14	75.61 ± 0.71	**78.70 ± 3.26**	49.09 ± 0.93	67.88 ± 3.90	51.99 ± 2.52	52.22 ± 0.95	48.81 ± 1.31	53.05 ± 1.09	48.21 ± 1.67	50.06 ± 2.65	57.15 ± 1.85
FordB	**63.83 ± 5.74**	62.17 ± 3.33	61.70 ± 3.19	48.54 ± 0.72	59.06 ± 1.85	46.75 ± 0.61	49.79 ± 1.45	47.24 ± 1.24	50.47 ± 2.25	47.33 ± 0.14	49.33 ± 1.67	55.60 ± 1.91
FreezerRegularTrain	86.27 ± 5.59	**89.20 ± 7.73**	79.00 ± 5.98	76.33 ± 0.75	80.48 ± 6.98	76.36 ± 0.25	61.75 ± 7.11	76.22 ± 0.12	68.27 ± 6.23	74.90 ± 1.40	66.57 ± 12.21	**89.20 ± 5.65**
FreezerSmallTrain	78.20 ± 4.17	80.08 ± 7.30	74.46 ± 2.32	76.11 ± 0.45	68.74 ± 3.00	73.18 ± 1.45	65.15 ± 4.43	71.92 ± 2.00	68.25 ± 2.23	70.97 ± 2.25	65.53 ± 10.69	**82.86 ± 3.49**
Fungi	93.98 ± 3.37	38.06 ± 6.98	67.85 ± 1.59	67.74 ± 4.64	69.57 ± 7.88	**95.88 ± 1.64**	72.94 ± 1.89	91.22 ± 1.73	74.73 ± 0.76	**95.88 ± 0.31**	12.47 ± 9.15	79.57 ± 1.20
GestureMidAirD1	49.38 ± 3.66	60.92 ± 1.48	49.38 ± 3.66	49.69 ± 1.40	48.31 ± 2.52	54.36 ± 1.94	26.15 ± 2.31	58.21 ± 2.22	44.15 ± 6.68	57.44 ± 1.60	48.31 ± 2.52	**67.08 ± 1.00**
GestureMidAirD2	38.00 ± 5.29	47.23 ± 3.34	38.00 ± 5.29	50.46 ± 0.69	50.46 ± 2.75	53.08 ± 2.66	11.28 ± 2.47	55.13 ± 1.18	34.00 ± 4.05	51.79 ± 1.18	50.46 ± 2.75	**55.69 ± 1.85**
GestureMidAirD3	20.15 ± 2.40	25.23 ± 1.38	20.15 ± 2.40	32.92 ± 0.34	29.23 ± 3.17	28.46 ± 0.77	15.13 ± 2.70	33.08 ± 2.04	29.38 ± 4.79	31.28 ± 0.44	29.23 ± 3.17	**33.69 ± 2.63**

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Table 10: Per-dataset 10-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
GesturePebbleZ1	76.86 ± 5.40	75.47 ± 8.13	77.67 ± 2.20	66.63 ± 2.78	68.84 ± 4.61	75.00 ± 1.16	22.29 ± 3.87	77.33 ± 2.10	38.95 ± 10.62	**81.40 ± 2.53**	68.84 ± 4.61	75.12 ± 5.19
GesturePebbleZ2	73.16 ± 4.51	63.42 ± 5.15	73.16 ± 2.82	60.38 ± 3.46	67.59 ± 4.18	69.62 ± 2.90	19.20 ± 0.73	74.26 ± 2.56	28.86 ± 9.85	73.00 ± 1.59	67.59 ± 4.18	**75.57 ± 4.64**
GunPoint	85.73 ± 9.32	82.13 ± 8.02	91.73 ± 11.16	68.27 ± 5.09	77.60 ± 4.39	71.78 ± 3.79	64.00 ± 6.57	72.00 ± 1.15	77.73 ± 11.90	76.67 ± 6.67	62.00 ± 16.53	**97.47 ± 2.23**
GunPointAgeSpan	89.56 ± 7.42	72.47 ± 9.28	74.87 ± 11.59	83.54 ± 2.13	82.22 ± 4.04	85.44 ± 1.67	52.22 ± 3.12	85.13 ± 2.39	71.08 ± 9.43	83.54 ± 3.02	53.61 ± 10.43	**92.72 ± 5.80**
GunPointMaleVersusFemale	92.78 ± 8.49	93.48 ± 5.65	92.22 ± 9.65	67.09 ± 4.97	95.57 ± 2.50	91.14 ± 4.91	60.97 ± 6.06	93.57 ± 3.96	74.05 ± 7.11	94.41 ± 2.22	67.22 ± 11.33	**98.54 ± 0.94**
GunPointOldVersusYoung	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	76.19 ± 1.96	79.17 ± 3.15	78.20 ± 2.30	57.46 ± 1.38	77.25 ± 5.77	61.90 ± 5.09	79.26 ± 2.47	85.78 ± 19.21	86.79 ± 4.71
Ham	60.76 ± 5.19	66.48 ± 2.64	65.52 ± 4.83	**71.24 ± 5.65**	62.86 ± 4.86	66.03 ± 7.40	63.49 ± 10.09	68.89 ± 6.69	61.71 ± 8.23	68.89 ± 7.02	61.52 ± 3.79	62.86 ± 3.37
HandOutlines	50.76 ± 12.61	50.92 ± 12.38	57.46 ± 2.17	81.57 ± 4.57	**82.65 ± 3.66**	81.62 ± 2.31	38.65 ± 4.09	78.47 ± 5.74	35.95 ± 0.00	81.62 ± 1.95	76.97 ± 8.74	71.51 ± 5.35
Haptics	29.68 ± 4.36	29.16 ± 4.62	33.31 ± 5.65	37.73 ± 3.74	31.95 ± 2.95	37.66 ± 2.89	23.81 ± 5.22	37.66 ± 2.77	25.45 ± 4.31	38.20 ± 2.76	35.32 ± 3.26	**39.22 ± 4.29**
Herring	55.00 ± 9.20	55.62 ± 5.91	55.00 ± 7.28	50.31 ± 6.57	**59.69 ± 3.39**	52.60 ± 3.93	47.92 ± 2.39	53.65 ± 8.02	49.06 ± 4.50	50.52 ± 1.80	57.19 ± 6.68	55.31 ± 5.01
HouseTwenty	**90.92 ± 1.10**	88.91 ± 1.82	89.58 ± 3.18	65.55 ± 5.48	81.51 ± 3.71	81.23 ± 2.43	60.50 ± 3.36	74.79 ± 1.46	64.20 ± 2.95	81.79 ± 4.63	70.76 ± 6.04	89.24 ± 2.81
InlineSkate	30.04 ± 2.77	24.04 ± 1.78	21.27 ± 3.03	25.16 ± 1.55	21.67 ± 3.27	28.42 ± 1.51	16.91 ± 0.91	28.73 ± 0.83	18.22 ± 1.24	29.09 ± 2.20	22.62 ± 1.07	**33.49 ± 2.77**
InsectEPGRegularTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	54.62 ± 4.82	79.12 ± 3.30	61.31 ± 3.90	42.70 ± 1.23	60.37 ± 1.29	55.18 ± 4.83	64.39 ± 5.02	96.63 ± 7.54	86.59 ± 4.06
InsectEPGSmallTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	60.00 ± 1.05	81.69 ± 6.46	69.08 ± 1.20	55.82 ± 1.06	69.88 ± 0.80	58.71 ± 4.87	66.40 ± 1.81	83.13 ± 0.00	85.06 ± 2.08
InsectWingbeatSound	34.13 ± 0.95	34.52 ± 3.87	30.58 ± 3.83	60.48 ± 0.59	47.36 ± 1.34	60.99 ± 1.05	39.46 ± 3.03	**61.67 ± 1.18**	49.66 ± 3.26	61.20 ± 0.34	46.64 ± 2.91	51.36 ± 2.05
ItalyPowerDemand	92.21 ± 1.02	92.56 ± 1.30	92.75 ± 1.07	53.43 ± 4.80	93.97 ± 0.64	87.66 ± 8.42	94.69 ± 1.03	95.59 ± 0.11	**95.82 ± 0.32**	94.33 ± 0.49	49.23 ± 14.84	92.19 ± 2.62
LargeKitchenAppliances	64.05 ± 4.75	66.13 ± 5.93	66.08 ± 5.44	40.85 ± 3.41	**67.95 ± 4.05**	46.22 ± 6.49	39.82 ± 1.74	43.11 ± 5.30	40.53 ± 2.82	45.33 ± 5.07	41.71 ± 0.52	55.09 ± 5.30
Lightning2	70.49 ± 3.84	64.92 ± 7.48	65.57 ± 4.34	58.03 ± 2.74	69.18 ± 5.84	71.04 ± 1.89	51.37 ± 4.13	63.93 ± 3.28	58.36 ± 5.26	68.85 ± 4.34	60.00 ± 3.77	**77.05 ± 4.18**
Lightning7	64.38 ± 5.97	66.03 ± 3.41	60.00 ± 4.37	60.27 ± 3.36	61.10 ± 9.76	74.43 ± 1.58	20.09 ± 4.81	65.75 ± 1.37	36.71 ± 3.41	**75.34 ± 2.37**	26.30 ± 5.76	66.85 ± 7.34
Mallat	68.22 ± 8.96	69.46 ± 5.32	**95.78 ± 0.57**	90.38 ± 0.87	87.49 ± 9.31	92.79 ± 0.83	66.92 ± 5.03	85.20 ± 1.21	79.12 ± 6.08	89.69 ± 0.98	25.09 ± 0.73	89.72 ± 1.50
Meat	53.00 ± 18.31	59.33 ± 6.08	85.00 ± 4.86	**87.00 ± 7.94**	60.00 ± 5.27	61.11 ± 4.19	68.89 ± 7.52	55.56 ± 19.25	55.33 ± 20.66	86.11 ± 7.88	38.67 ± 15.92	64.00 ± 6.08
MedicalImages	**56.61 ± 3.74**	54.08 ± 3.55	52.39 ± 5.17	34.84 ± 5.66	48.24 ± 3.52	46.27 ± 5.02	40.00 ± 3.41	45.09 ± 2.04	47.45 ± 1.06	47.72 ± 6.48	18.58 ± 8.91	54.68 ± 2.10
MelbournePedestrian	**84.60 ± 1.98**	81.55 ± 1.62	80.47 ± 2.59	30.74 ± 5.50	68.14 ± 1.82	73.08 ± 1.13	75.36 ± 1.75	78.01 ± 1.38	77.48 ± 0.93	76.77 ± 0.49	68.14 ± 1.82	70.88 ± 1.49
MiddlePhalanxOutlineAgeGroup	44.29 ± 12.74	54.68 ± 6.95	55.19 ± 4.97	54.16 ± 10.18	53.51 ± 5.82	**58.66 ± 1.35**	50.43 ± 7.47	54.11 ± 4.79	50.13 ± 8.45	56.49 ± 1.12	41.82 ± 18.66	51.04 ± 7.11
MiddlePhalanxOutlineCorrect	60.00 ± 5.84	**65.02 ± 2.98**	60.62 ± 1.86	54.23 ± 4.23	56.36 ± 2.69	50.86 ± 7.18	64.72 ± 6.49	54.98 ± 7.77	58.97 ± 7.04	52.23 ± 9.34	52.44 ± 10.59	59.18 ± 5.65
MiddlePhalanxTW	**49.74 ± 5.84**	48.05 ± 4.11	49.48 ± 2.12	45.71 ± 2.70	46.23 ± 6.05	46.10 ± 6.40	37.23 ± 5.86	42.64 ± 7.61	35.97 ± 2.62	41.34 ± 8.07	23.64 ± 8.52	44.42 ± 6.27
MixedShapesRegularTrain	67.65 ± 7.57	70.16 ± 5.08	72.53 ± 6.92	70.49 ± 2.59	73.35 ± 4.70	73.87 ± 1.26	24.40 ± 1.33	74.08 ± 2.13	28.76 ± 6.79	73.65 ± 2.23	63.69 ± 2.98	**79.60 ± 1.74**
MixedShapesSmallTrain	70.61 ± 10.30	69.46 ± 11.49	75.93 ± 3.55	73.56 ± 2.63	73.16 ± 8.70	78.08 ± 2.18	23.04 ± 1.80	77.03 ± 1.02	27.06 ± 6.78	77.29 ± 1.92	65.95 ± 4.04	**81.61 ± 2.47**
MoteStrain	89.38 ± 0.88	89.14 ± 3.04	90.18 ± 0.68	86.26 ± 0.20	86.58 ± 1.30	87.19 ± 1.69	77.40 ± 1.04	87.43 ± 0.48	85.32 ± 0.75	**90.65 ± 0.48**	67.41 ± 11.72	89.23 ± 1.53
NonInvasiveFetalECGThorax1	55.91 ± 3.81	50.28 ± 5.45	55.91 ± 3.81	**81.90 ± 0.59**	63.56 ± 4.40	77.13 ± 1.08	36.83 ± 0.46	78.18 ± 0.43	46.96 ± 3.44	75.71 ± 1.08	58.08 ± 1.54	76.84 ± 1.40
NonInvasiveFetalECGThorax2	62.50 ± 1.19	67.13 ± 2.71	62.50 ± 1.19	**86.79 ± 1.25**	75.21 ± 2.50	83.55 ± 0.48	57.79 ± 2.45	83.27 ± 1.00	69.23 ± 2.65	80.68 ± 0.66	70.22 ± 2.79	82.53 ± 1.33
OSULeaf	**72.64 ± 5.62**	69.26 ± 4.17	69.01 ± 3.75	34.88 ± 3.54	39.34 ± 3.85	46.42 ± 2.52	19.83 ± 4.60	40.50 ± 2.19	29.83 ± 2.82	37.19 ± 2.07	28.10 ± 5.12	59.42 ± 3.58
OliveOil	56.00 ± 14.79	40.00 ± 0.00	**66.67 ± 9.72**	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00	45.56 ± 9.62	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00
PLAID	28.04 ± 4.01	25.14 ± 3.42	22.46 ± 6.07	34.45 ± 1.13	54.53 ± 5.93	50.16 ± 2.34	32.59 ± 0.99	44.82 ± 3.36	37.77 ± 2.12	46.80 ± 0.84	54.53 ± 5.93	**59.89 ± 3.28**
PhalangesOutlinesCorrect	56.88 ± 6.98	56.74 ± 8.68	57.48 ± 7.09	54.27 ± 11.23	49.91 ± 6.62	**58.78 ± 4.01**	55.79 ± 3.74	55.36 ± 17.14	52.42 ± 4.71	54.23 ± 12.22	53.99 ± 12.03	55.73 ± 8.07
Phoneme	**28.10 ± 1.26**	25.96 ± 1.60	**28.10 ± 1.26**	8.58 ± 0.50	11.88 ± 0.54	13.82 ± 0.86	9.55 ± 1.16	11.71 ± 0.45	10.91 ± 1.85	10.88 ± 0.47	10.94 ± 0.56	21.95 ± 1.27
PickupGestureWiimoteZ	50.80 ± 15.01	68.40 ± 2.61	**73.60 ± 2.19**	51.60 ± 1.67	54.40 ± 6.84	56.67 ± 3.06	26.67 ± 1.15	54.67 ± 1.15	42.00 ± 5.10	60.67 ± 1.15	54.40 ± 6.84	70.40 ± 2.61
PigAirwayPressure	26.35 ± 2.05	12.88 ± 0.53	**50.58 ± 0.92**	4.81 ± 0.00	4.13 ± 0.43	4.81 ± 0.48	2.08 ± 1.00	6.25 ± 1.27	4.62 ± 0.87	6.73 ± 0.48	5.00 ± 0.65	9.62 ± 0.48
PigArtPressure	54.71 ± 1.57	25.19 ± 5.85	**69.81 ± 3.29**	8.27 ± 0.40	6.06 ± 2.62	8.97 ± 0.28	6.41 ± 1.00	9.78 ± 0.73	6.35 ± 0.92	8.65 ± 1.27	18.08 ± 2.06	31.73 ± 1.92
PigCVP	28.85 ± 2.68	19.62 ± 1.84	**61.35 ± 2.42**	7.12 ± 0.40	3.27 ± 1.61	7.37 ± 0.73	7.05 ± 1.21	8.33 ± 1.69	4.13 ± 0.80	5.93 ± 1.11	11.63 ± 1.24	19.52 ± 1.34
Plane	**100.00 ± 0.00**	99.43 ± 1.28	**100.00 ± 0.00**	96.57 ± 1.28	96.95 ± 1.24	98.10 ± 0.00	98.73 ± 0.55	97.14 ± 0.95	98.48 ± 0.52	97.14 ± 0.95	49.71 ± 5.36	99.24 ± 1.24
PowerCons	84.00 ± 3.74	85.67 ± 2.68	**88.11 ± 1.91**	73.22 ± 7.34	83.67 ± 3.78	85.93 ± 2.85	53.33 ± 0.56	86.30 ± 2.25	59.67 ± 5.39	84.63 ± 0.85	74.33 ± 14.06	82.33 ± 5.46
ProximalPhalanxOutlineAgeGroup	69.07 ± 5.10	80.49 ± 4.77	80.98 ± 2.76	80.98 ± 2.95	75.90 ± 8.78	81.30 ± 5.55	81.14 ± 3.66	**81.63 ± 4.89**	76.88 ± 9.31	77.56 ± 6.40	68.78 ± 16.36	79.80 ± 3.30
ProximalPhalanxOutlineCorrect	63.71 ± 11.74	64.60 ± 2.91	63.85 ± 3.21	56.01 ± 10.53	61.86 ± 4.47	64.60 ± 0.69	**66.55 ± 3.51**	64.26 ± 0.69	61.86 ± 6.47	65.98 ± 3.31	51.34 ± 15.85	60.07 ± 4.20
ProximalPhalanxTW	60.78 ± 11.91	70.05 ± 3.85	**72.39 ± 1.88**	51.12 ± 7.39	70.73 ± 1.42	64.07 ± 2.20	60.16 ± 2.69	65.53 ± 2.25	57.95 ± 5.18	59.84 ± 3.91	25.07 ± 15.12	69.66 ± 3.47
RefrigerationDevices	42.67 ± 7.00	44.27 ± 6.42	43.41 ± 5.31	34.72 ± 1.59	37.71 ± 6.85	44.00 ± 2.97	34.40 ± 1.33	34.76 ± 2.56	35.31 ± 4.05	**44.62 ± 2.22**	32.37 ± 1.67	41.23 ± 4.90
Rock	41.60 ± 6.84	34.40 ± 4.56	36.00 ± 1.41	81.60 ± 0.89	57.20 ± 9.01	82.67 ± 1.15	84.67 ± 6.11	**88.00 ± 0.00**	62.00 ± 6.16	86.67 ± 3.06	30.80 ± 4.82	76.40 ± 2.97
ScreenType	35.84 ± 5.22	36.48 ± 5.07	37.23 ± 5.64	36.37 ± 2.19	39.15 ± 5.32	**40.53 ± 1.87**	35.73 ± 2.12	36.09 ± 2.74	33.12 ± 2.51	39.82 ± 2.27	36.32 ± 3.91	37.12 ± 3.93
SemgHandGenderCh2	63.47 ± 5.58	**74.93 ± 6.24**	65.43 ± 6.18	67.83 ± 4.45	66.37 ± 2.41	62.89 ± 2.12	51.67 ± 5.43	65.78 ± 6.45	55.43 ± 2.94	60.72 ± 4.24	58.77 ± 7.13	63.80 ± 10.73
SemgHandMovementCh2	37.69 ± 5.41	36.27 ± 1.15	37.82 ± 1.58	37.42 ± 2.10	32.22 ± 2.74	30.67 ± 2.78	25.04 ± 0.56	34.52 ± 3.16	25.69 ± 1.85	33.04 ± 3.33	33.64 ± 4.19	**45.56 ± 2.93**

Continued on next page

Table 10: Per-dataset 10-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
SemgHandSubjectCh2	43.56 ± 2.78	45.20 ± 5.06	40.22 ± 5.31	62.80 ± 4.54	54.13 ± 4.47	60.37 ± 0.71	34.22 ± 3.27	**68.07 ± 2.65**	39.96 ± 2.48	60.96 ± 1.10	38.18 ± 2.42	66.71 ± 4.45
ShakeGestureWiimoteZ	82.00 ± 6.16	91.20 ± 2.28	88.80 ± 1.10	55.60 ± 2.97	68.40 ± 7.67	60.00 ± 4.00	33.33 ± 3.06	62.67 ± 3.06	45.20 ± 7.29	56.67 ± 1.15	68.40 ± 7.67	**91.60 ± 2.97**
ShapeletSim	86.33 ± 6.80	**100.00 ± 0.00**	**100.00 ± 0.00**	48.78 ± 0.82	95.22 ± 4.26	59.07 ± 5.56	48.70 ± 1.40	50.19 ± 2.80	51.00 ± 1.98	49.81 ± 1.95	52.33 ± 4.64	96.67 ± 1.96
ShapesAll	68.60 ± 2.31	66.63 ± 1.82	68.60 ± 2.31	62.47 ± 0.34	65.47 ± 2.49	72.11 ± 0.69	12.89 ± 4.16	73.28 ± 0.38	31.00 ± 5.80	65.56 ± 1.08	70.00 ± 1.20	**83.13 ± 0.32**
SmallKitchenAppliances	**67.20 ± 8.07**	61.87 ± 4.21	64.69 ± 7.08	36.59 ± 1.57	60.00 ± 3.85	55.11 ± 4.59	41.51 ± 2.40	38.76 ± 2.56	47.25 ± 3.08	51.02 ± 1.54	41.33 ± 5.03	57.60 ± 5.56
SmoothSubspace	**92.00 ± 1.56**	87.07 ± 3.67	**92.00 ± 3.02**	38.13 ± 6.59	67.73 ± 10.31	79.56 ± 2.78	80.67 ± 0.67	60.00 ± 11.02	86.13 ± 3.38	90.67 ± 4.00	43.60 ± 9.96	89.47 ± 2.60
SonyAIBORobotSurface1	95.11 ± 2.42	**96.01 ± 0.87**	92.45 ± 1.40	42.96 ± 0.07	87.95 ± 1.92	64.28 ± 1.83	67.05 ± 1.01	76.43 ± 0.69	64.86 ± 0.43	67.83 ± 1.06	45.09 ± 4.84	86.09 ± 3.97
SonyAIBORobotSurface2	**96.73 ± 3.11**	92.82 ± 3.91	94.67 ± 2.40	77.50 ± 0.71	90.95 ± 2.30	81.25 ± 2.70	83.07 ± 2.21	86.08 ± 2.22	82.71 ± 1.50	85.73 ± 2.18	61.15 ± 12.88	84.11 ± 2.29
StarLightCurves	71.47 ± 14.56	82.56 ± 3.87	75.10 ± 5.06	74.16 ± 3.22	77.51 ± 4.09	75.29 ± 4.53	29.60 ± 9.11	76.31 ± 3.24	30.20 ± 4.95	74.55 ± 5.89	70.71 ± 14.11	**85.26 ± 3.54**
Strawberry	79.73 ± 8.80	76.70 ± 3.26	76.70 ± 4.55	60.81 ± 1.94	79.89 ± 6.30	61.98 ± 1.58	**80.63 ± 1.13**	59.10 ± 1.09	77.51 ± 4.36	68.11 ± 2.15	50.54 ± 10.66	75.84 ± 0.99
SwedishLeaf	**88.03 ± 1.39**	83.68 ± 1.26	**88.03 ± 1.39**	66.11 ± 1.36	73.47 ± 2.54	75.95 ± 1.36	60.69 ± 4.66	81.39 ± 2.49	68.80 ± 1.87	74.72 ± 1.82	28.96 ± 3.20	85.89 ± 1.27
Symbols	**93.83 ± 2.55**	89.89 ± 7.08	92.52 ± 1.13	75.94 ± 0.93	85.87 ± 1.87	83.32 ± 0.52	34.77 ± 0.30	85.96 ± 1.03	61.21 ± 7.54	82.51 ± 0.70	21.11 ± 2.93	89.33 ± 0.82
SyntheticControl	95.60 ± 1.14	**97.87 ± 0.61**	97.60 ± 0.43	59.07 ± 3.98	94.20 ± 1.64	90.78 ± 1.64	59.44 ± 1.95	79.44 ± 4.86	68.40 ± 3.43	90.11 ± 1.35	22.60 ± 7.62	97.33 ± 0.85
ToeSegmentation1	**93.25 ± 1.60**	89.56 ± 3.33	91.58 ± 1.68	55.53 ± 3.14	68.95 ± 9.85	56.73 ± 5.97	50.29 ± 6.71	55.85 ± 3.55	49.39 ± 2.18	55.70 ± 4.98	50.53 ± 1.40	80.53 ± 2.75
ToeSegmentation2	81.38 ± 10.25	**87.08 ± 4.53**	85.85 ± 2.64	50.15 ± 3.62	73.38 ± 4.70	68.21 ± 10.44	56.67 ± 7.86	58.46 ± 6.30	46.77 ± 3.41	61.54 ± 3.35	43.54 ± 14.75	80.31 ± 4.70
Trace	92.20 ± 4.60	**100.00 ± 0.00**	95.20 ± 5.07	55.80 ± 5.72	86.20 ± 10.35	72.67 ± 1.53	55.67 ± 4.51	61.33 ± 4.62	74.80 ± 3.27	78.00 ± 4.36	46.80 ± 8.17	**100.00 ± 0.00**
TwoLeadECG	**99.86 ± 0.08**	98.79 ± 1.25	97.56 ± 2.37	58.05 ± 1.68	91.89 ± 5.00	70.85 ± 3.05	75.62 ± 1.63	65.26 ± 1.91	78.86 ± 1.56	78.99 ± 2.23	54.01 ± 3.73	99.68 ± 0.24
TwoPatterns	70.89 ± 4.77	77.56 ± 2.96	**78.86 ± 3.59**	42.28 ± 1.08	74.73 ± 6.48	42.55 ± 1.37	26.22 ± 0.85	42.35 ± 1.82	29.82 ± 3.35	37.95 ± 1.72	25.60 ± 4.32	70.06 ± 0.70
UMD	92.78 ± 3.35	98.33 ± 1.05	**98.75 ± 0.58**	64.17 ± 3.05	91.53 ± 4.77	76.62 ± 2.44	75.93 ± 7.23	89.58 ± 4.22	93.06 ± 1.47	87.27 ± 3.21	47.50 ± 8.96	98.67 ± 7.42
UWaveGestureLibraryAll	48.45 ± 3.31	48.15 ± 2.07	49.06 ± 3.81	80.57 ± 3.71	68.35 ± 4.39	84.19 ± 2.90	27.54 ± 1.68	**84.63 ± 3.19**	42.77 ± 5.45	83.32 ± 3.39	81.20 ± 2.71	52.41 ± 1.70
UWaveGestureLibraryX	51.98 ± 1.26	57.08 ± 2.26	52.99 ± 2.93	60.99 ± 2.04	57.69 ± 1.76	**62.37 ± 1.85**	17.30 ± 0.19	62.30 ± 0.55	36.29 ± 8.88	61.26 ± 1.95	56.81 ± 2.90	53.50 ± 1.91
UWaveGestureLibraryY	45.58 ± 1.39	47.42 ± 0.67	46.02 ± 2.85	55.44 ± 1.29	53.52 ± 3.23	**56.83 ± 0.99**	23.04 ± 3.42	56.65 ± 1.12	32.16 ± 4.98	55.22 ± 1.56	46.61 ± 1.40	52.03 ± 0.97
UWaveGestureLibraryZ	48.34 ± 2.78	51.60 ± 2.75	44.60 ± 1.67	53.18 ± 1.58	53.45 ± 2.14	54.99 ± 0.25	20.01 ± 2.19	54.36 ± 2.47	34.40 ± 4.42	51.61 ± 2.72	46.80 ± 1.81	**60.65 ± 2.84**
Wafer	84.60 ± 6.19	78.03 ± 11.28	81.53 ± 4.78	69.44 ± 4.07	**84.77 ± 11.10**	82.09 ± 7.52	76.30 ± 12.19	83.57 ± 7.70	81.81 ± 10.33	81.67 ± 9.12	78.96 ± 10.53	82.53 ± 8.58
Wine	**68.15 ± 6.33**	52.59 ± 5.80	64.07 ± 8.03	50.74 ± 1.66	50.00 ± 0.00	58.64 ± 5.35	50.62 ± 1.07	50.00 ± 3.70	50.00 ± 0.00	62.96 ± 7.41	47.78 ± 4.97	49.26 ± 2.11
WordSynonyms	35.14 ± 2.03	30.72 ± 3.50	35.14 ± 2.03	35.86 ± 2.07	39.44 ± 4.12	51.78 ± 3.11	15.15 ± 0.48	51.10 ± 4.21	25.64 ± 4.61	49.01 ± 4.47	33.92 ± 1.82	**58.90 ± 2.04**
Worms	50.39 ± 8.29	**51.95 ± 5.35**	45.71 ± 7.71	23.38 ± 5.11	39.74 ± 4.27	23.38 ± 6.49	26.84 ± 3.97	27.27 ± 9.09	25.45 ± 3.74	25.97 ± 9.09	20.78 ± 5.35	48.57 ± 4.91
WormsTwoClass	**64.94 ± 5.51**	63.12 ± 4.37	61.30 ± 7.20	49.09 ± 4.63	47.27 ± 1.97	50.65 ± 6.49	48.05 ± 4.50	48.05 ± 4.68	48.05 ± 3.56	51.95 ± 4.50	46.49 ± 5.15	56.36 ± 3.85
Yoga	59.91 ± 2.90	**59.97 ± 4.98**	59.88 ± 3.11	55.72 ± 1.71	57.07 ± 1.49	58.79 ± 3.68	53.03 ± 3.90	59.02 ± 1.75	51.43 ± 2.65	59.87 ± 3.99	51.89 ± 2.29	57.73 ± 2.05
Avg. Acc.	66.24	64.93	67.18	55.43	62.72	62.79	46.04	61.94	51.50	63.13	46.88	**69.16**
Avg. Rank	4.83	5.03	4.55	8.00	6.23	5.75	9.87	5.73	8.87	5.64	9.58	**3.91**
#Best	25	15	28	6	5	7	3	5	1	8	1	**40**

Table 11: Key few-shot adaptation settings for ChronoMorph.

Component	Setting
Base LLM	`meta-llama/Llama-3.2-1B`
Vision backbone	`facebook/dinov2-base`
Morphology-map construction	Overlapping-window grid transform with per-sequence robust normalization
Patch length / stride	16 / 8
Hidden size	128
Default downstream checkpoint	Stage II (TSQA transfer)
Warm-up / joint epochs	5 / 55
Batch size / eval batch size	8 / 8
Gradient accumulation	4
LR (encoder / projector / LoRA)	2×10^{-4} / 1×10^{-4} / 1×10^{-4}
Language-side adaptation	LoRA enabled in Phase II ($r = 16$, $\alpha = 32$)
Tokenizer training mode	`class_rows`
Prediction rule	anonymous class-token scoring; semantic-prior class-token scoring when label text is meaningful
Freeze vision backbone	true
Data augmentation	none

B Additional Experimental Details

We summarize the key implementation choices of ChronoMorph for the few-shot benchmark in Table 11. These settings are fixed across the main benchmark reported in Section 4.

B.1 Pretraining Stages and Data Sources

Table 12 summarizes the three training stages discussed in the main paper and clarifies which stage provides the default downstream checkpoint.

After stage-wise pretraining, downstream few-shot adaptation uses only the sampled support labels of the target dataset. For UCR targets contained in the Stage I pool, unlabeled TRAIN sequences may already have contributed to the encoder pretraining.

B.2 Code Release and Assets

We will release an anonymized code package with the submission. The release includes the ChronoMorph implementation, the few-shot adaptation and evaluation scripts, baseline launchers, reporting scripts for the tables in this paper, environment files, and README instructions for preparing the public datasets used in our experiments. The code is released under the MIT license. The package does not redistribute raw third-party datasets or gated pretrained model weights; users should obtain those assets from the original providers and comply with their licenses and terms of use. The released scripts document which experiments are directly reproducible from the package and which require separately downloaded datasets or pretrained checkpoints.

B.3 Asset and License Table

Table 13 summarizes the main existing assets used in this paper. We list the original source, role, version when available, license or stated terms, and whether the asset is redistributed in our release. For assets whose official source does not state a separate dataset license, we cite the official repository or archive page and do not redistribute the raw data.

Table 13: Existing assets used in this paper and their licenses or stated terms. URLs were checked during manuscript preparation.

Asset	Role	Source / version	License or stated terms	Redistribution in our release
UCR Archive 2018	Main few-shot benchmark and unlabeled Stage I source pool	TSC website; Zenodo v1	Zenodo record lists Creative Commons Attribution 4.0 International (CC BY 4.0); archive should be cited via Dau et al.	Not redistributed; scripts expect users to download from the original archive or cited mirror.

Table 13: Existing assets used in this paper and their licenses or stated terms (continued).

Asset	Role	Source / version	License or stated terms	Redistribution in our release
TSQA / Time-MQA	Stage I raw series and Stage II time-series question answering	Hugging Face dataset card	Apache-2.0 according to the Hugging Face dataset metadata	Not redistributed; scripts load from the original dataset source.
M4 Competition dataset	Stage I raw series and optional captioning / semantic enrichment data	Official M4 repository	The official repository publicly provides the dataset and asks users to cite the M4 competition resources; no separate dataset LICENSE file was found in the repository.	Not redistributed; users download from the official repository.
MIT-BIH Arrhythmia Database	External semantic-label evaluation	PhysioNet v1.0.0	Open Data Commons Attribution License v1.0; PhysioNet citation required	Not redistributed; preprocessing scripts require a local PhysioNet download.
Sleep-EDF Database Expanded	External semantic-label evaluation	PhysioNet v1.0.0	Open Data Commons Attribution License v1.0; PhysioNet citation required	Not redistributed; preprocessing scripts require a local PhysioNet download.
PhysioNet/CinC 2017 AF dataset	External semantic-label evaluation	PhysioNet v1.0.0	Open Data Commons Attribution License v1.0; PhysioNet and challenge citations required	Not redistributed; preprocessing scripts require a local PhysioNet download.
Llama 3.2 1B	Pretrained causal LLM backbone	Hugging Face model card	Llama 3.2 Community License and Acceptable Use Policy	Not redistributed; users must obtain access from Meta / Hugging Face.
DINOv2-base	Frozen visual backbone	Hugging Face model card	Apache-2.0	Not redistributed; loaded from the original model repository.
OpenTSLM base code	Starting codebase and time-series language-model components	Project repository and release package	MIT license in the repository license file	Modified code is released in anonymized form under MIT, preserving applicable notices.
Python ML libraries, including PyTorch, Transformers, PEFT, NumPy, pandas, scikit-learn, and related dependencies	Training, adaptation, evaluation, and reporting	Version constraints are listed in the released environment files	Governed by their respective open-source licenses	Not vendored; installed by the environment files.

B.4 Computational Resource Reporting

All reported downstream experiments were run on a local server with six NVIDIA GeForce RTX 3090 GPUs (24GB each), two Intel Xeon Silver 4314 CPUs, and 251GiB system memory. Unless otherwise stated, each training job used one GPU, and the batch launchers executed up to six single-GPU workers in parallel. Table 14 summarizes the compute used by the main experiment groups. GPU-hours are summed over single-GPU jobs; wall-clock time can be lower when multiple workers run concurrently. The efficiency measurements in Appendix B.7 additionally report peak memory, train-step latency, inference latency, throughput, total parameters, and downstream-updated parameters on the representative runtime subset.

Retained launcher summaries also include failed or exploratory jobs. These runs account for less than 2 additional GPU-hours in the preserved logs, and dry-run checks add a negligible amount of compute.

Table 12: Stage-wise data sources and roles for ChronoMorph.

Stage	Training data	Validation / monitoring data	Role in the main UCR few-shot benchmark
Stage I	Raw TSQA-train series, raw M4-train series, and official TRAIN-split raw series from a 98-dataset UCR training pool.	Raw TSQA validation/test series and raw M4 validation/test series. No UCR official TEST split is used.	Learns dual-view representations before the LLM interface; corresponds to `stage0_encoder_ssl` in the implementation.
Stage II	TSQA train split only, with standard language-model training on time-series question answering.	TSQA validation/test splits.	Provides the default downstream checkpoint used by all main few-shot results in Section 4; corresponds to `stage1_tsqa_transfer`.
Stage III	Optional mixture of TSQA, M4 captioning, and synthetic attribute supervision derived from TSQA/M4 raw series.	Corresponding validation/test splits of the same mixture.	Used only for additional semantic enrichment or caption specialization; not used for the default main benchmark initialization unless explicitly stated. This corresponds to `stage2_semantic_bridge` and later caption specialization in code.

Table 14: Compute-resource budget for the reported experiment groups. Downstream jobs use single RTX 3090 workers unless noted otherwise.

Experiment group	Scope	Compute workers	Mean per run	Total GPU-hours	Notes
Stage I–II pretraining	Curriculum pretraining checkpoint used to initialize downstream adaptation	RTX 3090 workers	–	–	Uses the same local GPU server; stage-wise timing is recorded with the pretraining checkpoint logs.
Optional Stage III enrichment	Semantic-enrichment training used outside the default UCR results	RTX 3090 workers	–	–	Not used for the default UCR benchmark.
ChronoMorph main UCR benchmark	128 UCR datasets, 1/2/5/10-shot, five runs per setting; 2,507 completed downstream runs	6 single-GPU workers	0.67 GPU-h/run	1,677	Batch size 8, gradient accumulation 4, 60 downstream epochs.
UCR baseline runs	ResNet, TapNet, COSCO-ResNet, PatchTST, GPT4TS, and TSLib-family baselines under the same few-shot protocol	6 single-GPU workers	0.40–0.49 GPU-h/run	10,937	Includes the retained main UCR baseline logs; GPT4TS coverage is reported for the completed 60-dataset run.
External-dataset ChronoMorph runs	MIT-BIH, SleepEDF, and CinC2017 under 10/20/30-shot protocols	single-GPU workers	1.69 GPU-h/run	153	Same downstream adaptation recipe as the external few-shot experiments.
External-dataset baseline runs	Reported baseline families on MIT-BIH, SleepEDF, and CinC2017	single-GPU workers	0.68 GPU-h/job	22	Launcher jobs cover all three external datasets and baseline families.
Top-level ablations	No-pretraining, temporal-only, visual-only, class-token decoding, and no-LLM backbone ablations	6 single-GPU workers	1.67 GPU-h/run	3,895	Includes the two retained no-LLM linear runs added together.
Semantic-prior ablation	External semantic-label interface runs	single-GPU workers	1.69 GPU-h/run	31	Uses the same 10/20/30-shot external evaluation protocol.
Efficiency probe	Representative UCR runtime subset with parameter, memory, latency, and throughput measurements	one RTX 3090	–	–	Detailed runtime and memory values are reported in Tables 4, 17, and 19.

B.5 External Dataset Results

Table 15 reports the full 10-, 20-, and 30-shot results on MIT-BIH, SleepEDF, and CinC2017. All methods use the same external few-shot settings and reported run count; we use these datasets as transfer checks beyond the primary UCR benchmark rather than as a claim of uniform dominance across external domains.

B.6 Semantic-Prior Decision Ablation

Table 16 expands the semantic-prior comparison summarized in Table 6. Both variants use one-pass class-token scoring; the semantic-prior variant additionally initializes and regularizes class-token

Table 15: Few-shot classification accuracy (%) on three external datasets. Each cell reports mean ± standard deviation over 10 runs under the same few-shot protocol. Best is bold and second-best is underlined within each dataset-shot column.

Model	MIT-BIH			SleepEDF			CinC2017		
	10-shot	20-shot	30-shot	10-shot	20-shot	30-shot	10-shot	20-shot	30-shot
ChronoMorph	55.03 ± 9.40	**63.95 ± 13.80**	**64.15 ± 5.32**	37.98 ± 3.69	48.80 ± 2.31	54.69 ± 3.67	**32.80 ± 9.08**	**36.98 ± 4.27**	34.13 ± 9.73
ResNet	37.97 ± 14.13	55.66 ± 8.58	57.07 ± 9.05	**60.27 ± 7.78**	**70.59 ± 4.14**	69.64 ± 3.64	30.95 ± 12.41	33.73 ± 7.43	**36.20 ± 4.84**
TapNet	31.94 ± 7.18	40.15 ± 10.28	44.11 ± 11.14	58.75 ± 9.67	70.50 ± 5.64	**70.86 ± 5.23**	30.05 ± 6.73	32.80 ± 8.81	31.01 ± 11.54
COSCO-ResNet	32.33 ± 14.25	35.08 ± 13.75	41.17 ± 8.94	57.86 ± 11.29	63.04 ± 8.19	67.32 ± 3.13	31.41 ± 4.48	31.04 ± 6.97	31.98 ± 8.37
PatchTST	38.94 ± 14.35	55.36 ± 8.84	57.08 ± 9.40	25.21 ± 11.89	29.47 ± 9.76	32.08 ± 8.05	26.07 ± 6.24	30.35 ± 7.51	31.21 ± 3.36
OneFitsAll	32.92 ± 16.63	42.48 ± 10.82	54.87 ± 10.04	26.98 ± 3.35	27.57 ± 3.81	26.60 ± 3.59	30.80 ± 6.07	27.56 ± 4.33	29.23 ± 3.03
TimesNet	51.01 ± 5.84	54.21 ± 7.81	56.15 ± 8.77	21.11 ± 2.91	23.42 ± 2.34	25.17 ± 1.82	27.25 ± 2.81	28.41 ± 4.24	29.43 ± 2.44
DLinear	55.57 ± 9.85	57.45 ± 7.95	58.84 ± 8.69	19.57 ± 1.32	19.94 ± 1.73	19.92 ± 1.69	24.92 ± 1.16	25.17 ± 1.22	24.53 ± 1.70
Autoformer	38.05 ± 5.65	42.49 ± 8.58	44.39 ± 7.25	26.25 ± 7.52	24.55 ± 4.24	28.64 ± 8.05	28.37 ± 5.44	27.67 ± 2.76	26.30 ± 4.36
Crossformer	53.48 ± 9.36	58.55 ± 10.43	58.24 ± 9.71	23.75 ± 1.74	24.53 ± 3.34	26.49 ± 3.02	24.02 ± 6.10	25.50 ± 2.86	27.35 ± 2.35
FEDformer	**58.88 ± 5.77**	56.80 ± 9.98	55.12 ± 10.09	15.15 ± 8.43	24.73 ± 6.55	25.47 ± 5.56	20.35 ± 8.32	26.90 ± 3.54	27.91 ± 4.87
Informer	53.19 ± 6.03	54.20 ± 7.61	52.35 ± 10.80	20.35 ± 12.77	22.18 ± 5.40	24.81 ± 8.94	26.93 ± 9.30	28.14 ± 4.01	30.12 ± 5.95

Table 16: Semantic-prior class-token ablation on three external datasets (accuracy, %). Each cell reports mean ± standard deviation over 10 runs under the same few-shot protocol. The better variant within each dataset-shot column is bold. Δ denotes Semantic prior minus Anonymous labels.

Variant	MIT-BIH			SleepEDF			CinC2017		
	10-shot	20-shot	30-shot	10-shot	20-shot	30-shot	10-shot	20-shot	30-shot
Anonymous labels	55.03 ± 9.40	63.95 ± 13.80	64.15 ± 5.32	37.98 ± 3.69	48.80 ± 2.31	54.69 ± 3.67	32.80 ± 9.08	**36.98 ± 4.27**	34.13 ± 9.73
Semantic prior	**57.05 ± 10.96**	**66.62 ± 6.54**	**68.87 ± 11.23**	**42.56 ± 3.24**	**52.69 ± 5.05**	**57.24 ± 4.34**	**34.72 ± 3.15**	35.24 ± 1.24	**38.86 ± 7.00**
Semantic prior − Anonymous	+2.02	+2.67	+4.72	+4.58	+3.89	+2.55	+1.92	-1.74	+4.73

rows from label-verbalizer prototypes. The semantic prior improves the averaged performance on all three datasets, while individual shot-level deltas remain dataset-dependent.

B.7 Efficiency Summary

Table 17 reports computational efficiency and adaptation cost under the 10-shot UCR protocol. Runtime and memory are measured on the predefined representative UCR subset with batch size 8 on a single RTX 3090, while parameter counts are reported separately for total deployed parameters and downstream-updated parameters.

B.8 Baseline Training Recipes and Hyperparameters

Table 18 summarizes the training rules used by the reported baseline families. The goal of this table is to make the few-shot comparison protocol explicit: all methods share the same sampled support/query split, but they do not all use the same checkpoint-selection rule or the same epoch schedule.

B.9 Computational Efficiency Probe Details

The efficiency probe is implemented by the repository's `efficiency_probe.py` script. It uses the same model families as the main benchmark and omits ChronoMorph-only ablations. The default runtime subset contains COFFEE, GUNPOINT, ECG5000, ELECTRICDEVICES, CROP, ACSF1, PHONEME, and NONINVASIVEFETALECGTHORAX1, covering short, medium, and long UCR series as well as different class counts. The script writes the LaTeX tables, CSV summary, and accuracy-versus-updated-parameters plot to the efficiency-report directory.

C Additional Plot

Figure 2 visualizes the shot-scaling trend over the 128 UCR datasets. ChronoMorph remains the strongest method on the averaged benchmark curve across the entire few-shot range, with the largest absolute advantage appearing at 10-shot.

Table 17: Computational efficiency and adaptation cost under the 10-shot UCR protocol. Total parameters count the full deployed model, while updated parameters count only the parameters optimized during downstream adaptation. Adapter size estimates the per-dataset trainable-state footprint. Runtime is measured on representative UCR datasets with batch size 8 on a single RTX 3090.

Model	Total params	Updated params	Updated %	Adapter MB	Peak GB	Train ms/step	Infer ms/ex.	Infer ex./s
ChronoMorph	1.28B	12.3M	0.96	46.7	13.75	1207.8	156.85	6.4
ResNet	629.1K	629.1K	100.00	2.4	0.33	13.2	0.26	4273.3
TapNet	912.1K	912.1K	100.00	3.5	0.22	15.0	0.47	3345.0
COSCO-ResNet	662.1K	662.1K	100.00	2.5	0.58	50.8	0.25	4315.1
PatchTST	665.4K	598.3K	90.80	2.3	1.54	32.6	0.75	2149.8
DLinear	1.2M	1.2M	100.00	4.6	0.04	2.7	0.07	14389.5
TimesNet	2.5M	2.5M	100.00	9.7	0.14	66.7	2.46	614.5
Autoformer	1.2M	1.2M	100.00	4.5	0.40	23.1	11.10	91.3
Crossformer	2.8M	2.8M	100.00	10.8	0.06	24.9	0.59	1709.6
FEDformer	1.8M	1.8M	100.00	6.7	0.26	244.1	5.17	201.9
Informer	1.4M	1.4M	100.00	5.3	0.22	18.0	0.62	1690.1
GPT4TS	82.2M	1.1M	1.34	4.2	0.56	22.6	0.86	1181.2

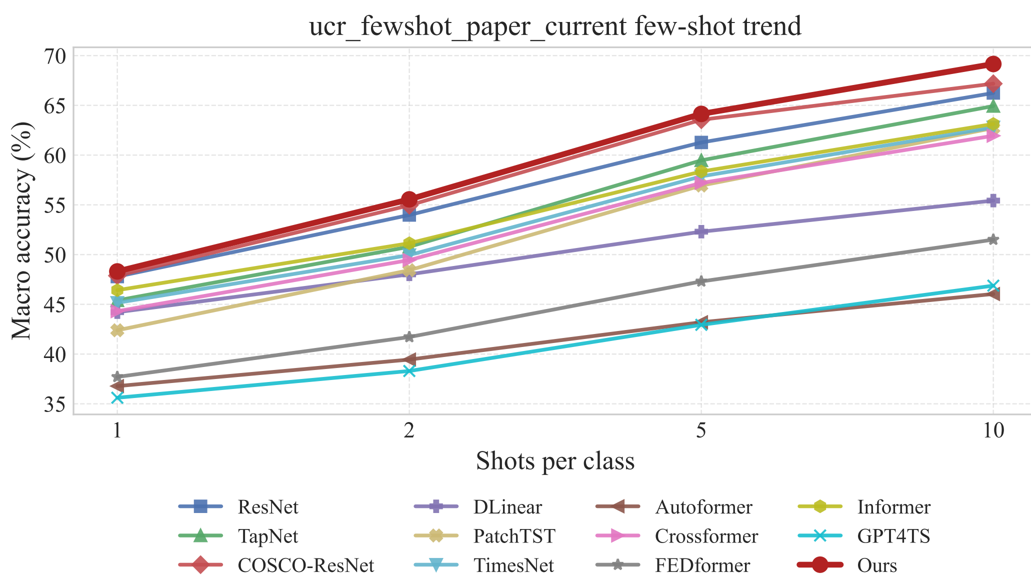

Figure 2: Shot-scaling trend on the 128 UCR datasets. Each point reports the mean accuracy over five runs. ChronoMorph remains the strongest method across all four shot settings.

Table 18: Few-shot training recipes for ChronoMorph and the reported baseline families. All methods use the same sampled support sets within each run and evaluate on all official TEST examples from the target dataset.

Method / family	Optimizer	Epochs	Batch / LR	Model selection rule	Val	ES	Few-shot notes
ChronoMorph	AdamW	5 warm-up + 55 joint	8 / 2×10^{-4} (encoder), 1×10^{-4} (projector, LoRA)	Report the last Phase II checkpoint	No	No	Class-token rows remain trainable; language-side adaptation uses LoRA in Phase II.
ResNet, TapNet	AdamW	Fixed schedule in the few-shot runner (60 in the benchmark configs)	model-specific / benchmark configs use 10^{-4}	Best support-set accuracy, tie-broken by support loss	No	No	Trained only on the sampled support set; the support set doubles as the model-selection set.
COSCO-ResNet	SAM with SGD base optimizer + prototypical loss	Fixed schedule (100 in the benchmark configs)	128 / 10^{-2} by default	Final trained model; support centroids computed after training	No	No	Query predictions are made by distances to support centroids rather than a learned classifier head.
PatchTST	AdamW + linear warmup	Two-phase fixed schedule (head warm-up + joint fine-tuning; 60 total epochs in the benchmark configs)	16 / separate backbone-head LRs	Report the last Phase II checkpoint	No	No	Uses the same support/query split and full dataset class set as the main protocol.
TSLib family (DLINEAR, AUTOFORMER, CROSSFORMER, FEDFORMER, INFORMER, TIMESNET)	RAdam + cosine schedule	Fixed model-specific schedule from the TSLib profile or benchmark override	usually 16 / 10^{-3}	Report the last checkpoint	No	No	Single-stage training; evaluation masks logits to the target dataset's class set. TimesNet uses a shorter profile than the other TSLib baselines.
GPT4TS	RAdam (default runner setting)	Fixed schedule (60 in the benchmark configs)	64 / 10^{-4} in the benchmark configs	Report the last checkpoint	No	No	Implemented with the One-Fits-All classification runner under the same sampled support/query protocol.

Table 19: Available per-dataset runtime details for the computational-efficiency probe. The full machine-readable runtime file is produced by the efficiency script; rows with missing measurements are omitted from the paper appendix.

Dataset	Model	Peak GB	Train ms/step	Infer ms/ex.	Infer ex./s
ACSF1	ChronoMorph	13.75	1262.4	173.87	5.8
ACSF1	ResNet	1.01	26.0	0.32	3167.0
ACSF1	TapNet	0.59	33.7	1.45	688.2
ACSF1	COSCO-ResNet	1.64	119.2	0.43	2303.7
ACSF1	PatchTST	6.50	107.6	2.23	447.7
ACSF1	DLinear	0.10	4.5	0.07	13898.9
ACSF1	TimesNet	0.27	121.8	4.54	220.5
ACSF1	Autoformer	1.19	40.5	12.53	79.8
ACSF1	Crossformer	0.12	25.0	0.58	1723.7
ACSF1	FEDformer	0.61	277.0	7.03	142.3
ACSF1	Informer	0.56	23.4	1.09	917.6
ACSF1	GPT4TS	1.00	29.1	0.89	1121.4

Table 20: Ablation of aggregate curriculum pretraining on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. Each value is mean accuracy (%) over five runs, and Drop denotes the average accuracy decrease relative to the full ChronoMorph model.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
w/o Curriculum Pretraining	44.70	51.81	60.65	66.60	55.94	3.35
ChronoMorph	**48.31**	**55.55**	**64.14**	**69.16**	**59.29**	–

D Ablation Tables

Table 5 in the main paper reports the compact ablation summary. Tables 20–22 expand that summary into three focused views under the same few-shot protocol. Table 23 summarizes the LLM-backbone ablation, and Table 24 reports its full per-dataset comparison on the 120 UCR datasets covered by both runs. Because UCR class identifiers are anonymous, the label-space ablation should be interpreted as testing closed-set class-token scoring; the semantic-prior class-token ablation is reported on external datasets with meaningful class names in Appendix B.6.

Table 21: Ablation of dual-view time-series-to-language modeling on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. These comparisons assess the effect of removing one top-level view at a time. Each value is mean accuracy (%) over five runs, and Drop denotes the average accuracy decrease relative to the full ChronoMorph model.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
Temporal only	43.51	50.55	58.40	64.39	54.21	5.08
Visual only	45.88	50.02	60.86	63.58	55.09	4.20
ChronoMorph	**48.31**	**55.55**	**64.14**	**69.16**	**59.29**	–

Table 22: Ablation of closed-set class-token label-space scoring on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. Each value is mean accuracy (%) over five runs, and Drop denotes the average accuracy decrease relative to the full ChronoMorph model. Since UCR labels are anonymous, this ablation does not evaluate semantic label priors.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
w/o Class-Token Label-Space Scoring	44.46	50.44	60.08	65.59	55.14	4.15
ChronoMorph	**48.31**	**55.55**	**64.14**	**69.16**	**59.29**	–

Table 23: Ablation of the LLM backbone on the 120 shared UCR datasets covered by both runs. Each value is benchmark-level mean accuracy (%) over the available runs recorded in the result files. Drop denotes the average accuracy decrease relative to full ChronoMorph.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
ChronoMorph	**47.93**	**54.98**	**63.27**	**68.48**	**58.67**	–
w/o LLM Backbone	42.95	48.20	57.06	61.01	52.31	6.36

Table 24: Per-dataset LLM-backbone ablation on 120 shared UCR datasets. Each value is mean ± standard deviation accuracy (%) over the available runs recorded in the result files.

Dataset	ChronoMorph				w/o LLM Backbone			
	1-shot	2-shot	5-shot	10-shot	1-shot	2-shot	5-shot	10-shot
ACSF1	41.40 ± 10.01	54.60 ± 3.58	62.80 ± 4.32	71.80 ± 2.17	36.00 ± 10.37	42.80 ± 3.96	51.60 ± 5.46	62.60 ± 3.44
Adiac	45.17 ± 5.28	56.01 ± 1.25	69.51 ± 2.42	76.42 ± 1.68	26.85 ± 1.68	21.53 ± 2.11	36.37 ± 4.24	33.66 ± 2.69
AllGestureWiimoteX	19.91 ± 1.48	26.51 ± 4.36	39.40 ± 2.86	52.31 ± 2.26	20.23 ± 4.10	23.34 ± 2.88	31.00 ± 1.71	35.34 ± 1.51
AllGestureWiimoteY	25.54 ± 4.42	29.43 ± 1.69	48.11 ± 2.89	56.34 ± 3.54	23.11 ± 3.11	26.46 ± 2.53	38.97 ± 1.87	44.00 ± 2.17
AllGestureWiimoteZ	22.26 ± 3.96	27.91 ± 5.21	36.14 ± 0.97	44.43 ± 2.91	22.26 ± 4.37	25.74 ± 5.60	31.91 ± 0.57	36.06 ± 1.75
ArrowHead	48.80 ± 4.79	57.03 ± 2.81	73.60 ± 3.71	81.49 ± 3.22	44.57 ± 6.72	47.20 ± 4.58	61.03 ± 6.78	58.74 ± 3.82
BME	66.40 ± 3.88	79.47 ± 11.40	97.73 ± 1.53	99.20 ± 0.30	66.80 ± 2.92	74.53 ± 11.00	93.47 ± 5.34	93.60 ± 0.37
Beef	30.00 ± 8.16	43.33 ± 5.27	51.33 ± 6.91	68.00 ± 1.83	31.33 ± 8.69	38.67 ± 6.06	36.67 ± 7.07	48.00 ± 3.80
BeetleFly	55.00 ± 14.58	61.00 ± 12.45	57.00 ± 11.51	59.00 ± 4.18	56.00 ± 5.48	58.00 ± 4.47	51.00 ± 9.62	59.00 ± 4.18
BirdChicken	68.00 ± 14.83	69.00 ± 22.19	85.00 ± 11.73	89.00 ± 5.48	74.00 ± 12.94	76.00 ± 9.62	86.00 ± 8.94	88.00 ± 5.70
CBF	67.51 ± 4.85	85.91 ± 4.23	93.76 ± 3.90	97.71 ± 0.69	69.18 ± 4.62	78.13 ± 11.07	87.40 ± 4.28	90.04 ± 1.83
Car	48.00 ± 5.06	58.67 ± 3.80	69.67 ± 3.98	76.33 ± 5.45	43.33 ± 12.86	53.00 ± 7.21	63.33 ± 8.50	66.33 ± 3.80
Chinatown	67.17 ± 13.59	85.31 ± 4.44	82.45 ± 9.21	88.05 ± 3.11	57.26 ± 10.98	72.36 ± 12.11	78.02 ± 7.25	75.39 ± 3.06
ChlorineConcentration	34.73 ± 3.81	34.25 ± 5.73	36.45 ± 2.05	33.98 ± 7.65	34.41 ± 3.06	34.99 ± 5.96	36.05 ± 1.89	36.44 ± 3.56
CinCECGTorso	32.32 ± 4.12	44.68 ± 5.88	49.61 ± 6.52	71.09 ± 3.22	31.30 ± 5.37	40.86 ± 7.99	37.54 ± 6.84	49.80 ± 0.39
Coffee	75.71 ± 13.69	88.57 ± 4.66	94.29 ± 1.96	95.71 ± 2.99	68.57 ± 11.68	82.14 ± 15.15	72.86 ± 20.45	87.86 ± 9.65
Computers	46.00 ± 10.27	52.72 ± 10.03	55.92 ± 7.37	59.36 ± 8.46	53.20 ± 7.38	58.96 ± 7.32	58.00 ± 7.81	64.00 ± 4.03
CricketX	26.26 ± 5.29	37.33 ± 3.78	45.74 ± 2.60	56.77 ± 3.51	25.59 ± 5.37	31.90 ± 2.09	38.97 ± 3.38	47.74 ± 2.29
CricketY	25.54 ± 2.45	34.51 ± 6.10	45.64 ± 1.99	54.31 ± 1.68	25.13 ± 3.56	33.28 ± 2.21	36.62 ± 1.65	41.95 ± 1.59
CricketZ	29.28 ± 4.82	33.59 ± 5.51	49.59 ± 2.62	57.38 ± 4.03	28.36 ± 1.29	29.44 ± 4.90	42.87 ± 1.47	46.62 ± 1.96
DiatomSizeReduction	85.82 ± 8.55	90.39 ± 5.62	93.66 ± 1.97	92.61 ± 2.00	86.73 ± 8.56	85.62 ± 7.19	77.78 ± 3.00	75.23 ± 1.32
DistalPhalanxOutlineAgeGroup	40.72 ± 17.98	67.77 ± 3.39	69.35 ± 1.20	70.65 ± 2.18	43.31 ± 18.16	70.22 ± 4.60	70.65 ± 1.56	71.80 ± 1.79
DistalPhalanxOutlineCorrect	53.91 ± 13.35	55.80 ± 14.95	63.77 ± 11.63	65.51 ± 11.85	60.58 ± 6.43	51.23 ± 12.26	64.49 ± 7.76	59.06 ± 13.65
DistalPhalanxTW	54.53 ± 1.72	51.37 ± 8.97	56.40 ± 4.87	54.10 ± 3.11	51.51 ± 3.83	47.48 ± 13.90	57.27 ± 3.58	55.11 ± 1.40
DodgerLoopGame	56.96 ± 11.69	61.01 ± 13.03	72.61 ± 10.25	83.62 ± 0.65	50.29 ± 8.64	54.93 ± 10.24	60.58 ± 4.46	73.19 ± 1.36
DodgerLoopWeekend	85.36 ± 4.30	84.64 ± 2.73	93.62 ± 2.37	97.83 ± 0.00	87.68 ± 3.80	84.35 ± 2.83	88.26 ± 2.14	92.32 ± 0.40
ECG200	60.60 ± 6.91	73.00 ± 3.46	67.40 ± 8.91	77.40 ± 4.88	62.80 ± 12.30	67.80 ± 5.22	68.60 ± 2.97	77.60 ± 2.70
ECG5000	67.18 ± 10.57	67.68 ± 9.76	73.89 ± 8.21	73.78 ± 6.18	64.82 ± 0.00	39.33 ± 0.00	85.18 ± 0.00	80.11 ± 0.00
ECGFiveDays	61.77 ± 6.61	59.35 ± 9.08	70.96 ± 3.91	75.28 ± 1.71	58.07 ± 0.00	71.31 ± 0.00	70.85 ± 0.00	76.42 ± 0.00
EOGHorizontalSignal	29.06 ± 4.16	37.35 ± 4.07	46.35 ± 2.27	49.17 ± 1.62	31.22 ± 0.00	33.15 ± 0.00	45.30 ± 0.00	44.20 ± 0.00
EOGVerticalSignal	27.35 ± 1.90	31.82 ± 3.47	36.74 ± 2.24	42.21 ± 2.26	22.10 ± 0.00	33.70 ± 0.00	34.53 ± 0.00	38.95 ± 0.00
Earthquakes	55.54 ± 11.57	55.25 ± 14.33	54.68 ± 9.31	54.68 ± 7.63	50.36 ± 0.00	61.87 ± 0.00	39.57 ± 0.00	42.45 ± 0.00
ElectricDevices	33.14 ± 10.96	47.14 ± 3.99	47.06 ± 4.18	49.42 ± 5.02	17.31 ± 0.00	46.54 ± 0.00	47.30 ± 0.00	47.89 ± 0.00
EthanolLevel	26.68 ± 1.86	27.00 ± 3.11	25.92 ± 1.40	28.24 ± 0.50	26.20 ± 0.00	27.20 ± 0.00	27.00 ± 0.00	24.00 ± 0.00
FaceAll	32.84 ± 6.86	39.09 ± 1.28	54.54 ± 1.26	63.01 ± 1.34	32.49 ± 0.00	28.58 ± 0.00	40.30 ± 0.00	44.02 ± 0.00
FaceFour	54.77 ± 8.25	64.55 ± 4.91	75.68 ± 4.37	79.09 ± 1.90	46.59 ± 0.00	53.41 ± 0.00	65.91 ± 0.00	73.86 ± 0.00
FacesUCR	50.37 ± 2.98	55.08 ± 2.32	68.58 ± 2.57	79.08 ± 1.41	46.83 ± 0.00	46.68 ± 0.00	54.83 ± 0.00	58.29 ± 0.00
FiftyWords	36.48 ± 2.66	47.30 ± 2.89	61.58 ± 2.55	71.34 ± 2.10	31.87 ± 0.00	33.41 ± 0.00	45.05 ± 0.00	54.51 ± 0.00
Fish	34.74 ± 4.91	48.91 ± 10.25	64.57 ± 5.92	78.29 ± 3.92	36.00 ± 0.00	34.29 ± 0.00	54.86 ± 0.00	61.71 ± 0.00
FordA	50.71 ± 0.98	53.35 ± 2.84	54.00 ± 2.22	57.15 ± 1.85	51.44 ± 0.00	51.36 ± 0.00	50.61 ± 0.00	48.26 ± 0.00
FordB	49.75 ± 2.56	50.42 ± 2.72	52.32 ± 2.78	55.60 ± 1.91	52.84 ± 0.00	49.75 ± 0.00	51.60 ± 0.00	52.84 ± 0.00
FreezerRegularTrain	61.90 ± 22.25	79.14 ± 7.94	85.11 ± 3.83	89.20 ± 5.65	76.35 ± 0.00	69.68 ± 0.00	81.09 ± 0.00	78.95 ± 0.00
FreezerSmallTrain	66.53 ± 10.64	64.30 ± 13.28	75.05 ± 3.10	82.86 ± 3.49	67.30 ± 0.00	75.09 ± 0.00	77.82 ± 0.00	82.63 ± 0.00

Table 24: Per-dataset LLM-backbone ablation on 120 shared UCR datasets (continued).

Dataset	ChronoMorph				w/o LLM Backbone			
	1-shot	2-shot	5-shot	10-shot	1-shot	2-shot	5-shot	10-shot
Fungi	79.35 ± 5.30	80.97 ± 3.24	81.18 ± 3.16	79.57 ± 1.20	40.32 ± 0.00	37.10 ± 0.00	40.86 ± 0.00	51.61 ± 0.00
GestureMidAirD1	32.00 ± 3.94	44.92 ± 6.31	60.31 ± 2.35	67.08 ± 1.00	22.31 ± 0.00	36.92 ± 0.00	44.62 ± 0.00	50.00 ± 0.00
GestureMidAirD2	33.85 ± 3.03	43.85 ± 4.03	54.62 ± 1.96	55.69 ± 1.85	25.38 ± 0.00	34.62 ± 0.00	44.62 ± 0.00	42.31 ± 0.00
GestureMidAirD3	18.46 ± 2.61	21.08 ± 1.85	32.15 ± 3.33	33.69 ± 2.63	15.38 ± 0.00	20.00 ± 0.00	20.00 ± 0.00	23.08 ± 0.00
GesturePebbleZ1	35.81 ± 4.48	37.91 ± 6.35	63.84 ± 8.99	75.12 ± 5.19	40.70 ± 0.00	36.05 ± 0.00	49.42 ± 0.00	64.53 ± 0.00
GesturePebbleZ2	33.29 ± 5.85	41.39 ± 4.24	60.89 ± 8.06	75.57 ± 4.64	24.68 ± 0.00	47.47 ± 0.00	54.43 ± 0.00	67.72 ± 0.00
GunPoint	58.00 ± 7.79	79.33 ± 16.25	90.13 ± 10.49	97.47 ± 2.23	60.67 ± 0.00	60.67 ± 0.00	85.33 ± 0.00	90.00 ± 0.00
GunPointAgeSpan	60.51 ± 10.32	64.49 ± 7.90	81.52 ± 9.45	92.72 ± 5.80	61.71 ± 0.00	52.22 ± 0.00	71.52 ± 0.00	71.84 ± 0.00
GunPointMaleVersusFemale	81.01 ± 10.12	87.66 ± 5.66	94.62 ± 6.24	98.54 ± 0.94	64.56 ± 0.00	77.53 ± 0.00	98.10 ± 0.00	97.15 ± 0.00
GunPointOldVersusYoung	57.97 ± 10.78	70.92 ± 5.05	80.44 ± 8.63	86.79 ± 4.71	54.29 ± 0.00	71.11 ± 0.00	76.83 ± 0.00	80.32 ± 0.00
Ham	52.76 ± 9.30	50.86 ± 8.07	60.57 ± 9.91	62.86 ± 3.37	61.90 ± 0.00	64.76 ± 0.00	65.71 ± 0.00	58.10 ± 0.00
HandOutlines	61.73 ± 6.31	64.59 ± 10.14	68.54 ± 6.21	71.51 ± 5.35	62.43 ± 0.00	66.49 ± 0.00	72.16 ± 0.00	55.68 ± 0.00
Haptics	25.65 ± 4.99	26.62 ± 4.64	26.95 ± 4.88	39.22 ± 4.29	19.81 ± 0.00	27.92 ± 0.00	32.79 ± 0.00	33.77 ± 0.00
Herring	50.00 ± 10.00	55.00 ± 4.34	45.00 ± 4.87	55.31 ± 5.01	53.12 ± 0.00	59.38 ± 0.00	45.31 ± 0.00	62.50 ± 0.00
HouseTwenty	62.18 ± 8.04	71.93 ± 10.30	84.37 ± 7.51	89.24 ± 2.81	65.55 ± 0.00	56.30 ± 0.00	62.18 ± 0.00	85.71 ± 0.00
InlineSkate	17.13 ± 2.24	18.98 ± 2.54	28.07 ± 1.19	33.49 ± 2.77	13.45 ± 0.00	19.64 ± 0.00	23.27 ± 0.00	25.45 ± 0.00
InsectEPGRegularTrain	55.34 ± 14.47	68.76 ± 9.60	80.24 ± 3.80	86.59 ± 4.06	54.22 ± 0.00	57.03 ± 0.00	75.90 ± 0.00	77.51 ± 0.00
InsectEPGSmallTrain	50.76 ± 7.86	67.79 ± 2.45	83.05 ± 2.80	85.06 ± 2.08	34.14 ± 0.00	67.47 ± 0.00	71.49 ± 0.00	77.51 ± 0.00
InsectWingbeatSound	25.27 ± 2.20	31.49 ± 3.72	41.44 ± 2.25	51.36 ± 2.05	25.86 ± 0.00	30.00 ± 0.00	35.30 ± 0.00	38.43 ± 0.00
ItalyPowerDemand	58.52 ± 15.27	65.03 ± 13.70	84.26 ± 5.94	92.19 ± 2.62	86.98 ± 0.00	53.35 ± 0.00	85.33 ± 0.00	92.03 ± 0.00
LargeKitchenAppliances	45.44 ± 5.60	42.56 ± 5.00	50.13 ± 3.07	55.09 ± 5.30	44.53 ± 0.00	43.20 ± 0.00	50.93 ± 0.00	50.67 ± 0.00
Lightning2	59.67 ± 8.87	63.28 ± 8.25	69.51 ± 10.47	77.05 ± 4.18	55.74 ± 0.00	63.93 ± 0.00	62.30 ± 0.00	68.85 ± 0.00
Lightning7	41.64 ± 7.60	46.03 ± 6.17	66.30 ± 3.83	66.85 ± 7.34	47.95 ± 0.00	58.90 ± 0.00	63.01 ± 0.00	69.86 ± 0.00
Mallat	68.24 ± 7.54	80.67 ± 4.06	90.88 ± 1.15	89.72 ± 1.50	58.68 ± 0.00	70.02 ± 0.00	66.10 ± 0.00	60.00 ± 0.00
Meat	55.67 ± 9.62	57.67 ± 17.30	64.00 ± 11.82	64.00 ± 6.08	51.67 ± 0.00	56.67 ± 0.00	75.00 ± 0.00	73.33 ± 0.00
MedicalImages	27.66 ± 4.41	31.63 ± 5.59	42.03 ± 2.75	54.68 ± 2.10	25.53 ± 0.00	35.39 ± 0.00	38.68 ± 0.00	53.29 ± 0.00
MelbournePedestrian	39.16 ± 5.22	49.78 ± 4.68	65.81 ± 3.77	70.88 ± 1.49	37.15 ± 0.00	41.94 ± 0.00	52.85 ± 0.00	65.23 ± 0.00
MiddlePhalanxOutlineAgeGroup	39.48 ± 7.40	42.99 ± 12.95	49.74 ± 4.72	51.04 ± 7.11	43.51 ± 0.00	31.17 ± 0.00	60.39 ± 0.00	52.60 ± 0.00
MiddlePhalanxOutlineCorrect	51.68 ± 5.00	48.87 ± 3.56	48.45 ± 7.93	59.18 ± 5.65	41.92 ± 0.00	54.98 ± 0.00	62.20 ± 0.00	64.26 ± 0.00
MiddlePhalanxTW	37.14 ± 16.84	44.81 ± 7.78	43.64 ± 5.71	44.42 ± 6.27	27.27 ± 0.00	46.75 ± 0.00	22.08 ± 0.00	48.05 ± 0.00
MixedShapesRegularTrain	56.06 ± 5.34	67.98 ± 4.06	76.68 ± 5.11	79.60 ± 1.74	48.62 ± 0.00	69.44 ± 0.00	68.49 ± 0.00	74.68 ± 0.00
MixedShapesSmallTrain	63.30 ± 3.34	68.35 ± 3.26	78.71 ± 1.70	81.61 ± 2.47	64.54 ± 0.00	65.81 ± 0.00	61.90 ± 0.00	71.88 ± 0.00
MoteStrain	54.36 ± 12.77	58.56 ± 15.12	80.37 ± 4.50	89.23 ± 1.53	68.61 ± 0.00	50.00 ± 0.00	82.27 ± 0.00	83.39 ± 0.00
NonInvasiveFetalECGThorax1	39.16 ± 2.28	50.58 ± 2.13	67.42 ± 0.56	76.84 ± 1.40	24.53 ± 0.00	36.49 ± 0.00	46.41 ± 0.00	51.30 ± 0.00
NonInvasiveFetalECGThorax2	47.48 ± 2.06	59.72 ± 2.76	75.31 ± 0.82	82.53 ± 1.33	39.64 ± 0.00	46.87 ± 0.00	55.73 ± 0.00	66.77 ± 0.00
OSULeaf	29.83 ± 5.82	36.28 ± 6.97	49.92 ± 6.46	59.42 ± 3.58	33.47 ± 0.00	29.75 ± 0.00	43.80 ± 0.00	49.17 ± 0.00
OliveOil	28.00 ± 14.26	23.33 ± 9.72	28.67 ± 11.69	40.00 ± 0.00	16.67 ± 0.00	40.00 ± 0.00	16.67 ± 0.00	40.00 ± 0.00
PLAID	32.10 ± 5.72	40.19 ± 5.06	52.96 ± 5.39	59.89 ± 3.28	21.60 ± 0.00	36.13 ± 0.00	53.63 ± 0.00	58.66 ± 0.00
PhalangesOutlinesCorrect	51.93 ± 9.54	50.84 ± 2.93	53.61 ± 8.69	55.73 ± 8.07	61.89 ± 0.00	54.55 ± 0.00	36.60 ± 0.00	65.27 ± 0.00
Phoneme	6.87 ± 1.15	10.02 ± 0.59	15.63 ± 1.56	21.95 ± 1.27	7.12 ± 0.00	10.50 ± 0.00	15.82 ± 0.00	17.25 ± 0.00
PickupGestureWiimoteZ	47.20 ± 7.29	58.80 ± 3.03	68.00 ± 2.45	70.40 ± 2.61	26.00 ± 0.00	54.00 ± 0.00	60.00 ± 0.00	52.00 ± 0.00
PigAirwayPressure	5.48 ± 2.19	9.81 ± 0.43	9.81 ± 0.73	9.62 ± 0.48	5.77 ± 0.00	7.21 ± 0.00	7.69 ± 0.00	4.81 ± 0.00
PigArtPressure	15.38 ± 4.07	30.67 ± 1.38	33.08 ± 1.33	31.73 ± 1.92	17.79 ± 0.00	25.00 ± 0.00	23.56 ± 0.00	24.52 ± 0.00
PigCVP	11.44 ± 2.08	19.33 ± 0.99	19.52 ± 0.55	19.52 ± 1.34	10.58 ± 0.00	16.83 ± 0.00	15.87 ± 0.00	18.27 ± 0.00

Table 24: Per-dataset LLM-backbone ablation on 120 shared UCR datasets (continued).

Dataset	ChronoMorph				w/o LLM Backbone			
	1-shot	2-shot	5-shot	10-shot	1-shot	2-shot	5-shot	10-shot
Plane	93.90 ± 6.92	99.24 ± 0.80	99.24 ± 1.24	99.24 ± 1.24	91.43 ± 0.00	90.48 ± 0.00	100.00 ± 0.00	99.05 ± 0.00
PowerCons	60.89 ± 11.09	66.78 ± 15.52	76.11 ± 6.55	82.33 ± 5.46	77.78 ± 0.00	48.89 ± 0.00	52.22 ± 0.00	78.33 ± 0.00
ProximalPhalanxOutlineAgeGroup	48.98 ± 23.27	66.24 ± 17.89	76.88 ± 5.79	79.80 ± 3.30	78.54 ± 0.00	81.46 ± 0.00	83.90 ± 0.00	76.10 ± 0.00
ProximalPhalanxOutlineCorrect	52.23 ± 12.03	59.93 ± 12.82	65.84 ± 4.70	60.07 ± 4.20	59.79 ± 0.00	64.26 ± 0.00	62.89 ± 0.00	62.89 ± 0.00
ProximalPhalanxTW	55.22 ± 6.63	57.76 ± 7.59	66.24 ± 3.19	69.66 ± 3.47	60.98 ± 0.00	39.02 ± 0.00	73.17 ± 0.00	45.37 ± 0.00
RefrigerationDevices	29.76 ± 11.02	33.39 ± 7.13	40.96 ± 5.59	41.23 ± 4.90	31.20 ± 0.00	26.40 ± 0.00	41.60 ± 0.00	40.00 ± 0.00
Rock	39.20 ± 13.75	64.00 ± 7.07	76.00 ± 3.16	76.40 ± 2.97	20.00 ± 0.00	66.00 ± 0.00	56.00 ± 0.00	64.00 ± 0.00
ScreenType	32.53 ± 4.84	35.63 ± 4.40	34.51 ± 4.14	37.12 ± 3.93	33.60 ± 0.00	33.60 ± 0.00	35.73 ± 0.00	33.33 ± 0.00
SemgHandGenderCh2	51.90 ± 14.92	59.73 ± 10.87	66.80 ± 7.25	63.80 ± 10.73	42.00 ± 0.00	68.17 ± 0.00	60.33 ± 0.00	70.67 ± 0.00
SemgHandMovementCh2	28.13 ± 4.95	31.16 ± 4.70	36.31 ± 5.92	45.56 ± 2.93	28.89 ± 0.00	26.00 ± 0.00	32.00 ± 0.00	37.78 ± 0.00
SemgHandSubjectCh2	37.42 ± 8.82	43.20 ± 6.72	54.31 ± 3.37	66.71 ± 4.45	41.56 ± 0.00	44.22 ± 0.00	37.33 ± 0.00	56.89 ± 0.00
ShakeGestureWiimoteZ	60.40 ± 7.92	81.60 ± 2.61	92.00 ± 0.00	91.60 ± 2.97	70.00 ± 0.00	72.00 ± 0.00	82.00 ± 0.00	84.00 ± 0.00
ShapeletSim	62.44 ± 7.44	67.44 ± 10.92	92.00 ± 3.88	96.67 ± 1.96	52.78 ± 0.00	61.11 ± 0.00	86.67 ± 0.00	96.67 ± 0.00
ShapesAll	50.20 ± 3.03	63.03 ± 2.22	76.20 ± 1.83	83.13 ± 0.32	46.50 ± 0.00	52.67 ± 0.00	64.00 ± 0.00	71.83 ± 0.00
SmallKitchenAppliances	42.88 ± 15.27	55.68 ± 8.74	51.63 ± 12.25	57.60 ± 5.56	54.93 ± 0.00	56.00 ± 0.00	45.07 ± 0.00	54.40 ± 0.00
SmoothSubspace	68.53 ± 8.95	85.07 ± 5.20	88.00 ± 2.87	89.47 ± 2.60	58.67 ± 0.00	77.33 ± 0.00	76.00 ± 0.00	87.33 ± 0.00
SonyAIBORobotSurface1	57.20 ± 8.62	71.85 ± 8.28	84.86 ± 2.67	86.09 ± 3.97	67.22 ± 0.00	69.72 ± 0.00	70.88 ± 0.00	78.04 ± 0.00
SonyAIBORobotSurface2	66.09 ± 10.78	68.21 ± 5.28	77.21 ± 3.67	84.11 ± 2.29	62.12 ± 0.00	67.68 ± 0.00	73.87 ± 0.00	75.87 ± 0.00
StarLightCurves	63.03 ± 14.23	74.47 ± 15.08	81.86 ± 7.93	85.26 ± 3.54	64.24 ± 0.00	77.25 ± 0.00	85.65 ± 0.00	82.39 ± 0.00
Strawberry	51.46 ± 5.21	59.41 ± 6.38	58.05 ± 7.04	75.84 ± 0.99	54.05 ± 0.00	54.59 ± 0.00	52.97 ± 0.00	59.73 ± 0.00
SwedishLeaf	57.98 ± 2.72	64.99 ± 3.83	79.78 ± 2.56	85.89 ± 1.27	52.96 ± 0.00	56.48 ± 0.00	72.64 ± 0.00	74.24 ± 0.00
Symbols	80.94 ± 7.54	82.69 ± 3.43	88.04 ± 1.30	89.33 ± 0.82	86.03 ± 0.00	80.10 ± 0.00	91.16 ± 0.00	76.98 ± 0.00
SyntheticControl	84.20 ± 6.54	87.93 ± 3.95	95.07 ± 1.62	97.33 ± 0.85	75.00 ± 0.00	80.67 ± 0.00	90.33 ± 0.00	90.67 ± 0.00
ToeSegmentation1	58.16 ± 3.70	60.26 ± 8.54	70.53 ± 10.39	80.53 ± 2.75	67.11 ± 0.00	67.98 ± 0.00	67.54 ± 0.00	76.75 ± 0.00
ToeSegmentation2	75.23 ± 12.13	76.15 ± 12.44	78.00 ± 6.36	80.31 ± 4.70	69.23 ± 0.00	70.00 ± 0.00	66.92 ± 0.00	70.77 ± 0.00
Trace	99.20 ± 1.79	98.40 ± 3.05	100.00 ± 0.00	100.00 ± 0.00	97.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
UWaveGestureLibraryZ	46.44 ± 2.61	52.88 ± 2.03	55.54 ± 2.61	60.65 ± 2.84	38.11 ± 0.00	41.57 ± 0.00	51.20 ± 0.00	56.31 ± 0.00
Wafer	46.27 ± 7.11	48.01 ± 15.62	71.65 ± 9.58	82.53 ± 8.58	68.14 ± 0.00	67.07 ± 0.00	86.02 ± 0.00	80.50 ± 0.00
Wine	51.11 ± 7.48	48.89 ± 5.34	51.48 ± 2.03	49.26 ± 2.11	61.11 ± 0.00	57.41 ± 0.00	50.00 ± 0.00	48.15 ± 0.00
WordSynonyms	20.91 ± 2.73	32.92 ± 2.64	47.05 ± 3.87	58.90 ± 2.04	13.64 ± 0.00	21.79 ± 0.00	25.24 ± 0.00	30.09 ± 0.00
Worms	24.68 ± 8.95	38.70 ± 4.63	44.42 ± 6.26	48.57 ± 4.91	42.86 ± 0.00	35.06 ± 0.00	42.86 ± 0.00	44.16 ± 0.00
WormsTwoClass	47.53 ± 10.69	54.03 ± 7.72	54.81 ± 7.59	56.36 ± 3.85	54.55 ± 0.00	45.45 ± 0.00	51.95 ± 0.00	55.84 ± 0.00
Yoga	50.42 ± 4.51	50.64 ± 4.90	51.31 ± 4.34	57.73 ± 2.05	51.50 ± 0.00	55.20 ± 0.00	53.20 ± 0.00	53.63 ± 0.00
Average	47.93	54.98	63.27	68.48	46.95	51.20	57.06	61.01

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